AIDPATH

Artificial Intelligence-driven, Decentralized Production for Advanced Therapies in the Hospital

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[1] RO: research organization; SME: small or medium enterprise; MNO: multinational organization

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1 Excellence

AIDPATH (**A**rtificial **I**ntelligence-driven, **D**ecentralized **P**roduction for **A**dvanced **T**herapies in the **H**ospital) is a high-energy EU consortium, dedicated to enable and to augment the next-generation of personalized medicine with gene-engineered immune cells at EU hospitals through the use of AI technology. The exemplary embodiment of gene-engineered immune cells in AIDPATH will be T cells modified to express a synthetic chimeric antigen receptor (CAR-T) that are already a revolutionary novel treatment in hematology and oncology, and will also be useful for treating infections and autoimmune diseases. Conventional CAR-T therapy is complicated by complex logistics and supply from centralized manufacturing facilities, and inflexible manufacturing and clinical use schemes that disregard patient and cell product characteristics, thereby limiting patient access and therapeutic outcome.

AIDPATH will apply top-notch AI technology to integrate patient-specific data and biomarkers in CAR-T therapy, and apply flexible manufacturing schemes to obtain CAR-T cell products with optimal fitness and anti-tumor potency. AI technology will also be applied in pre- and in-process controls to optimize scheduling and resource planning to reduce cost and hospital resource utilization and to augment patient access. AIDPATH will establish a role model that integrates the hospital environment in a smart manner, incorporating also aspects of logistics, capacity planning, data management and cybersecurity by design. The project is designed to enable the smart 'bedside' provision of personalized treatments directly at the 'point-of-care', and to accomplish end-to-end automation of hospital-based CAR-T manufacture. A key deliverable in AIDPATH will be an integrated software and hardware package that allows rapid dissemination of the 'smart CAR-T manufacturing and clinical use' concept to hospitals throughout EU member states.

AIDPATH brings together a world-class consortium, consisting of key opinion leaders in science and translational medicine, innovative biotech SMEs and commercial partners, patient advocates, policy makers and regulatory experts. The consortium members are well-positioned to engage into a productive dialogue with regulators and patients to ensure acceptance and rapid implementation, adoption by EU hospitals and industry. A strategic aim in AIDPATH is to contribute to the sustainability and well-being of the EU health care systems, to place the EU at the fore-front (ahead of the US and Asia) of the global CAR-T movement and to ensure that EU citizens will continue to have access to excellent health care.

Personalized cell & gene therapies are transformative novel treatments, but at the same time introduce unprecedented challenges in manufacture and clinical use. The most clinically advanced embodiment of these advanced therapies are CAR-T cells. In this cell therapy, immune cells (T lymphocytes) are engineered to express a synthetic chimeric antigen receptor (CAR) and are thereby redirected to seek and destroy cancerous cells in the patient's body. CAR-T cell therapy is indeed a transformative new treatment in haematology with clinical proof-of-concept in patients with acute leukaemia, lymphoma (lymph node cancer) and multiple myeloma (bone marrow cancer)[2]. A conceptual appeal of this treatment is that the patients' own immune cells are redirected – providing a personalized treatment - and that a single infusion of CAR-T cells can be sufficient to eliminate the cancerous cells and to establish immunological memory that protects the patient from tumour relapse ('living drug paradigm'). CAR-T cell therapy in acute leukaemia and lymphoma is now an approved treatment that is highly demanded by patients and caregivers, and in the focus of attention of bio entrepreneurs, policy makers and regulators. Intriguingly, the success of CAR-T cell therapy as a role model for gene-engineered cellular immunotherapy in haematology and oncology, has spurred the development of this treatment for novel applications in infectious diseases (targeting viral infections and fungal infections), chronic and autoimmune diseases.

The challenge with engineered T-cell therapies is that they require a sophisticated process of manufacturing (i.e. sourcing, gene-engineering and expanding the cell product) and clinically use (i.e. preparing the patient, administering the cell product, monitoring the patient). Small alterations in either manufacturing and/or clinical use typically trigger a round of regulatory re-review and approval because they can change the safety and efficacy of the manufactured cells. This poses an enormous challenges for the hospitals that find themselves in the epicentre of this CAR-T & cell therapy revolution: with their specialized health care infrastructure and highly trained health care providers they are the prime centres of choice for administration of the novel treatments. Accordingly they are suddenly responsible for the complex tasks and logistics while still being

[2] Nature. 2017 May 24; 545(7655): 423–431. doi:10.1038/nature22395.

subject to enormous pressure to operate efficiently and economically. A 'smart hospital of the future' offering engineered T-cell therapy to their patients has to provide a sophisticated infrastructure, internal operations scheme and highly-trained dedicated staff that is proficient both in the manufacture and clinical use of these novel treatments. Only this way, clinical sites can ensure patient safety, positive clinical outcomes, and compliance with the socioeconomic demands of EU health care systems.

A critical and vulnerable component for a 'smart hospital of the future' in cellular immunotherapy is the ability to carry out the GMP (good manufacturing practice) process. In conventional approaches, manufacturing is done manually with the assistance of some automation platforms. There is also the fact that variability from patient to the patient means that the optimal manufacturing procedures can vary from batch to . Unlike centralised manufacturing there is no economy of scale and there are multiple individualised manufacturing efforts happening in parallel[3]. The difficulties in decentralized engineered T-cell manufacturing has led to the consideration of 'off-the-shelf' allogeneic third party cell products and cell lines that can be administered to any patient. However, these allogeneic approaches carry a significant risk of graft versus host disease where the foreign immune cells attack non-malignant cells. The patient's immune system may also recognize the therapeutic cells as foreign which may lead to product rejection and limited therapeutic efficacy. Product rejection in the allogeneic setting may be mitigated by the use immune system suppressing medications – which is however undesired as this would compromise the patient's immunocompetence and increase the rate of infections.

Artificial intelligence (AI), combined with process automation technology, has the potential to increase the efficiency and the quality of engineered cell therapy to a point the equals or excels what can be achieved with centralized manufacturing. AI approaches also have the potential to substantially increase the personalization of therapies through deep learning and the use of digital twinning concepts. This changes the role of hospitals: instead of just receiving and administrating drugs, they assume an active role in the manufacturing of therapeutics. Smart hospitals of the future will evolve into decentralized but interconnected hubs that offer development, manufacture and administration of personalized advanced therapies such as CAR-T cells from a single source.AIDPATH introduces a trustworthy, AI powered approach that allows automated and intelligent production of the therapy in the hospital and bolstering the capacity of centralized manufacturing while ensuring manufacture is reproducible, safe and economically viable. This project paves the way towards a future where engineered cell therapies are not limited by the challenges and inefficiencies of a complicated manufacturing process.

Vision for the smart hospital of the future

It is April 2027 and Stephanie, a young soccer player develops a nose bleed during a match. The ball or any other player has not hit her. The coach and her parents rush to her side when they see that it is not letting up. Her parents have noticed that she has been more fatigued and complaining of feeling cold all the time. Later that night in the Emergency Room, her parents are startled when not just the nurse enters the room. A whole group enters, including a kindly doctor. The blood test shows that their daughter has leukaemia. A day later, once their confusion and fear have transformed into a firm resolve to overcome this leukaemia, they are back in the hospital clinic.

Further evaluation shows that the young girl has acute lymphoblastic leukaemia. They listen intently as the doctor explains to them what her diagnosis means. The concerned look on the faces of her parents eases a bit when they hear that there is an effective therapy consisting Stephanie's own immune cells (T cells) engineered to express a chimeric antigen receptor (CAR) that recognizes a surface molecule on the leukaemia cells and destroys them. A nurse draws more of her blood. That blood sample is packed in ice and taken down the hall to the AIDPATH system. The blood samples are attached to a machine which takes them up, and the technician enters in the patient's name which pulls up her medical record which the technician checks to make sure that all the relevant data fields are complete. Unbeknown to the patient, a complex AI analysis and logistical manufacturing chain is set into operation, just a couple of stories below her. The artificial intelligence that examined the patient and cell characteristics suggests a process sequence that is tailored precisely to the patients profile to provide the much desired CAR-T cell therapy. At the same time, all material and resources needed to produce the CAR-T cell batch is requested from central storage and allocated for by an intelligent software program. A trained person equips the machinery, which checks that everything is in place and correctly inserted. The process is triggered, T cells are extracted, and the

[3] Wang Molecular Therapy — Oncolytics (2016) 3, 16015; doi:10.1038/mto.2016.15

process of modifying using a viral-vector free process starts. Then, the CAR-T cells are expanded in a bioreactor equipped with sensors that performs completely autonomously - even samples are taken and analysed automatically in a small robotic chamber. The results are fed back into an intelligent system control, which makes sure that the cells are manufactured fast and efficiently, and are provided in the highest quality. The only thing that the trained personnel has to do is to stand by, supervise the suggestions of the AI displayed through the system software and to monitor the process performance online. Finally, the modified set of CAR-T cells is produced and then undergoes a set of standard analytical testing which have been developed and promoted by the Innovative Medicine Initiative (IMI) projects imSAVAR and T2EVOLVE. Fourteen days after the blood was drawn, the automated system produces a CAR-T cell therapy product that is ready to be infused. The system also notifies the treating haematologist that the young girl has an elevated risk for cytokine release syndrome. Here therapy is administered in the ICU with close monitoring and appropriate treatment ready in the case she develops CRS. She does not and she has a complete response to the therapy. Her response and her data, clinical and manufacturing, are recorded and placed in a database.
The treating haematologist presents Stephanie's data along with a whole series of other data from this and other hospitals as part of the AIDPATH stakeholder community meeting. In addition to results, hospital administrators are presenting how the AIDPATH system has been implemented in their hospital and is saving costs. Most exciting are the presentations about new types of therapies and new indications for CAR-T cell therapy where the attendees brainstorm on how the AIDPATH system needs to be adapted to broaden out the success seen in haematology to other areas of medicine.

1.1 Objectives

The overall AIDPATH objectives are to:

1. Define and develop an automated robotic, modular manufacturing platform for hospital-based production of multiple batches in parallel of engineered T-cell therapies that is (i) adaptable (ii) incorporates extensive fully automated at-line analytics and (iii) allows for GMP compliant production that is efficient and reliable;
2. Produce a set of artificial intelligence solutions capable of supporting the manufacturing process, as well as the resource and product management using the concept of 'digital twins' so that engineered cell therapies for cancer treatment are optimized and personalised for patient characteristics increasing therefore efficiency of hospital operations and ensuring viability.
3. Establish a data architecture to support the AI solutions and a continual learning and optimization process of automated engineered cell therapy manufacturing providing an end-to-end capability
4. Convene a stakeholder community to provide insights on stakeholder perspectives and to gather regulatory input to gain and understanding of how to make the adoption and regulatory approvals of AI-enhanced automated advanced therapy manufacturing system as efficient as possible;
5. Develop and implement a plan to sustain project assets by validating business models that take into consideration the health economics of engineered T-cell therapy.

This translates into the following innovation, impact and exploitation objectives:

Innovation Objectives

- Automated and robot-assisted cyberphysical manufacturing pilot to pave the way towards autonomous self-adaptive CAR-T manufacturing flow operating with hospital infrastructure
- Compilation of a comprehensive data-base for patient derived variability of CAR-T performance, from existing data of our clinical partners and high throughput experimentation (e.g. Capturing impact of age, sex, genetic and disease background).
- Intensification of data-generation though employment of robotics for analytical assays and sampling
- Establishment of an AI-based digital twin for each batch capable of catching donor-derived variability
- Implementation of an AI-based control of bioreactor operation using multiple targets i.e. optimising bioreactor operation in terms of quantity of CAR-Ts and quality as defined through our CQA panel
- Implementation of an AI powered smart scheduling and logistics for resources and adjacent processes

- Holistic IoT infrastructure for AI enabled manufacturing and logistic processes as well as data storage and tacking, which integrates into the hospital environment in a smart and seamless manner
- Demonstration of quality and efficacy of CAR-T products manufactured in the AIDPATH pilot

Impact Objectives

- Demonstrate that AI based technologies are effective for resource planning and for automating the process of manufacturing advanced medical therapies by fully implementing an AIDPATH system in a hospital and delivering CAR-T cell therapies to at least ten patients during the project thereby providing the scientific, clinical outcome, and cost effectiveness rationale for supporting wider adoption of the 'smart manufacturing' hospital concept.
- Build the basis for a deployable platform in alignment with regulatory requirements and a cybersecurity by design approach so that pilots can be initiated in multiple hospitals towards the end of and after the project's funding period.
- Transform the stakeholder community developed during the project into a European led innovation ecosystem that attracts funding to build upon the outputs of the project thereby driving the concept of the 'smart manufacturing' hospital forward

Exploitation and Dissemination Objectives

- Development of standards, building blocks and guidelines for AI in the Hospital, that allow wide-spread adoption of AI and AIDPATH technology
- Leveraging the network of the innovation ecosystem consisting of companies, clinics, associations, regulatory bodies and healthcare policy makers to communicate and educate about the AIDPATH results, gain visibility, trust and acceptance.

1.2 Relation to the work programme

AIDPATH aligns excellently with the work programme topic "H2020-DT-2018-2020: AI for the smart hospital of the future":

Call topic themes	***AIDPATH approach***
Health systems need to undergo adaption if they are to adequately respond to future population health needs. (…). In particular, ***artificial intelligence*** *and* ***robotics****, have the potential to transform health and care facilities*	The focus of AIDPATH on AI and robotics to support hospital-based manufacturing of advanced medical therapies addresses this challenge directly. One of the biggest future population needs is the ability to deploy advanced medical therapies for which society has invested heavily on a wide scale. The process of deploying many advanced medical therapies such as CAR-T cell therapy are complex. The complexity is further complicated by the need for the use of GMP processes. These factors taken together are exactly the types of processes that AI, and robotics make better, more efficient, and more cost effective.
Piloting at scale is needed to prove the transformative impact of AI. Pilots need to be embedded in operational health and care settings.	By the end of the project there will be a fully functional AI automated CAR-T manufacturing fully operational in one of the leading CAR-T cell therapy clinical centres in Europe. Furthermore, the efforts in stakeholder engagement will be focused in identifying and readying other hospitals to begin their own pilots based upon the processes, software, and know how that is developed.
Pilots should enable or support clinical, diagnosis and treatment, etc. carried out with clinical outcomes ***comparable to human delivered procedures*** *and with comparable results.*	The process of deploying many advanced medical therapies such as CAR-T cell therapy are complex is currently a predominantly human delivered procedure. The complexity is further complicated by the need for the use of GMP processes. CAR-T therapy in particular is particularly relevant because to deliver the therapy to a patient requires clinical, diagnostic, care organization, and logistical aspects. AIDPATH intends to deliver a platform that will make use of AI to make the process of delivering CAR-T cell therapy efficient and scalable. The technology will be tested and validated at the clinical site by production of CAR-T cells and results will be compared with the state-of-the art process.

AI based systems (....) have shown considerable promise so far, however their effective use (...) depends on their ***successful integration (and acceptation)***	AIDPATH places particular emphasis on building a participatory stakeholder community that will then be transformed into an innovation ecosystem in recognition of the fact that multiple different types of stakeholder have to accept and be engaged in attaining the future vision of personalized therapies being manufactured and administered at a SMART hospital. In addition, the AIs will be trustworthy by design, incorporating aspects such as transparency and accountability in accordance with the EU guidelines.
Address any aspect of health facility operations (e.g. diagnostics, treatments, logistical aspects)	AIDPTAH will deliver a holistic approach to smart manufacture in the hospital. Therefore, multiple AIs will be developed addressing four important areas: CAR-T cell production (1) process prediction, (2) process control, (3) manufacturing scheduling and (4) resources planning
Solution performance measured against health care metrics	After deployment of the AIDPATH IoT platform at UKW, it will be tested and validated the by performing trial runs on the actual platform. Bioprocess results such as yield, duration and doublings will be compared to the performance of state of the art processes. The generated product will be tested *in vitro* and *in vivo* for growth potential, phenotype, qualify, functionality, transfection rates, efficacy and bio distribution for benchmarking against manual processes.
Development in cooperation with health and care facility partners	The AIDPATH consortium includes (1) world-class leaders in CAR-T therapies, including two clinical sites (UKW, FCRB) providing their insights from the hospital infrastructure standpoint and (2) a company with extensive background in hospital and healthcare data management (ORT). The project will also set up an advisory board and build a stakeholder community which includes other hospital sites, healthcare policy makers and related associations.
Consideration of wider dimensions (e.g. impact on resources, staff)	The overall target of the project will be to minimise the need for specialised operators during CAR-T production. Potential expert profiles will be needed to also interact and interface with medical doctors (regarding decision making) and nurses (operational tasks). Efficient resource management will allow a high resource availability and low waiting times.
AI-based systems that combine ***digital and physical*** *services*	The project will implement an AI-integrated cyberphysical infrastructure that connects all devices of the CAR-T cell production plant to the AIDPATH software and data solutions in the sense of an internet of things. Services provided by the system include physical production and testing of the product, intelligent manufacturing control and scheduling up to data storage and resource optimization.
Individualised and integrated *care solutions*	AIDPATH enables the production of truly personalized therapies completely in proximity to the patient integrated into the smart hospital of the future, including connection to hospital IT and logistics. AIs will allow tailoring of the process towards individual patient history and requirements.
Integration of autonomous ***AI-driven smart components***	AIDPATH process control is smart and autonomous in a sense that it operates without human intervention necessary and governs all operations in the system (e.g. bioreactor parameter adaption, resource rescheduling) through direct integration of the AI decisions.
Direct and positive impact on ***effectiveness and efficiency*** *in the health care context*	AIDPATH will allow provision of CAR-T cell products in a better quality, tailored to the patient and produced in shorter time. The installation of the smart infrastructure at the hospital site reduces transportation times and enables a lean production and resource planning for just-in-time logistics. The collection of extensive data sets on process and products allows further optimization and extraction of insights on CAR-T therapy in the future

Integration of health and care partners *in the design of the pilot, the development of performance indicators and access to the relevant operational environment*	A demonstrator pilot will be implemented at the clinical site at partner UKW with additional input from Partner FCRB to ensure a technology that is deployable across sites. The partner ORTEC is currently engaged in supporting hospital resource planning with AI. Development of the technologies will integrate feedback from the projects advisory board and stakeholder-centric design principles.
Demonstrate likely "at scale" ***benefits in efficiency or cost reduction***	Cost reduction is crucial for this type of product due to its current pricing (>200K/treatment). Our solution will directly impact this area since automation through robotics and optimization through the use of AI will result in products that will be produced with lower cost. The possibility to parallelize production and intelligently schedule processes will ensure intelligent utilization of resources. In addition to the integration with the hospital organisation through AI will allow for efficient use of the overall facility reducing costs associated with facility and resource utilization, personnel and operator costs, length of stay and recovery time of the patient, product quality and rejected batches due to poor product quality. Demonstration of cost effectiveness will be one core output from the project.
Effectiveness of any novel service models *in providing economic justification for scale-up investment*	For smart hospitals to create, deliver and capture value by means of the innovative AIDPATH platform, they need to innovate their business models. This project will develop a business model that will serve as a basis for discussion with industry, investors and stakeholders and will help to shape an action plan for wide-scale technology implementation.
Development of ***European standards*** *that enable wide spread adoption and new market creation*	The build-up of the AIDPATH IoT infrastructure will be based on state-of-the-art standards and approaches in context Industry 4.0. AIDPATH will ensure knowledge gained will be cycled back towards the general public, by publishing new standards, building blocks and guidelines for AI in the smart hospital, thus providing a big step towards hospital digitalisation.
Privacy and cybersecurity	Due to the sensitivity of patient and medical product data, ensuring privacy and cybersecurity is a core element of the AIDPATH technology. Both will be incorporated into all AIDPATH components from the beginning by employing a security by design approach. Moreover, the AIDPATH consortium will engage with targeted communities in order to promote dialogue with policy-makers, the broader cybersecurity and technology ethics communities.
Alignment with the European Digital Innovation Hub	AIDPATH will closely align with the DIH Hero, mainly through the coordinator IPT who will interlink at the hub both through the regional node (Aachen) and on international level (see LoI from DIH in the Appendix to Section 4-5 of this proposal). Coordinator IPT intends to directly associate with the DIH Hero, in order to get insight from the community and in turn educate on and disseminate project results.

1.3 Concept and methodology

1.3.1 Concept

The AIDPATH imperative is to adopt advances from the field of Artificial Intelligence, robotics and CAR-T manufacture and for the first time bringing together individual cutting edge solutions into a holistic, AI-powered cyber physical approach to enable intelligent manufacture of advanced therapies in the hospital. In the following, we will describe the overall AIDPATH infrastructure for smart manufacturing and the automated platform itself, which is embedded into the overall framework.

AIDPATH infrastructure in the smart manufacturing hospital

The concept of AIDPATH (see Figure 1) consists of an infrastructure installed at the clinical site for the automated, data-driven CAR-T production considering all physical assets (e.g. patients, materials, devices and specialized staff) and existing IT and logistic systems in the hospital:

- On the physical level, the **automated CAR-T cell production** consists of heterogeneous machines and devices, which will be connected via standardized interfaces in the so called *integration framework*. All devices are governed through a service-oriented *control software* on top of the integration framework, allowing a centralized management and *supervision* of the process. The control software will be supported through an AI-based digital cell twin defining initial process parameters and an AI process control adjusting these parameters during the operation.
- The **manufacturing and resource planning systems** optimize the contiguous logistic processes via AI solutions adjusted to the AIDPATH context assuring unobstructed, efficient production processes. The *production scheduling* is closely linked to the control software generating schedules based on durations and characteristics of the current bio processes. The *resource management* considers demands of the *hospital IT and logistics systems* regarding the process-relevant requirements such as material supply, staff availability and hospital utilization and aligns them with the production schedule.
- The **LogiqCare Platform** will take care of a secure and reliable cloud-based data management in the AIDPTH project interacting with the *hospital data management* and the AIDPATH components, especially the **Artificial Intelligence**. LogiqCare will store and process historical data for the AI training as well as establish a continuous data pipeline for the AI application. Also, LogiqCare will lay the foundation for data exchange between different hospital sites in the future.

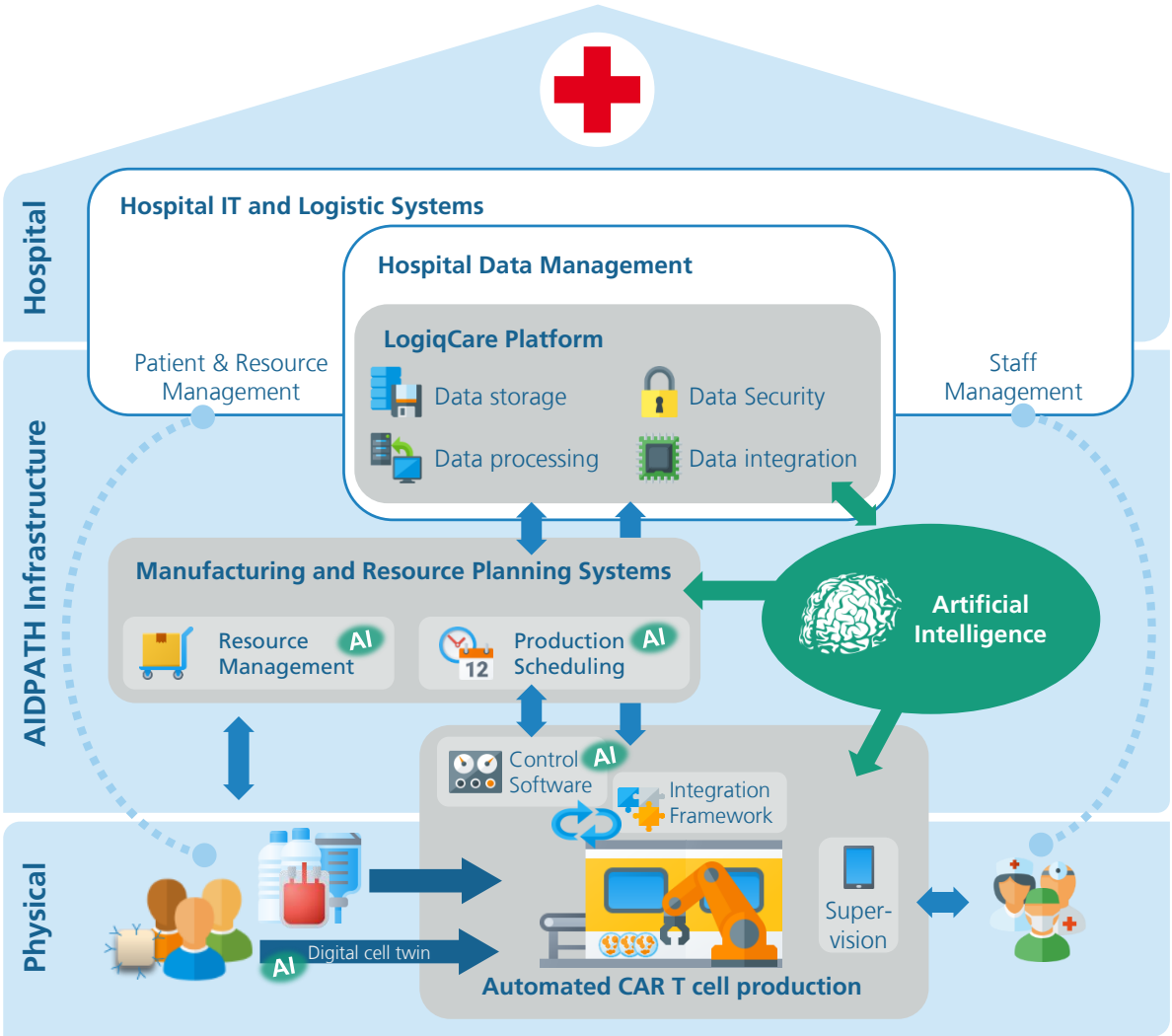

Figure 1: AIDPATH infrastructure integrated in the health care environment and connected to all physical assets needed for the automated CAR-T cell production including adjacent logic processes

Automated CAR-T manufacturing platform

If we zoom in onto the physical production level, the AIDPATH approach will deploy an automated and robot-assisted pilot plant for AI controlled manufacture of CAR-T cells. The AIDPATH manufacturing platform (Figure 2) will consist of 1) a manufacturing module and 2) a robotic quality control module (QC module), which are both controlled via the same smart control software:

- The **manufacturing module** integrates and connects state-of-the-art sensors and devices for completely closed production of CAR-T cells in a modular manner. This for example includes the Facer bioreactor (Aglaris) and a nucleofection device. The bioreactor thereby comprises of several modular cartridges that allow parallelized production of multiple cell batches. The devices are autonomously operated by the AIDPATH control software rendering human interaction only necessary for set up and process oversight.
- The **QC module** is a robot-assisted chamber that integrates a liquid handling robot as well as automated analytical devices such as a cell counter and flow cytometer. Coordinated through the central control software, the QC module processes samples from the manufacturing module fully automatically, runs complex assays and feeds back the results into the AI-powered process control. Thanks to the robot and intelligent process scheduling, no human intervention is required.

All data generated on the physical production level are gathered through the control software which ensures storage and communication with the AIs and the overarching AIDPATH infrastructure. This way, comprehensive data sets are generated to allow process optimization and collection of extensive processing records on each cell batch.

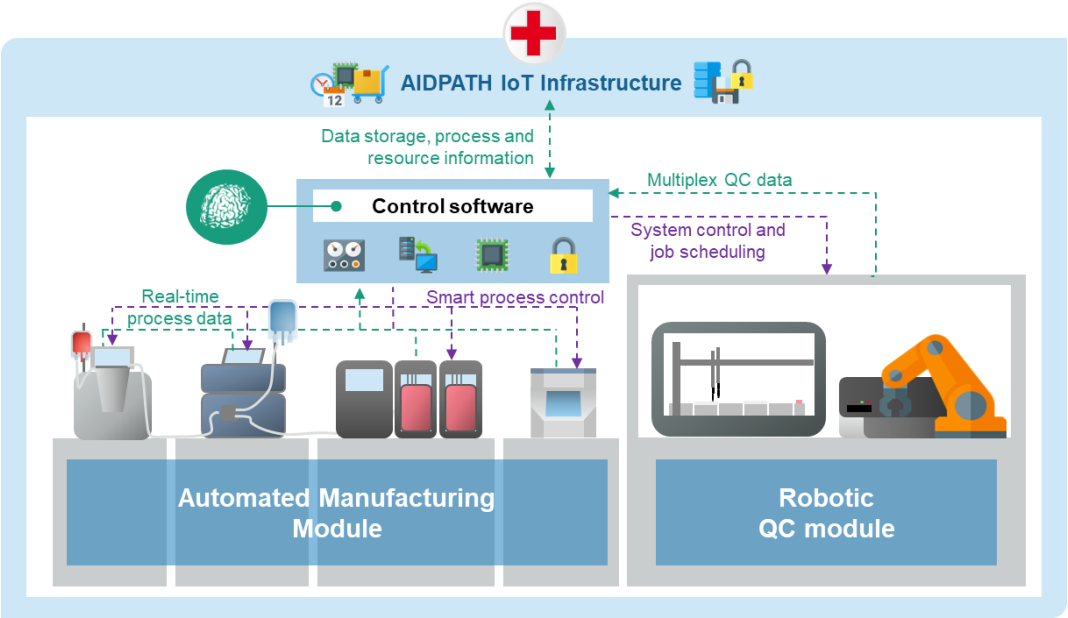

Figure 2: Automated CAR-T manufacturing system to be installed physically in the hospital, consisting of a manufacturing and a QC module. Both are governed through a centralized control software that integrates the bioprocessing AIs and communicates with the overarching AIDPATH infrastructure.

1.3.2 Project positioning

The recent regulatory approvals of CAR-T therapies and the impressive remission rates in cancer applications of between 69-90% for previously untreatable patients have demonstrated the clinical potential of such advanced therapies. However, the significantly high cost of the therapies ($300-$450k and above) dictates that the future challenges are associated with improving affordability and accessibility. However, a major translational barrier currently limiting the clinical and commercial translation of these products and their accessibility has to do with current manufacturing processes which are inherently inefficient, difficult to scale, resource and labour intensive, requires skilled personnel, and necessitates significant human intervention which can result in errors, contamination and final product variation. What is more, therapies are patient-specific, which means that the industry of scale that works for many other biotechnological products is not applicable. Current manufacturing processes are mostly still manual, non-supervised containing multiple open events limiting robust production of CAR-T products. Production in the clinics is mostly still associated with

small scale production that is only feasible to treat low numbers of patients. In order to make the therapies available for the patients on a wide scale, a disruptive approach to production is urgently needed.

AIDPATH is critical in addressing these elementary challenges in providing access to the therapy by reimagining the manufacturing system to bring costs down through automation and minimisation of expert operators' time, optimising the use and occupancy of expensive clean rooms, scale-out through decentralization as well as improved consistency and robustness. Thus, AIDPATH will deliver the first decentralized, AI-powered manufacturing approach for CAR-T therapy production (and ultimately other cell and gene therapies) in the hospital, that is smart, automated and in close proximity to the patient, thus reducing lead time and logistical efforts. Moreover, through AI-powered production the speed of production, approval will increase while the high risk which is inherent in such a personalized context will be quantified and mitigated.

The stated aim of the project is to develop a fully-validated, end-user tested, GMP-ready automated prototype installed in the hospital which can generate viral vector-free engineered CAR-T cells from multiple donors in parallel. The scope and destination of the project is to demonstrate feasible and economic production of efficacious CAR-T cells from more than one patient in a completely automated manner and to validate the technology demonstrator deployed at the hospital at scale. The holistic AIDPATH IoT infrastructure will be a global exemplar for how decentralised, point-of-care manufacturing approaches can be achieved, intertwining the key regulatory considerations necessary for a decentralised manufacturing process to be approved.

Critical for such an approach to be commercially viable are the development of corresponding operational and business models. AIDPATH has a specific and dedicated focus to examining the potential commercialisation, reimbursement and cost of goods strategies that can be employed to ensure that point-of-care processing is not just clinically viable but also commercially attractive. The approach followed and developed in this proposal will allow the use of the 'digital twin' as a decisional tool that will enable the comparison of key strategic investment and process scenarios, including centralised vs decentralised manufacturing, enabling to drive the cost of this value product down, while enabling to manage and mitigate, inherently high risks entailed in personalised manufacturing.

Translation: The AIDPATH project has a highly translational character, reflected by the composition of the partners, of which many are SMEs. In addition, our consortium has close proximity to the regulatory bodies through the partners UKW and IZI. This will facilitate a fast and sustainable translation of the project results to market and clinic. AIDPATH will elevate existing platforms, technologies and processes from current TRLs 5-6 through to TRLs >6, which is pivotal to ensure that Europe remains a global leader in this sector, not just with respect to innovation but also translation and exploitation. The novel AGL bioreactor is currently is in the system prototype stage (TRL 5). By the end of the project, this bioreactor will not only be production-ready, but it will have been integrated into the AIDPATH manufacturing system and tested and validated by UKW as an end-user (TRL 7). In addition, the final AIDPATH platform will be established as a pilot unit in the hospital by the end of the project.

1.3.3 Linked Innovation Activities

Types of activities	Initiative	Type	Partner(s)	Description/Website
Align, knowledge, and data sharing	imSAVAR	EU	UKW, IZI	Immune therapy toxicology including CAR-T models
Align, knowledge and data sharing	CARAMBA	EU	UKW	First-in-man clinical trial of CAR-T therapy in multiple myeloma using virus-free gene transfer
Align and share knowledge	AutoCRAT	EU	IPT	Robotic system for human mesenchymal stromal cells
Align and share knowledge	AUTOSTEM	EU	IPT	Patient specific gene editing for CAR-T
Align and share knowledge	JointPromise	EU	IPT, FORTH	Tissue engineered implant manufacturing
Align and share knowledge	BioManu II	Fraunhofer	IPT, SZTAKI	AI in automated stem cell production

Align and share knowledge	FoFeBat	National	IPT	Development of an AI-Toolbox to optimize battery production
Align and share knowledge	everPro & ACOP	Regional	IPT	AI application in optics production
Align and share knowledge	EPIC	EU	IPT, SZTAKI	AI in different domains in Hungary
Awareness and network	Future Targeted Healthcare Manufacture Hub	National	UCL	Hub of 35 companies
Awareness	EU-CARAT	EU	UCL	Improve CAR-T constructs
Awareness	REGMEDxB	Crossborder National	FORTH	Organoid based tissue manufacturing
Align and share knowledge	Mathematical Foundations of Artificial Intelligence	Regional	SZTAKI	Designing AI methods with theoretical guarantees for supervised and reinforcement learning problems.
Align and share knowledge	MIRACLE	EU	IRIS	Development of support medical decision system for cartilage status evaluation
Align and share knowledge	Openmind	EU	IRIS	On-demand production of entirely customized minimally invasive medical devices
Align and share knowledge	DataMine 4.0	EU	IRIS	Advanced Data Modelling and Analysis Applied to next Generation Industry 4.0 settings and the Internet of Things
Align and share knowledge	Ecobulk	EU	IRIS	Circular Process for Eco-Designed Bulky Products and Internal Car Parts
Align and share knowledge	MIRACLE	EU	IRIS	Development of support medical decision system for cartilage status evaluation
Align and share knowledge	Openmind	EU	IRIS	On-demand production of entirely customized minimally invasive medical devices
Align and share knowledge	AutoCart	Regional	AGL	Autonomous platform for T-cells culture
Align and share knowledge	Aglaris neotec	Regional	AGL	Develop a fully-automated bioreactor for cell therapy, tissue engineering, and pharmaceutical development

1.3.4 Methodology

AI Driven Manufacturing

AIDPATH aims to automate "**Value**" extraction from multiple sources of data through intelligent algorithm design, mathematical modelling and advanced control implementation leading to enhanced monitoring and operation of CAR-T cell manufacturing. Fundamental to all activities is the manipulation of the highly varied and large volume of data generated in WPs 3 and 4. This will include multiple data sources recorded from multiple devices (e.g. bioreactors, analytical devices, patient records…), measured at different frequencies (e.g. second, hour and daily, weekly), containing both numerical and textual information (e.g. sensor information, patient records and meta-information). The analysis of these data will governed by a **Data Analysis Methodology**[4] developed by UCL enabling data understanding, data preparation, modelling,

[4] Goldrick et al. Biotech. J. (2020)

evaluation and deployment of control algorithms for CAR-T manufacturing operations[5]. The algorithm selection will combine the applicable of proven statistical tools for data dimensionality reduction including Principal Component Analysis (PCA) and Partial Least Square (PLS) and will leverage innovative Machine Learning (ML) algorithms such as Support Vector Machines, Neural Networks, Rule Induction, Genetic Algorithms (optimize with hill climbing/gradient descent/simulated annealing) and Multi-agent systems (for modelling stochastic systems). These algorithms will be utilised in WP 3 and 4 enabling advanced classification, regression analysis and automated decision making. Some preliminary data is shown in Figure 3 evaluating the performance of a selection of algorithms to predict the antibody concentration from 200 Raman spectroscopy measurements recorded on a micro-bioreactor system (<15 mL). This novel sensor technology and advanced modelling approach will play a critical role in the CAR-T HT experiments planned in WP 3. The digital twin (***AI-CAR$_{Twin}$***) developed in WP 3 will make use of Genetic Algorithms (GA) to automatically and efficiently generate a mimic of individual patient CAR-T cells. The model parameters of ***AI-CAR$_{Twin}$*** will be initialised based on correlations generated from patient clinical information (WP 5) and adjusted in real time based on cell performance and growth using on-line bioreactor sensors (e.g. DO_2, *pH*, gas flow rates...). These metrics will be further coupled to metabolomics analysis that will provide predictive biomarkers for the potency of the CAR-T cell product. For this a robotic protocols will enable assay automation and will be employed in WP5 as an at-line quality control that will be linked with a metabolome library active at FORTH[6]. The development of the digital twin will build upon the significant work published in this area by the consortium members outlined in www.industrialpenicillinsimulation.com[7,8] and in the paper describing Multi-Agent Modelling Simulation of In-Vitro T-Cells for Immunologic Alternative described in Nettleton et al. (2020).[9] A great emphasis will be given to the trustworthiness of the developed AI's assuring that they are lawful, ethical and robust and fulfil the requirements of the *ethics guidelines for trustworthy AI*[10].

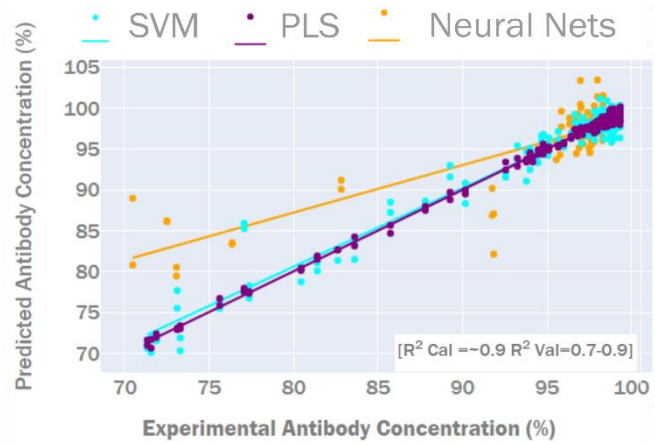

Figure 3: Raman spectra ML Prediction Performance

AI resource management and production scheduling

One of the challenges of CAR-T cell therapy is the efficient allocation of medical resources over time, where the resources include the key personnel, the scarce medical equipment, and materials. This is a kind of *resource constrained project scheduling problem*, where the projects are determined by the patients' treatments. Each project is a network of tasks that represent the main steps of the treatment, which have to share on the medical resources. The tasks are temporarily related, i.e. CAR-T harvesting and preconditioning chemotherapy have precedence constraints, while the latter can run in parallel with CAR-T manufacturing. There are conditional tasks as well: handling of adverse effects is needed only if recovery monitoring requires intervention into the therapy (see Figure 4). The deterministic variants of the problem can be solved optimally by *mathematical programming* techniques, see e.g., Kis, 2005[11]. Unfortunately, our problem is further complicated by the fact that the exact duration of the tasks, for example the production of CAR-T cells in sufficient amount and quality, cannot be determined in advance, they are uncertain. Creating a rigid resource allocation plan in such situations would either involve significant risks of not meeting the schedule or would be too conservative. It is much more efficient to manage the resources by taking feedbacks into account.

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[5] Nettleton Book – Data Mining Commercial Data Mining: Processing, Analysis and Modelling for Predictive Analytics Projects, book

[6] Vernardis S et al. Metabolic Engineering 2013

[7] Goldrick S et al. J. Biotech. 2015

[8] Goldrick S et al. Computers & Chem. Eng 2019

[9] Nettleton et al. (2020) ICAART 2020, Valletta, Malta

[10] https://ec.europa.eu/digital-single-market/en/news/ethics-guidelines-trustworthy-ai

11 Kis, T.: A Branch-And-Cut Algorithm for Scheduling of Projects with Variable Intensity Activities, Mathematical Programming, 103, 2005

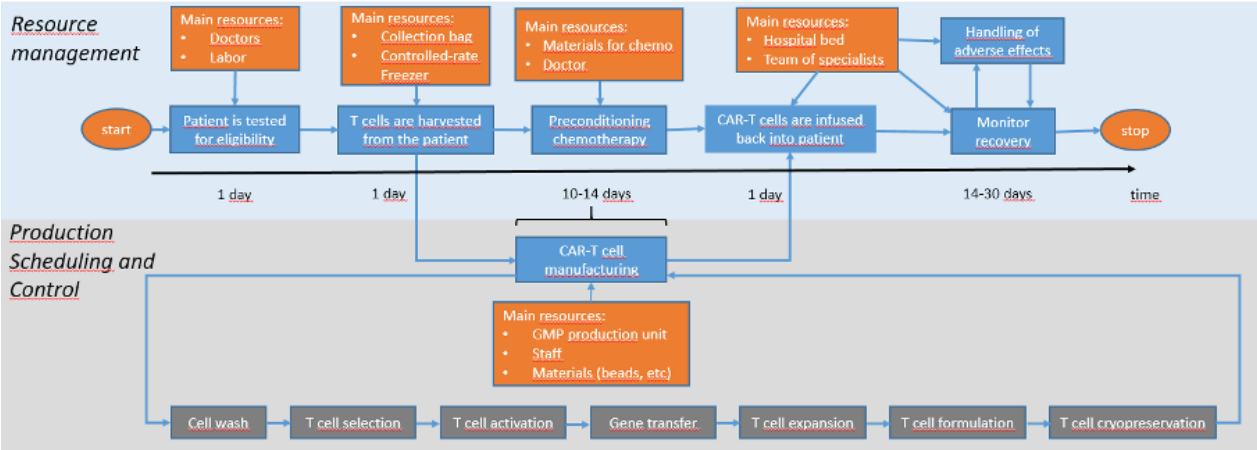


Figure 4: Two decision levels: planning for resource management, and production scheduling and control.

This results in a *resource allocation policy*, which is a dynamic rule that also considers the actual evolution of the controlled process.

Resource allocation problems in uncertain and changing environments can be efficiently modelled by *Markov decision processes* (MDPs). *Reinforcement learning* (RL) is a class of AI methods for MDPs which, in contrast to classical dynamic programming approaches, can work without a perfect model of the environment, as they can iteratively improve their solutions via sequential interactions with the (uncertain, dynamic) system. Naturally, in our case, it is not possible to directly apply a trial-and-error style approach in the real hospital; nevertheless, such techniques can be applied in a simulated environment. This leads to efficient resource allocation solutions[12], which not only scale well and handle uncertainties w.r.t. task durations but are also able to cope with changes in the dynamics of the underlying system[13]. There is a wide range of RL methods available, such as Q-, TD-learning and policy gradient, depending on the addressed problem. In one of our earlier projects, a policy gradient method combined with simulation was applied to optimize the controller of an automated stem cell production platform, in order to maximize the expected throughput (mean cell yield over a specified horizon). The approach resulted in 30% improvement[14].

The *scheduling* of the CAR-T cell production facility has similar characteristics as that of managing the ordinary hospital resources, but its decision horizon is shorter, and its variation in durations stems from the uncertainties of the various production steps, such as CAR-T cell expansion. Hence, a deterministic scheduling strategy would not work properly, we combine methods of *robust scheduling* with prediction techniques based on AI driven manufacturing of personalized therapies.

Since resource management and production scheduling are strongly linked, in the course of plan generation and adaptation, the scheduling of the production facility has to be taken into account. This calls for specific *aggregation* and *decomposition* techniques well-proven in traditional manufacturing, as described e.g. in Váncza et al, 2004[15] and Kis and Kovács, 2012[16].

We will also develop two distinct *simulation models* for the planning and scheduling levels: (1) one for the medium-term resource management and, (2) another for the short-term CAR-T cell production process. In both applications we will use *agents* for modelling the key resources and the projects. The input of the simulation consists of the demand of the projects, the resource availability and supply, the skill sets of the staff and the scheduled treatment and production processes including the uncertainties. The output is the outcome of the

12 Kis, T.: A Branch-And-Cut Algorithm for Scheduling of Projects with Variable Intensity Activities, Mathematical Programming, 103, 2005, pp. 515-539

13 Csáji, B. Cs.; Monostori, L.: Value Function Based Reinforcement Learning in Changing Markovian Environments, Journal of Machine Learning Research (JMLR), MIT Press and Microtome Publishing, Vol. 9, 2008 (b), pp. 1679–1709

14 Egri, P.; Csáji, B. Cs.; Kis, K. B.; Monostori, L.; Váncza, J.; Ochs, J.; Jung, S.; König, N.; Schmitt, R.; Brecher, C.; Pieske, S.; Wein, S.: Bio-Inspired Control of Automated Stem Cell Production, 13th CIRP Conference on Intelligent Computation in Manufacturing Engineering, Naples, Italy, 2019, in print.

15 Váncza, J., Kis, T., Kovács, A.: Aggregation - the Key to Integrating Production Planning and Scheduling, Annals of the CIRP, 53 (1), 2004, pp. 377-380.

16 Kis, T.; Kovács, A.: A Cutting Plane Approach for Integrated Planning and Scheduling, Computers & Operations Research, 39 (2), 2012, pp. 320-327.

processes, which helps evaluate in a risk-free environment how efficiently the resources are used. The possible applications are prediction and support of machine learning. We have already used *agent-based simulation* (ABS) in several projects to evaluate the benefit of resource sharing in cooperative production networks[17][18]. We have also successfully applied it in combination with machine learning techniques in the field of *stem cell production* [14].

IOT infrastructure, cybersecurity and data management

AIDPATH purpose is to build an end-to-end autonomous manufacture of advanced therapies, with the goal of improving availability, accessibility and thus accelerate adoption of patient-specific treatments. The novel approach of shifting the CAR-T cell manufacturing into the hospital and automating it requires a continuous data integration and a standardized involvement of all hardware and software components as well as the AI-based decision support. Since these data are considered electronic healthcare records (e.g. donor-derived cells, blood type, medical history, test results, etc.) they are qualified to be sensitive in terms of privacy and security at least to their owners or the different stakeholders involved.

The implementation of the **IoT infrastructure** is based on existing approaches and standards (RAMI 4.0, IIRA), which enable a reliable and structured build-up[19]. A level-based architecture approach is used to define the connection between all AIDPATH components and the connection to the hospital environment. The centralized control and monitoring of the CAR-T cell production plant will be managed by a control software whose decisions are supported by real-time capable AI solutions. With the IPT's extensive knowledge in setting up production plants in the life sciences area, which consist of a multitude of heterogeneous machines and devices, an efficient, service-oriented connection is achieved by an agent-based approach[20,21]. An additional layer of software will be implemented that manages resources (materials, cells, staff etc.) in an ERP-like manner and enables their monitoring and tracking over the treatment process including hospital supply chains.

The Inner IT core will be built on the **Logiq platform** for medical use cases, which is GDPR compliant, NEN7510 certified, secure-by-design and privacy by design. The Logiq platform has a basis in a service-oriented cloud architecture (SOCA) running on Microsoft Azure hosted in the EU. The Logiq Platform has various service solutions that could be used in combination: LogiqCare, Big Data Portal and Clinical Decision-Making Tools.

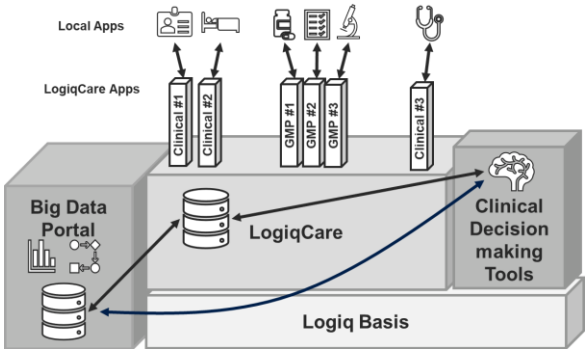


Figure 5: Logiq Platform

LogiqCare apps are designed to easy assemble a clinical software solution with a variety of apps in different domains. Each app will be used for handling of personalized clinical care data or pseudomized Good Manufacturing procedure (GMP) and laboratory data. Clinical care data are structured in the patient-case structure with an underlying hierarchy for the clinical diagnosis and treatment follow-up. The GMP facility will be based on pseudomized identification to allow sample identification, with minimal sacrifice for privacy. Data will be shared when need to know using role- (RBAC) and attributed-based access control (ABAC). Relevant data from this platform will synced anonymously through a one-way function to the Big Data Portal. The *Big Data Portal* has modules for data analytics and data processing. Analytics can be performed with both easy to use tools for direct analytical and advanced data tooling (R, Python) for machine learning and other artificial intelligence techniques. The Big Data Portal is also designed to exert business logic and coordinate batch processing of data. The data scientists will have access to their data sets to develop artificial intelligence solutions. The data processing module can orchestrate data processing in LogiqCare apps and/or batches in the

17 Kádár B., Egri P., Pedone, G., Chida T.: Smart, Simulation-Based Resource Sharing in Federated Production Networks. Annals of the CIRP - Manufacturing Technology, 67(1), 2018, pp. 503–506

18 Szaller Á., Egri P., Kádár B.: Trust-Based Resource Sharing Mechanism in Distributed Manufacturing. International Journal of Computer Integrated Manufacturing, 33(1), 2020, pp. 1-21

19 Weyrich and Ebert, C (2016): Reference Architectures for the Internet of Things. In: IEEE Softw. 33 (1)

20 Jung et al. (2018): Service-oriented Communication and Control System Architecture for Dynamically Interconnected Assembly Systems. In: Thorsten Schüppstuhl, Kirsten Tracht und Jörg Franke (Hg.): Tagungsband des 3. Kongresses Montage Handhabung Industrieroboter. Berlin, Heidelberg: Springer Berlin Heidelberg

21 Jung et al. (2018): Highly modular and generic control software for adaptive cell processing on automated production platforms. In: Procedia CIRP 72

Big Data Portal.The *Clinical-decision making* tools sections is prepared to productize medical device regulation (MDR)-compliant AI tools, that could be used within LogiqCare and/or the Big Data Portal.

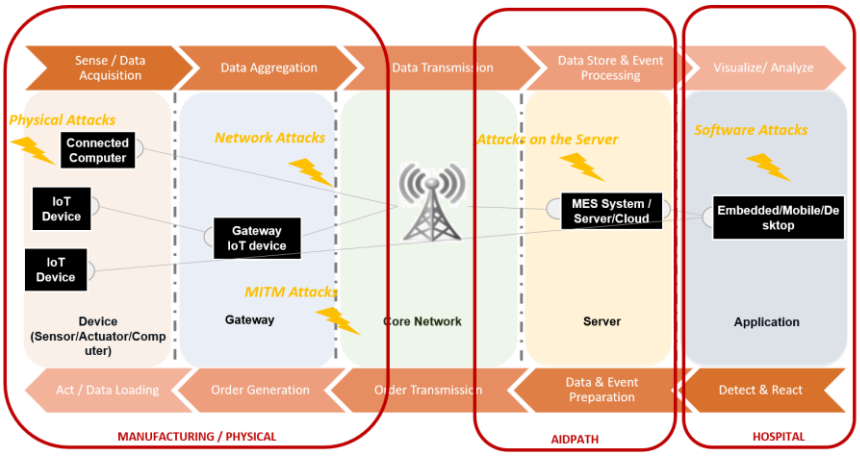


Figure 6: Cybersecurity by design

For **security and privacy issues** cybersecurity experts specialized in connected infrastructures are involved in this project and are planning to deploy an end-to-end methodology to secure all data and any potential sensitive asset that could be impacted by cyber-attacks. The methodology includes a full security risk analysis highlighting the security risks and defining the adequate countermeasures and a security architecture approach focusing on security countermeasures by design allows data protection at the three stages: data at rest, data in use and data in transit. Security Requirements will consider applicable standards and regulations such as EU Cybersecurity Act, GDPR, eIDAS, MDR, FDA, ISO 62304, ISO 14971, ISA/IEC 62443, ISO 15408, and Eurosmart IoT SCS. A reliable and secure digital environment will be achieved by including all network, service, device and application levels allowing to counter cyber-attacks such as local network attacks (e.g. sending bad data to network interface inside the gateway, man-in-the-middle (MITM) attacks between any equipment and its server.) or at the system level against physical (hardware access) and logical attacks such as server specific attacks (e.g. cross site scripting, SQL injection, brute force attacks, malware, etc…). Robustness and sufficiency validation will be gained through security assessment procedures (e.g. testing with automated tools, pen-testing, source code review, etc.).

Virus-free CAR-T manufacturing / Clinical translation and first-in-man clinical use of CAR-T

Providing the End-user perspective, the hospitals will bring in their research based and clinical CAR T cell experience to ensure that the AIDPATH manufacturing platform is based on state of the art technologies and fulfils the high quality requirements of adoptive T cell products. Therefore, the AIDPATH consortium comprises key experts in CAR technology and cancer immunotherapy, e.g. UKW has previously developed CARs specific for the B cell antigen CD19, the ROR1 tumour antigen that is expressed on solid tumours like breast and lung cancer[22], FLT3 which can be targeted on Acute lymphoblastic leukaemia (Jetani et al, DOI: 10.1038/s41375-018-0009-02018) and the SLAMF7 molecule on multiple myeloma[23] that are now successfully used in clinical trials in the US and Europe and have been adopted by biotech and pharmaceutical industry for commercial development.

With the AIDPATH consortium, we are developing a highly innovative platform that connects AI with hospital needs to create the smart hospital of the future. We are including truly cutting-edge technology into this approach: a variety of new targeting molecules with exquisite specificity and reduced immunogenicity, a novel modular multifunctional CAR design for enhanced safety and efficacy, and our revolutionary *Sleeping Beauty (SB)* virus-free gene-transfer strategy from minicircles[24] or SB transposase protein (Querques et al, 2019; DOI: 10.1038/s41587-019-0291-z). All technologies are protected by submitted/granted patents to provide a basis for successful subsequent marketing of the AIDPATH platforms as a GMP compliable CAR T cell manufacturing platform.

The stellar methodologies in AIDPATH:

i) Enhanced Sleeping Beauty transposition for high-level CAR gene-transfer into safe genomic loci:

Sleeping Beauty transposition in CAR-T cell engineering has long been explored, but was complicated by very low gene-transfer rates and substantial toxicity when SB transposase and CAR transposon were introduced into T cells as large circular DNA plasmids. We have solved both problems through the use of minimal DNA

22 Hudecek et al. 2010, 2013

23 Gogishvili et al, 2017, Miskey et al, 2019

24 Monjezi et al. 2016

expression cassettes called minicircles (MCs) instead of plasmids and could show that transfection of MC- (or mRNA-) encoded SB100X transposase and MC-encoded CAR transposon results in very high gene-transfer rates at low levels of toxicity (Figure 7). Recently, we have developed a highly efficient approach by using soluble SB-protein for transfection, which provides additional safety to the process[25]. Importantly, CAR-T cells engineered with this new strategy are superior to CAR-T cells transduced with lentiviral (LV) vectors due to lower genotoxicity and a highly safe genomic integration profile. MCs and SB protein can be produced either in-house or through commercial vendors, are easier to handle than LV (MCs are classified S1; LV is classified S2), and significantly reduce the time, effort and cost of generating CAR-T cells compared to conventional CAR-T products that require much more expensive viral gene-transfer vectors – which is a major milestone accomplished for the development of CAR therapy. Furthermore, we are now able to combine different gene modification strategies, e.g. SB-based transposition and CRISPR based techniques in a complete virus free, one-step protocol.

ii) GMP-manufacturing of CAR-T cells using virus-free Sleeping Beauty gene-transfer

We have established the manufacturing process for CAR-T cells using our novel Sleeping Beauty-based gene-transfer strategy under GMP-conditions (i.e. to produce clinical-grade CAR-T cells), and recently gained the

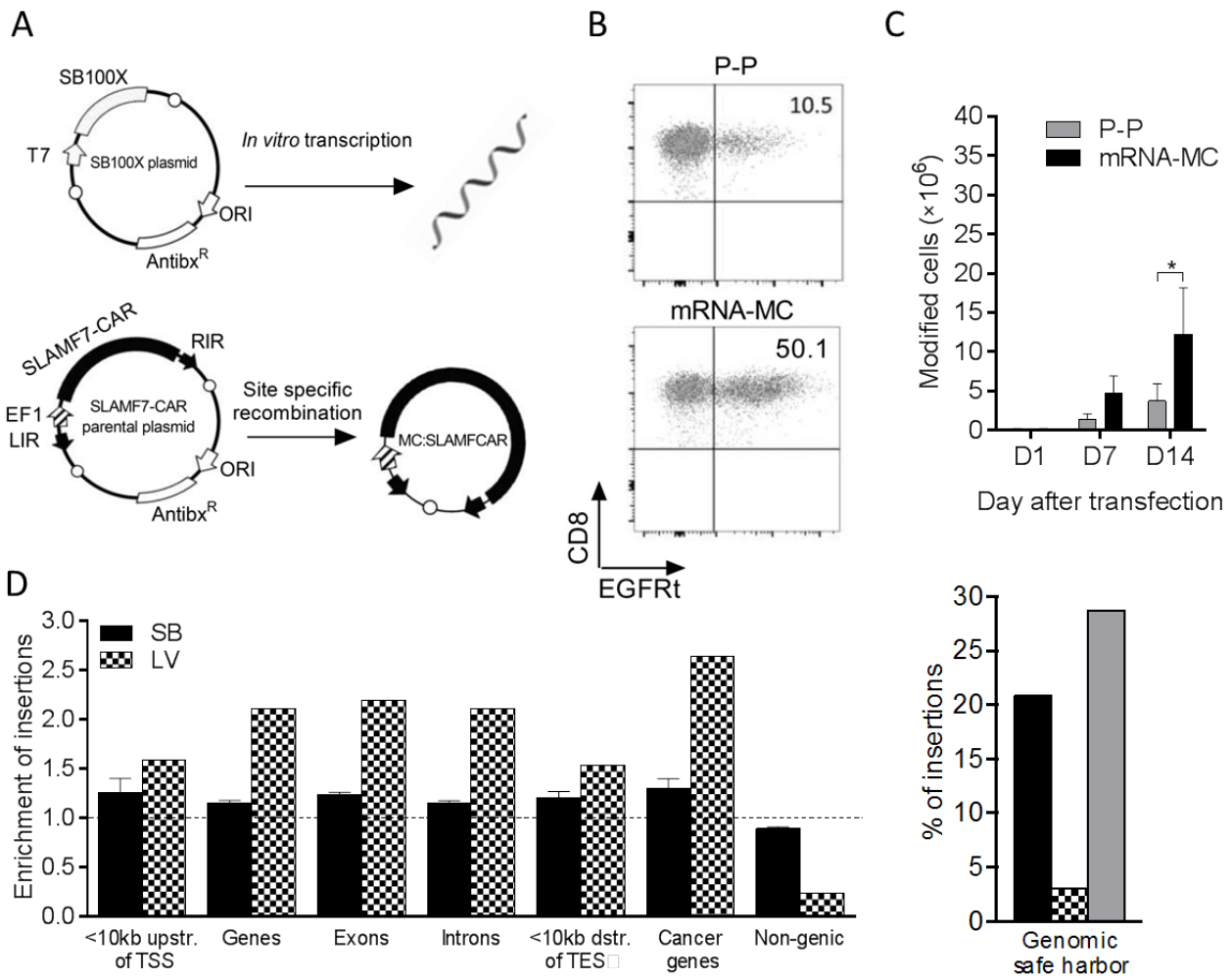


Figure 7: Virus-free Sleeping Beauty CAR gene-transfer. (A) Schematic of SB100X transposase mRNA and SLAMF7 CAR minicircle (MC) DNA as transposon donor vector, prepared from parental conventional plasmids through in vitro transcription (mRNA) and site-specific intramolecular recombination (MC). SLAMF7 CAR MCs contain exclusively the transgene and its promotor, but no bacterial origin of replication and antibiotic resistance genes. (B) Stable Sleeping Beauty CAR gene-transfer using CARAMBA's proprietary technology with SB100X mRNA and CAR minicircle (mRNA-MC) compared to the conventional approach with DNA plasmids (P-P). FACS plots show CAR-modified (based on EGFRt expression) CD8+ T cells on day 14 after SB-modification. (C) Higher yield of CAR-modified CD8+ T cells with mRNA-MC approach compared to the conventional plasmid-plasmid approach due to high gene-transfer rate and lower toxicity. (D) Comparison of SB transposon and lentivirus genomic insertions after CAR gene-transfer in human T cells. Fold enrichment of SB and LV insertion sites over the random expected frequency are plotted and show a close-to random integration profile for SB (left) and a significantly higher rate of insertions into genomic safe harbours for SB compared to LV (right).

[25] Querques et at, 2019

1st manufacturing license for SB-generated CAR T cells in the EU as part of the CARAMBA project. Multiple CAR-T manufacturing runs with T cells from healthy donors and patients have been performed successfully. These optimized processes will serve as a background to create the fully automated AIDPATH based manufacturing process. SOPs for the manufacturing process, as well as all assays for quality control and release testing are established, and can be adapted to AIDPATH.

iii) Target epitope selection and CAR design to derive a unique CAR with optimal anti-tumour function:
We have demonstrated in previous work that the choice of epitope on the targeted surface molecule and specific CAR design critically affect tumour recognition and CAR-T cell function (Hudecek et al. 2013 & 2015). Further, additional advanced functionalities will be incorporated into the CAR design which allow selection, detection, expansion and deletion of CAR T cells. By equipping our CARs with an EGFRt marker we are adding an additional layer of safety to our CAR T cell products. This EGFRt safety switch can be triggered with anti-EGFR antibodies to deplete CAR-T cells in case of toxicity or after a therapeutic window once tumour clearance has been obtained[26].

iv) Defined CD8+ killer and CD4+ helper T cell composition to enhance safety and efficacy:
We have shown that formulating CAR-T cell products with defined proportions of CD8+ killer and CD4+ helper T cells is advantageous to the use of undefined bulk T cell populations[27] due to several reasons. First, it is possible to use lower total doses of CAR-T cells to achieve therapeutic efficacy because CD8+ and CD4+ CAR-T cells work synergistically. Second, CAR-T cell products with defined composition ensure consistent pharmacokinetic between patients (time to and level of CAR-T cell engraftment, peak proliferation) which is the basis for predicting and eventually preventing side effects such as cytokine release syndrome. Further, we have demonstrated that the addition of specific cytokines during cell culture (e.g. IL-7/IL- 15) can polarize CAR-T cells toward stronger anti-tumour function and persistence[28]. We will apply both concepts into the automated manufacturing process by AIDPATH.

v) Phenotypical and functional evaluation of CAR T cell products
At the end-user site, we have cutting edge technologies available for phenotypic characterization and functional validation of the CAR T cells produced in the AIDPATH approach. This incorporates in depth analysis in vitro, e.g. real time measurement of target cell lysis (xCELLigence), multiplex analysis of cytokines, RNA-sequencing approaches to thoroughly determine the fitness (NanoString technology), activation- and exhaustion state in a multiplex assay format (CyTOF) as well as clonality of the final product. High-resolution microscopy (dSTORM) is available to perform in depth-analysis of surface expression of highly interesting proteins, e.g. the amount and distribution of single CAR molecules on the T cell surface. Additionally, we have all tools available to perform integration site analysis and copy number determination, which is frequently requested by regulatory authorities. Using the closely located animal facility and highly trained staff, we are able to engage state of the art in vivo studies using recently published xenograft models addressing the safety and efficacy of the generated CAR T cell product[29]

1.3.5 Sex and Gender

The AIDPATH approach delivers an integrated, intelligent and holistic platform technology that will enable enhanced production for CAR-T therapeutic products in particular, but also with prospect of expediting other personalized cell therapy products. Thus, ATs will become available faster and safer for a broad spectrum of patients, regardless of sex and gender. Nonetheless, this element will be actively integrated in the experimental and modelling design of the project.

At the current state of science, there is little known about the influence of gender on the manufacturing capability, e.g. whether gender is linked to shorter or longer expansion times. To date, there has been no variation according to sex in the efficacy of CAR-T therapies noted. Thus, we do not anticipate any implications on the manufacture or application of the ART products. Nevertheless, the gender of all samples with human origin as well as of all pre-clinical models will be recorded to allow for analysis of any (unexpected) influence it may have. The data regarding sex, gender and where available also other patient data (age, health, ethnicity, etc.) in combination with the associated results from the production processes will serve as a valuable dataset and allows us to assess distribution of slow and fast expanders in different subpopulations.

26 Wang et al. 2011, Paszkiewicz et al. 2016
27 Sommermeyer & Hudecek et al. 2016
28 Cieri et al. 2013; Casucci et al. 2013
29 Mestermann et al, 2019, Norelli et al, 2018

In order to avoid generation of gender-specific biases, we will make sure that the cell samples are balanced. Should there be any gender-specific requirements or characteristics of the manufacturing process, the AI will be capable of detection and adjustment of the process towards providing optimal conditions. In any case, by applying enhanced and adaptive expansion protocols and patient-tailored media, we will be able to adapt processes specifically to the patient's need, whether due to gender or other variations in the population.

1.4 Ambition

1.4.1 Current state-of-the-art (SOTA) and progress beyond SOTA

Novel bioreactor to enable the simultaneous T cell expansion of multiple patients in parallel

Current SOTA: The manufacturing process for an autologous CAR-T process begins with the collection of peripheral blood from the patient. This process requires the patient to have their blood continuously circulated through a device in order to **isolate** the T lymphocytes (T Cells), a process known as apheresis. The blood minus the relevant cell type is then returned to the patient. Many systems exist to isolate T Cells from peripheral blood, including the Fresenius Kabi LOVO® Cell Processing System, the Terumo COBE2991 and Elutra Separation systems, the Miltenyi CliniMACS system, and the Biosafe Sepax system. However, the cost and time for T cell **isolation** using such systems is significant, and they require trained and experienced operators. There is also inter-machine variability that impacts the starting material. Moreover, only the Miltenyi CliniMACS enables the **activation** of the T cells isolated from the blood. This usually requires a separate procedure and additional equipment.

The scalable and consistent production of autologous T cells suitable for clinical administration is a major manufacturing challenge. Many pre-clinical and clinical processes currently rely on sterile cell **culture bags** which lack support for process control, automation and homogeneity. This approach is also not suitable for treating large patient numbers due to the requirement for **manual manipulation** of the bag. Other issues include unpredictable cell yield due to the generation of gas and mass transfer gradients within the culture bags.

Bioreactor platforms such as the Wave™ systems offer an attractive alternative because they include embedded sensors for process monitoring and control and can be automated. Similarly, the Miltenyi Prodigy system has the benefit of combining multiple processing steps (e.g. cell selection, activation and expansion) in one device. However, the **scalability of such systems is limited** when producing autologous therapies for larger patient populations. The Prodigy is limited by its 'one-patient-at-a-time' approach: the expansion of T cells can take 10 – 21 days, and therefore during this period, the system cannot be used to produce cells for any other patient. To produce cells for 10 patients in parallel, a gene therapy development company, or a hospital administering cell therapies, would need to purchase a minimum of 10 separate devices. This greatly increases the cost and requires significant lab space.

Progress beyond the state of the art: A novel cell expansion system will be developed by partner AGL and specifically designed for CAR-T manufacture thereby achieving high density, controlled, automated expansion of multiple CAR-T cells in the same bioreactor platform, with xeno-free media.

Success metrics: Demonstration of the ability to prepare at least two CAR-T preparations in parallel

Novel modular, GMP-ready, AI enhanced automated manufacturing platform

Current SOTA: Although platforms exist that enable combination of multiple cell processing steps (such as the Miltenyi Prodigy), there is no platform currently available that allows for the *integration of multiple stand-alone devices within a closed, flexible, automated platform*. This is important for true flexibility, because each cell and gene therapy development and production process will be different, requiring different technologies, approaches and ultimately different products. The closest state-of-the-art technologies include Tokyo Electron's "Smart Cell Processing Technologies"[30] and Sartorius' CompacT SelectT robotic liquid handlers; these are solely research apparatus, are **not scalable**, and cannot be adapted to **new or additional technologies**. The partner IPT is one of the leading experts in developing robot-aided laboratory automation systems, such as the StemCellFactory, the StemCellDiscovery or the Autostem platform, which are representative for a number of robotic platforms that have been built both in Europe and worldwide in recent years[31].

[30] https://ct.catapult.org.uk/news-media/regulatory-news/cell-therapy-catapult-working-tokyo-electron-smart-cell-processing

[31] *Daniszewski* et al (2017): Automated Cell Culture Systems and Their Applications to Human Pluripotent Stem Cell Studies. In: SLAS technology, DOI: 10.1177/2472630317712220.

Progress beyond the State of the Art: Modular, flexible automated GMP-Ready CAR-T production platform that is made more efficient and through AI enhancement of the manufacturing process and resource planning.

Success metrics: 30% gain in efficiency, implementation of fully operating system that is capable of treating 40 or more patients per year, orthogonal log for quality controls enabling regulatory compliance and continuous verification.

AI enhanced data platform for precision medicine

Current SOTA: Precision medicine is an idea that has garnered sometimes intense interest. The concept is that by conducting a 'deeper' characterization of a patient or a tumour the therapeutic regimen can be tailored to be more specific and more likely to be successful in a given patient. The impetus for precision medicine is the simple fact that most clinical measurement and classification schemes tend to lump patients together into broad categories. This means that within any given disease category there is a substantial degree of clinical and biological variability. As therapeutics, such as CAR-T therapy, target sophisticated biological domains such as the immune system it is more likely that a given therapy will be more or less effective in treating smaller subsets of patients. Typical precision medicine approaches involve the molecular profiling of patient or tumour to identify specific molecular receptors or pathways and then tailor therapy accordingly. Autologous engineered cell therapies add an extra dimension because the process of manufacturing therapies from a patient's own blood means that the patient characteristics and manufacturing processes can affect the efficacy and safety of a therapy. This is why the production of cell therapies requires the conduct of analytical measurements to predict the safety and efficacy of a given batch of cells. The IMI project T^2EVOLVE, a project coordinated by the partner UKW, is examining ways to improve and standardize that analytical process. Yet the data generated from the analyses is not typically organized and analysed to learn and improve the clinical decision making around engineered cell therapies.

Progress beyond the state of the art: Deploying a platform that uses AI to organize and make use of the potential large network of sites administering cell therapies to analyse data at scale to create a learning system of continual improvement while expanding knowledge thereby fuelling new therapeutic research.

Success metrics: establishment of a platform that combines manufacturing data with individual patient characterization, and outcome data, achieving agreement of at least five other hospitals to provide data to the data platform.

Smart 'manufacturing' hospitals

Current SOTA: The Smart hospital is defined as "a hospital that relies on optimised and automated processes built on an ICT environment of interconnected assets, particularly based on the internet of thing (IoT), to improve existing patient care procedures and introduce new capabilities." Most Smart hospital concepts centre upon patient flow, improved diagnostics and surgical capability, enhanced patient safety, cyber resilience and trustworthiness.[32] The manufacturing of autologous engineered cell therapy is a complex process that is highly amendable to the ICT and automation process, yet this activity remains, particularly in Europe something that is only performed in specialised centres with dedicated facilities and teams.

Progress beyond the state of the art: Applying the concept of a smart hospital to the manufacturing of engineered cell therapies – the smart 'manufacturing' hospital allowing them to scale specialised procedures

Success metrics: Generation of a multi-stakeholder community with at least 50 organisations interest in the smart manufacturing concept, production of a set of guidelines, procedures and software.

Improved patient outcomes and health care / Improved socioeconomic impact of CAR-T therapy

Current SOTA: At present, there is a high rate of initial responses with CAR-T therapy in cancer patients that have failed conventional treatment with chemoradiotherapy; in many patients (approx. 50% in acute leukaemia), these responses are durable, still a significant proportion of patients ultimately relapses – and the reasons for this include attributes of cell product (suboptimal fitness/exhaustion) and tumour (enhanced resistance/antigen loss escape). At the same time, CAR-T therapy in responding patients is accompanied by significant side effects, including cytokine release syndrome (CRS) that can – in most patients – be managed clinically but increase the use of health care infrastructure and overall treatment cost.

Progress beyond the state of the art: AIDPATH will address these challenges and use AI to identify patient and cell product biomarkers to optimally tune manufacturing and clinical use to augment the rate of durable

[32] Smart Hospitals Security and Resilience for Smart Health Service and Infrastructures
November 2016, European Union Agency for Network and Information Security www.enisa.europa.eu

responses while at the same time reducing the incidence of severe CRS. This effort will lead to reduced hospitalizations and emergency room consultations, thereby reducing cost and improving socioeconomic impact.

Success metrics: Reducing treatment failure due to imperfect CAR-T product characteristics by at least 50%; reducing the incidence of severe CRS by at least 50% with CAR-T in haematology indications.

1.4.2 Innovation Potential

AIDPATH will utilize state-of-the-art AI technologies in order to (1) predict and improve continuous manufacturing process outcomes based on the highly variable patient input cell material and (2) integrate the system into the physical and IT infrastructure of the hospital by governing various discrete tasks such as logistics, scheduling and capacity planning. Thus, AIDPATH for the first time will combine available key innovations from different industries to pilot a holistic approach for smart, patient-specific, safe and effective therapy manufacturing in the hospital.

Innovation potential is defined by three innovation characteristics: novelty, technical feasibility, and market demand. Below is a table that describes the innovation potential of a 'smart manufacturing hospital' rated by their innovation characteristics.

The following innovations will be provided by AIDPATH:

Smart manufacturing hospital

Innovation characteristics	Rating	Rationale
Novelty	Very High	To our knowledge there are currently no 'smart manufacturing hospitals' that for AI driven manufacturing of CAR-T cells. There are parts of the manufacturing process that are automated but only on a limited scale and not in conjunction with the concept of smart hospitals
Technical feasibility	Very High	All the elements of manufacturing are well established as well as the elements of automation, and AI.
Market demand	Very High	CAR-T cell therapies are well established and a rapidly growing field. One of the biggest problems from the point of view of the hospitals is setting up the necessary GMP processes. From the point of view of a contract manufacturing centre the biggest problem is being able to scale up operations. The AIDPATH system solve both of these problems

The smart manufacturing hospital is comprised of a number of component innovations:

Innovation	Novelty	Feasibility	Demand
Modular bioreactor for cartridge-based, parallelized production	high	feasible	high
Robotic and modular manufacturing system	high	feasible	high
AI digital CAR-T twin for manufacturing	high	very feasible	high
AI for intelligent bioprocess control	very high	feasible	very high
AI models for resource planning	high	Very feasible	high
End-to-End AI pipeline integrating automated machine learning	high	very feasible	very high
Compilation of extensive CAR-T data sets	very high	feasible	very high
Holistic IoT infrastructure for hospital-based therapy production	very high	very feasible	medium
Manufacturing data platform for precision medicine	high	very feasible	high
High Quality personalized CAR-T products	medium	very feasible	very high

2 Impact

2.1 Expected impacts

AIDPATH will enhance the **emergence of European-led AI based pilots for the smart hospital of the future enabled by open system platforms** by producing and then demonstrating the effectiveness AI driven automated modular system to manufacture personalised CAR-T cell therapy. This will be achieved by implementing a pilot system within the hospital at partner UKW and then using the system to manufacture and deliver CAR-T cell therapies in a clinical setting. The AIDPATH solution will fundamentally be AI-based, embedding AI and machine learning approaches across all aspects of the system, including:

1. **A big-data approach to characterise the cellular starting material** by extract key and specific data with respect to age, gender, pre-existing conditions for donor and patient datasets to ensure the optimal bioprocess and manufacturing strategy is employed to yield a high-quality, efficacious therapy. By understanding the impact of key patient factors and variables in the starting material through a big-data approach, it will be possible to establish the appropriate process settings from the outset, including selection of cell growth medium and other control parameters. This will be complemented and validated by a high-throughput experimental approach which will be used to accelerate the development of using these different culture media, process conditions, different donor starting material and will be used to establish the basis for the AI tools.
2. **Modelling solutions and real time analytics** including mass spectrometry data will create an intelligent control solution ready to facilitate AI deployment within the bio-manufacturing sector, particularly for patient-specific cell therapies where each product batch is for an individual patient. This solution would allow adjustments to be made in real-time to maximize the yield, and the "AI-ready" nature of the system would enable AI to be applied in the future to determine optimal batch conditions based on patient information (e.g. age, height, weight, previous treatment regime).
3. **The Digital Twin** (DT) will capture not only the biochemical, medical and engineering background knowledge about the cell production system an its environment but will also provide passages between the real and virtual (i.e., modelled) worlds of manufacturing. The DT will be updated repeatedly by using incoming flows of data to reflect real production and its broader environment, whereas it will be used in designing the structure as well as in scheduling, monitoring and controlling the behaviour of production system. The challenge for AI is to find the right and efficient representations and methods to support this bi-directional exchange of data, information and knowledge. We are going to facilitate the standardized industrial take up of DT so that its full potential can be exploited by using the appropriate combination of intelligent faculties such as sensing, interpretation, representation, planning and decision making, scheduling, execution control, simulation and forecasting, learning, and visualization. Furthermore, non-representable biochemical, medical and engineering knowledge, experience and intuition can also produce very successful insights and heuristic solutions, thus human involve be integral in our approach. It will be also a guarantee to high-level conformance, transparency, explanatory power, and confidence of models used in all levels of decision making
4. **Establish machine-learning algorithms for automated process models to facilitate a data-driven approach to optimisation and autonomous live process control**. This will be complemented with experimental studies that will adopt a risk-based approach to identify the complex relationships between parameter controls/inputs and process outcomes to optimise the manufacturing process. This will involve systematic ranking of key variables after which a full factorial design of experiment study will be undertaken to investigate the impact on the model parameters. Such an approach is necessary to address the complexity associated with patient-specific therapies, as there is recognition within the AT field that outcomes will vary based on input variation (patient cells and reagents) in a manner that is highly dependent on process operation.
5. **AI solutions for production scheduling:** Production scheduling will facilitate accurate planning of patients' therapies and cost effective exploitation of the pilot which will be enabled by an efficient relation of AI prediction and robust or stochastic scheduling techniques. The AI prediction will be based on the process model developed under point 2 and will be optimized during the process based on real-life events.
6. **Automated clinical resource management:** Clinical resource management will be centred on the patient. Beyond managing the essential equipment and materials, staff management is also of key

importance. When implementing CAR-T cell therapy, unit and staff preparation will include all members of the multidisciplinary team including nursing, haematology staff and residents, neurology, emergency department and ICU. The problem will be modelled as a Markov Decision Process (MDP) and advanced Reinforcement Learning (RL) methods, such as policy gradient, will be used to learn an efficient resource management policy that is able to cope with the uncertainties and disturbances arising during the process. The solution will exploit the power of the Digital Twin (DT) developed within the project.

The UKW pilot will serve as a **demonstration of the effectiveness of AI based technologies in a range of healthcare tasks.** The AIDPATH platform will integrate smart robotics, machine learning and autonomous systems to develop a comprehensive solution embedded into an operational healthcare setting, **delivering an integrated digital** and **physical service for the hospital**. The system will incorporate Artificial Intelligence into several different areas of the hospital workflow. Firstly, through process prediction for the CAR-T cell production, by successfully modelling the complex stochastic environment represented by the bioreactor. Secondly, the incorporating smart sensors, which not only capture monitoring data, but can also perform sophisticated processing on it (e.g. Raman Spectroscope) to control the process in real time. Thirdly, Artificial Intelligence algorithms will be used in production scheduling and resource management, incorporating smart algorithms, which are able to manage and prioritize complex requirements and dependencies, such as the patient diagnostics and prognostics.

The AIDPATH platform will not be limited to the manufacture CAR-T cell therapies. It will be designed to be **flexible** in terms of the raw cell source it accepts and the potential operations (purification, isolation, gene augmentation/editing, expansion, testing, characterisation, monitoring and much more) that it can carry out. Any technology that can be modularised can be included in the platform. Different forms of bioreactors can be supported including emerging breakthroughs in gene engineering, or new forms of sensor technology. The platform will be an **open platform system** agnostic to type of technology or device providers. AIDPATH will utilize open software and standardized interfaces where feasible to avoid barriers that could arise from licensing or the use of proprietary systems. Where interfaces are established, protocols will be standardized, documented and published to allow future technology providers to easily integrate into the system. At the same time, we will ensure that the stored data and communications are secure by design and that privacy is ensured.

Data from the effort to produce and implement the pilot will be used to demonstrate cost-effectiveness through cost modelling that integrates both perspectives from the clinical side through the multiple clinical partners as well as the industrial perspective provided by the contract manufacturing organization Hitachi Chemical Solutions (HCTAS), which provide fundamental insight into the industrial cell therapy production landscape. The aim is to show that 'smart manufacturing' hospitals will not only reduce manufacturing and logistics costs; they will also increase patient access to advanced therapies. The process of achieving the pilot will also build up knowledge that is both technical and practical in terms of resolving the issues that arise from deploying such a system. This knowledge will be combined with the different stakeholder perspectives derived from members of the **AIDPATH stakeholder community**, which will include multiple types of stakeholders (see stakeholder impact map below)**,** to produce a societal guide for the implementation of AI driven automated advanced therapy manufacturing.

Results of the pilot and the guide will form an **effective basis for developing deployable applications** of the 'smart manufacturing' hospital concept. The deployment of the AIDPATH outputs will be fostered through the launch of an innovation ecosystem that is evolved out of the stakeholder community. The intent is that the innovation ecosystem serves as a catalyst for the further development of the concepts and assets of AIDPATH. The goal will be that at the close of the funding period of the project a number of hospitals (at least ten) are preparing to run smart 'manufacturing' hospital pilots. The innovation ecosystem will also serve as a multi-stakeholder collaboration network that will endeavour to **reach a high leveraging effect on other sources of funding, in particular regional and national funding.** This will happen through engagement of stakeholders in explicit efforts to work together to develop new projects that can serve as overarching structures that connect together separate regional and national funding efforts. For example, additional funding is likely to be made available by the German Ministry for Research and Education (BMBF). In fact, in a recent press release, the Federal Minister Karliczek announced to massively invest into an activation program to fund "Autonomy in the growth market of health" and "Investment in education and research infrastructures" among other[33]. These are calls with potentially excellent synergies with AIDPATH. In addition, with two Fraunhofer Institutes in the consortium, additional monetary resources can be secured through application for Fraunhofer funding programmes. Future Targeted Healthcare Manufacturing Hub which focuses on personalised therapy

[33] https://www.bmbf.de/de/karliczek-jetzt-massiv-in-bildung-forschung-und-innovation-investieren-11671.html

production and is integrating novel digital tools. Dr Rafiq and Dr Goldrick (UCL) are Co-Investigators of the Future Targeted Healthcare Manufacturing Hub and can leverage funding from the Hub given the alignment and complementarity of objectives. Similarly, there is funding available through organisations such as MedCity and the AT Network, which provide funds for translational activity and support key activity such as testing and feasibility studies. In addition, the personalised medicine fund from the Greek General Secretariat for Research and Technology could be harnessed towards directions related to personalised medicine manufacturing strategies. At the same time, given the transformative nature of AIDPATH, this is something that the consortium can benefit from and ensure that AIDPATH utilises actively leverages funding from such sources. For example, AIDPATH will strongly liaise with the DIH Hero, who provide additional funding opportunities through their own calls for proposals. At the end of the project, technologies such as the robotic platform or AI based bio manufacturing control will be in a good position to apply for additional funding through this and other DIH in order to exploit and roll out the technologies.

2.1.1 Stakeholder Community

The intent of AIDPATH is to build a robust participatory stakeholder community that will help to **contribute to trust and acceptance building in the AI technology.** This community will be a key resource of input for the development of the AIDPATH platform and it will form the basis of the innovation ecosystem which is meant to help achieve a critical mass of stakeholder collaboration that will drive the concept of the smart manufacturing hospital forward. It is therefore of vital importance that the stakeholders perceive the value of the impact AIDPATH will achieve to fulfil their own needs. The stakeholder impact map, depicted above, shows the different stakeholder types in relation to the objectives and outputs of AIDPATH and the different impacts relevant to them. AIDPATH will interact with a number of different stakeholders through established organisations which many of the consortium members have connections.

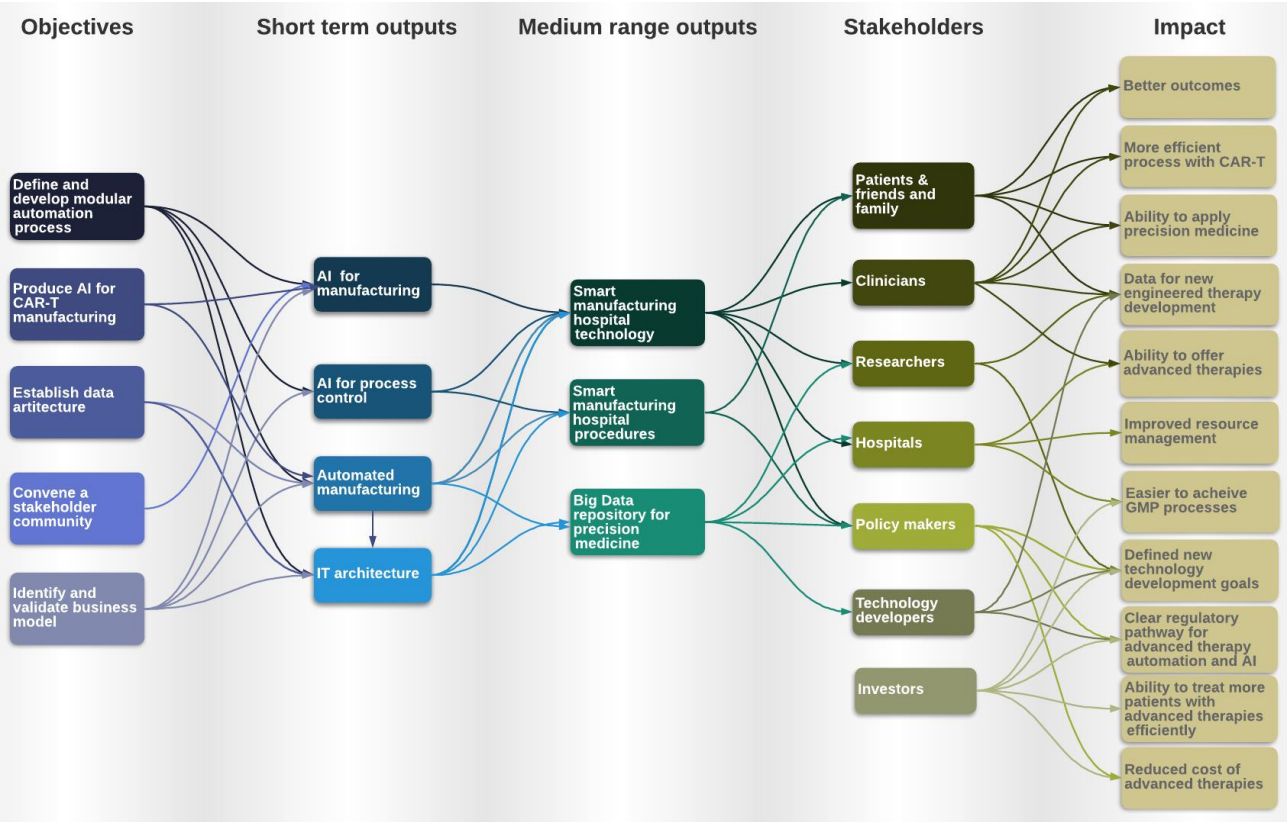


Figure 8: Stakeholder impact map

Stakeholders	Organizations
Technology providers	• Alliance for Regenerative Medicine (ARM) • UK EPSRC Future Targeted Healthcare Manufacturing Hub
Hospitals	• International hospital networks, i.e. HOPE

	• VUD (Verband der Universitätsklinika Deutschlands)
Researchers	• International & European Society for Cell and Gene Therapy (ISCT / ESCGT) • European Haematology Association (EHA) • European Society for Blood and Marrow Transplantation (EBMT)
Patients	• European Cancer Patient Coalition (ECPC) • European Patient's Forum
Clinicians	• AIDPATH clinical partners • European Haematology Association (EHA) • European Society for Blood and Marrow Transplantation (EBMT)
Regulators	• European Medicines Agency, EMA (EU) • Paul Ehrlich Institute, PEI (GER) • MHRA (UK), AIFA (ITA), AEMPS (ESP), FDA (U.S.)
Investors	• VC Investors (e.g. Syncona, Bain Capital, Angels in MedCity, HBM, 4Bio) • Corporate Investors (Novartis Venture Fund, BIVF, JNJ Innovations)
Policy makers	• European Haematology Association (EHA) • European Cancer Patient Coalition (ECPC)

2.1.2 Economic impact

Engineered T cell therapies like CAR-T carry unprecedented medical value (considering that a single-dose cell product can replace several years of repetitive chemotherapy and bone marrow transplantation in patients with acute leukaemia – the value is in excess of 2 Mio €). The cost of CAR-T therapy is dictated by the complex manufacturing, logistics, in-patient administration and foremost – the dependence of conventional CAR-T manufacturing on viral gene-transfer vectors as the major bottleneck in the supply chain. AIDPATH will focus on virus-free gene-transfer and point-of-care manufacturing to alleviate most of the complexity and cost associated with CAR-T therapy and create a sustainable way of using this treatment across all EU member states.

The market prize for CAR-T is approx. 300.000 € for CD19 CAR-T to treat acute leukaemia/lymphoma in Germany, and this prize varies between EU member states depending on agreements that have been made with the commercial vendors and the national health care systems. According the European Society of Bone Marrow Transplant (EBMT) registry[34] there were 418 patients treated with CAR-T therapies in Europe in 2019. When you consider that in Europe there were 209.898 new cases of either Leukaemia or Non-Hodgkin's lymphoma[35] approximately 0.1% are being treated with CAR-T cell therapy. Given the effectiveness of the therapy it would not be unreasonable that in the short term future this percentage would rise to 1% or even 5%. At a cost of 320K per patient treating 1% of the patients would cost 1.28bn euros and 6.4bn if 5% were treated. A modest 10% decreased in cost gained from the AIDPATH platform would result in a 128 or 640m euro savings each year. This also highlights that at the current cost levels patient access to CAR-T therapies is likely to be very limited. Therefore, the cost savings that can result from an AI driven automated system could have a substantial economic impact. Technology providers who can help deliver that cost savings either in isolation or as a component in a comprehensive platform as envisioned in this project would be able to gain market share.

In addition to CAR-T therapies, today, hundreds of companies and research groups are exploring **other advanced cell therapies for cancer**, notably Tumour-Infiltrating Lymphocytes (TILs) for solid tumours and T-Cell Receptors (TCRs) for squamous cell cancers. The **global market** for these therapies, which are ideal applications of AIDPATH, will be in the tens to hundreds of thousands. As costs and prices fall, and cell therapies are adopted as first-line treatments, this will grow into the millions.

Meeting the needs of these growing populations of patients with autologous therapies will be very challenging. AIDPATH, with its ability to **produce an autologous treatment for several patients at once**, by running multiple expansion modules, isolation modules or gene delivery modules in parallel, offers a realistic solution to this challenge in the short to medium term. Moreover, given its **inherent flexibility and equipment interoperability**, the platform can also be used in the future for allogeneic therapies.

[34] https://www.ebmt.org/ebmt/news/growing-community-car-t-cells-activities-europe

[35] https://gco.iarc.fr/today/data/factsheets/populations/908-europe-fact-sheets.pdf

2.1.3 Key impacts and metrics

Key Impacts	Metrics
Emergence of European-led AI based pilots for the smart hospital of the future, enabled by open system platforms	Successful comprehensive CAR-T cell production pilot and identification of at least ten hospital prepared to run pilots.
Demonstration of effectiveness, in use, of AI based technologies, such as smart robots, in a range of healthcare tasks	Implementation of an AI powered pilot in the hospital (UKW), validation of both logistic and bioprocess use cases, at least three validation runs with equal or exceeding cell number and quality as in the manual state of the art
Engagement of healthcare policy makers, investors, stakeholders and through the pilot	Assembly of an advisory board, at least four meetings with the advisory board, innovation ecosystem with at least fifteen members from diverse stakeholder groups
Effective basis for developing deployable applications	Generation of a guide that incorporates supporting data, technical and clinical know-how, as well as the societal perspective for implementing a SMART manufacturing hospital
Reaching a high leveraging effect on other sources of funding, in particular regional and national funding	Application for at least three synergetic projects and acquisition of at least one additionally funded projects synergetic to AIDPATH
Contributing to trust and acceptance building in the AI technology among all stakeholders (including patients, their formal and informal caregivers, decision makers, etc.)	Assembly of an advisory board with at least four meetings in total, publication of a societal guide, At least six publications in (scientific) journals and participation in at least four conference presentations with outreach to stakeholders
Growth of 'smart manufacturing' hospital pilots	Successful comprehensive pilot, identification of at least ten hospital prepared to run pilots
Reduced cost of autologous CAR-T therapy	Realization of at least a factor of four reduction in costs for autologous CAR-T cell manufacturing
Surpassing the cell expansion bottleneck	Demonstration of the ability to process at least two donor-specific CAR-T products at the same time in parallel using the AIDPATH system
Increased access to CAR-T cell therapies	At least 1.000 patients receiving autologous CAR-T manufacturing using at least some elements of the AIDPATH system within five years of the end of the funding period.

2.1.4 Barriers, obstacles and framework conditions

Healthcare and public acceptance for hospital-based therapeutic manufacture: Advanced therapies, such as CAR-T, are traditionally manufactured by pharmaceutical organisations who have a significant track record, capability and expertise in manufacturing therapeutics. However, long before pharmaceutical companies took control of CAR-T, hospital and clinical centres of excellence made these treatments. Centres like the University of Pennsylvania, Memorial Sloan Kettering Cancer Center, UCL and University of Wurzburg, and others figured out how to engineer CAR-T cells and ran the initial trials to test them. Drug companies were later involved mainly as a means to scale up production. However, there are significant benefits to enabling hospitals to manufacture CAR-T therapies, specifically the fact that hospitals can deliver the CAR-T therapies quicker because they do not need to ship cells to and from a drug company plant. This is a critical issue, because most patients currently treated with CAR-T have not responded to other treatments and their health is seriously compromised; some patients die between the time stem cells are removed from them and the time they are supposed to come back to the hospital for re-infusion.

Privacy Requirements: AIDPATH is expected to handle Personal Identifiable Information (PII) and Electronic Healthcare Records (EHR) of patients, users, employees and third-party entities. The General Data Protection Regulation (GDPR) requires to protect such data and therefore specific security functional

requirements must be taken into consideration by design. This aspect must be considered carefully to find a balance between the business functional needs and the privacy regulations.

Cost of CAR-Ts: the high cost of CAR-T therapies remains a major obstacle to their widespread use, particularly as first-line or second-line therapies (both Kymriah and Yescarta are approved for third-line only at this time). However, a major benefit of AIDPATH will be a radical reduction in the development and production costs of CAR-Ts and other cell therapies. Thus, we are confident that this obstacle actually represents an opportunity. This is evidenced from experience in making CAR-T in leading medical centres show that locally engineered CAR-T cells can be made for less than half of what they are currently priced by pharmaceutical companies.

Regulatory approval: AIDPATH proposes an important change in the development, production and delivery of advanced therapies – the location of the entire process ('vein to vein') at a large hospital. This contrasts with more centralised production strategies used by Novartis and Gilead, where patient blood is frozen, shipped to a central location (or one of a network of locations), and the resulting therapy then shipped back to the patient. Regulatory approval for this 'local' approach will be essential, and we anticipate extensive interactions with the relevant regulatory authorities including PEI, MHRA, HPRA, EMA and the US FDA. We have extensive regulatory experience within the team, as well as a plan for engagement (see WP 10). While our decentralized model is different, we believe it is also better – it **removes several points of failure** (transport, handling and logistics remain important challenges when working with fragile biologics), as well as reducing both time-scale and costs of production. The closed, modular nature of the AIDPATH manufacturing platform means that there is **no requirement for highly-trained scientist operators**. Note that because all processes are automated, monitored and documented, this will increase consistency and cell quality, and provide much more process documentation. All these improvements will help with regulatory approval.

Lack of standards: There are currently no universally-accepted technical standards for modular cell therapy production platforms. The most relevant standards (or best practice models) are in the robotic domain, with open platforms such as FiWare and UniversAAL and in the device connection area OPC-UA and SiLa. Our software will support interfaces to those modules, but the platform is not directly relevant enough that it makes sense to implement our software on it directly. In order to be flexible towards any kind of technology, the software utilized by IPT is capable of integrating devices with different standards and interfaces via software agents independently from communication protocols. In addition, we will strongly encourage usage of existing and development of new open standards, especially in the area where AI is utilized in the hospital. These standards will include existing initiatives from the AI perspective as well as domain oriented standards from hospital and medical care.

Financing the roll-out: It is to be expected that additional development, refinement, testing and accreditation/validation work will be needed, to take project end-result and make it into a product that can be installed into a hospital laboratory setting. However, if the approach and technologies are successful it is envisaged that a consortia of public/private enterprises and commercial organisations will be keen to develop this further. Importantly, the AIDPATH manufacturing platform will be developed with the regulatory framework in mind and developments will be informed by the relevant regulations through expertise from consortium partners and engagement with PEI, MHRA, EMA and US FDA. A specific deliverable in WP10 is the preparation and organisation of a scientific advice meeting with the EMA and US FDA in the early and latter stages of the project. Multiple partners in the AIDPATH consortium have significant experience of designing and running clinical trials, therapy development, clinical translation and gaining regulatory approval (e.g. partners WUR, UCL, HCATS), as such, the platform will be developed with the regulatory process in mind.

2.2 Measures to maximise impact

a) Dissemination and Exploitation

Dissemination outputs: The establishment of localized, hospital-based automated manufacture and delivery of CAR-T therapies will require novel AI-based tools and technologies, and will position European hospitals as exemplars for smart hospitals of the future. This will provide lower-cost CAR-T therapies for patients and this approach and outcome will be of great interest to a broad spectrum of audiences, from healthcare providers to patients, researchers to clinicians. Subject always to appropriate intellectual property management, we will communicate the aims, approach, technologies and results of AIDPATH to these audiences. The **message**, and the optimum communication **channel**, will vary from one **target audience** to another. The target audiences,

and how we plan to reach them, are summarised here; the relevant activities of the project are also described in WP10. In order to achieve a continuous and comprehensive dissemination through the project we going to publish at least **four dissemination items** (articles for conferences and journals, industry-focussed publication, white paper, etc.) per year in average and participate in at least **two event for dissemination and exploitation per year** in average.

Audience	Channels	Key Messages
Cell therapy researchers; cancer researchers; biomedical engineers	Papers, conferences, web, networking	Availability of excellent new technologies for producing cell therapies; innovative AI powered decentralized manufacturing infrastructure available for testing validating further new technology inventions; cell therapy research material of unprecedented quality; avoidance of preparation variation and multiple points of failure; facilitation of replication and validation of research results; new research opportunities; value of AIDPATH beyond CAR-T.
AI, Digital Technologies and Data Security Communities, Digital Innovation Hubs	Papers, conferences, open-source code, networking, web	Advances in robot-aided cell therapy manufacture and digital twins to capture the biochemical, medical and engineering background knowledge about the cell production system and its environment, and also provide passages between the real and virtual (i.e., modelled) worlds of manufacturing.
Researchers in automation and advanced manufacturing	papers, conferences, web, networking;	Information about modular cell therapy manufacturing platform, with high configurability and flexibility, as well as support for new technologies as they emerge. Potential for replication and re-use.
Technology and vendor companies	publications, meetings, conferences, personal and professional networks	Excellent opportunity to test and validate new technologies in a controlled close-to-market environment; invitation to develop technology fit for implementation into AIDPATH for a future-oriented, sustainable platform; availability of an environment that links their technology seamlessly to others.
Clinicians	publications, conferences, web, patient groups, media	Prospect of lower cost CAR-T and other cell therapies; enabling of use of cell therapies earlier in the treatment cycle; strong possibility of translation of exciting research results to the clinic; building confidence in high-value, safe and effective cell products.
Patients and patient groups	briefings, publications, conferences, websites, networking, media	prospect of clinical availability of CAR-Ts at much lower cost, potentially for a wider range of cancers; reduction of major obstacle to routine use of cell therapies; reducing the risk of manufacturing failure
Policy and regulators	publications, briefings, media, patient groups	A key obstacle to wider use of cell therapies (particularly CAR-Ts) has been overcome; cell therapy costs will fall; prospect of greater demand for cell and gene therapies – policy may need to adjust; encourage the dialogue on regulation of cell manufacturing, as it is disrupted by advanced technologies, facilitate regulatory approval for new products due to standardized manufacturing.

Other cell therapy and advanced therapy projects	personal networks, web, publications;	Scope for synergy, open publication of processes, technologies, training materials, engage in discussions and exchange learnings in this specific field, etc.
Projects in complementary domains (automation, bioreactors, advanced manufacturing, pharmaceuticals, sensors)	publications in relevant domains and across domains, conferences, web, networking	Important research developments in these domains; exemplars of multi-disciplinary collaboration; demonstrate potential to integrate novel technology into platform
Healthcare funders	publications, media, briefings, web	Prospect of reduced cost of cell therapies, especially CAR-Ts; prospect of increased demand for cell therapies; prospect of being able to serve larger populations of patients
Media	media staff, press releases, web	Wider access to CAR-Ts and other cell therapies, at lower cost, now a realistic prospect
Future industrial partners, including contract manufacturing organisations (CMOs) and therapy development companies	meetings, media, web, publications	Commercial opportunity; novel business models

2.2.1 Preliminary business model for sustainability

The preliminary business model for sustaining the outputs of AIDPATH will be that of an innovation ecosystem. An innovation ecosystem is defined as:

"…the evolving set of actors, activities, and artefacts, and the institutions and relations, including complementary and substitute relations that are important for the innovation performance of an actor or population of actors."[36]

This definition is depicted graphically in the figure the right. One of the best examples of an innovation ecosystem is Apple. Apple provides the smartphones, computers, podcasting, streaming, and cloud storage and invites others to innovate such as content providers within that ecosystem. It is both collaborative and competitive. One of the most important aspect of an innovation ecosystem is the underlying network of actors. An innovation ecosystem can also be viewed as a collection of layers made up of different innovation systems.

The innovation ecosystem developed out of AIDPATH will consist of seven innovation system. Each layer is described in the table below in terms of the relevant actors (stakeholders), activities and artefacts.

Actors (stakeholders)	Activities	Artefacts
Technology providers	• Implementing technology and systems • Collaborating on delivery • Developing project funding	• AIDPATH system • Standards • Guide • Data
Researchers	• Collaborative analysis of data from the AIDPATH system • Developing project funding	• Dataset
Clinicians & patients	• Educational activities on cell therapies in alignment with T^{2}EVOLVE • Developing local funding to implement pilot AIDPATH systems	• Educational materials • Promotional materials

[36] https://doi.org/10.1016/j.technovation.2019.102098

Investors	• Pitch and match for investment in technology developers	• Business models • Cost effectiveness analysis
Hospital	• Pilot system implementations	• Guide • AIDPATH System
Policy makers	• Defining policy to enable 'smart manufacturing hospitals'	• Policy briefs • Data on outcomes
Regulators	• Defining regulations	• Standards • Data • Guide

The foundation of the AIDPATH Innovation Ecosystem will be the stakeholder community that is developed during the project. This community will be developed as a participatory and self-directed community. The individual stakeholders will be encouraged to participate in the activities listed above. To foster community building members will be allowed to propose new efforts in line with the activities defined above. This stakeholder community will serve as the test bed for business model development. They will form the source of initial interviewees for interview-based business model element testing. Business model testing will take place both for individual assets and for the concept of the innovation ecosystem. There are multiple possible revenue streams for the innovation ecosystem including: membership fees, technology marketplace, access to system data, standards development, conferences/education sessions, local and international project funding, charities, public-private partnerships, and SME development advice.

A business model development process will also be carried out for each of the individual assets produced in during the project. This will form part of an asset sustainability and exploitation plan which will detail how each asset will be maintained after the project funding period ends and how it will be made accessible and/or form the basis for exploitation.

The broad competitive strategy for the platform will be a combined cost leadership (through DtC) - differentiation approach (currently, there are no commercially available integrated platforms capable of precision, automated manufacturing of functional cell therapy products). In terms of the AIDPATH platform's value proposition towards potential customers, two options must be considered. Each option, while related, targets a distinct business domain and could follow a separate route for exploitation (Figure 9).

For example, the customers for a precision, automated manufacturing platform (e.g. a small cell therapy product company) are different from the 'customers' for the cell product implant (patients, healthcare providers and payers of health care services) or the AI pipeline (personalised medicine companies for other

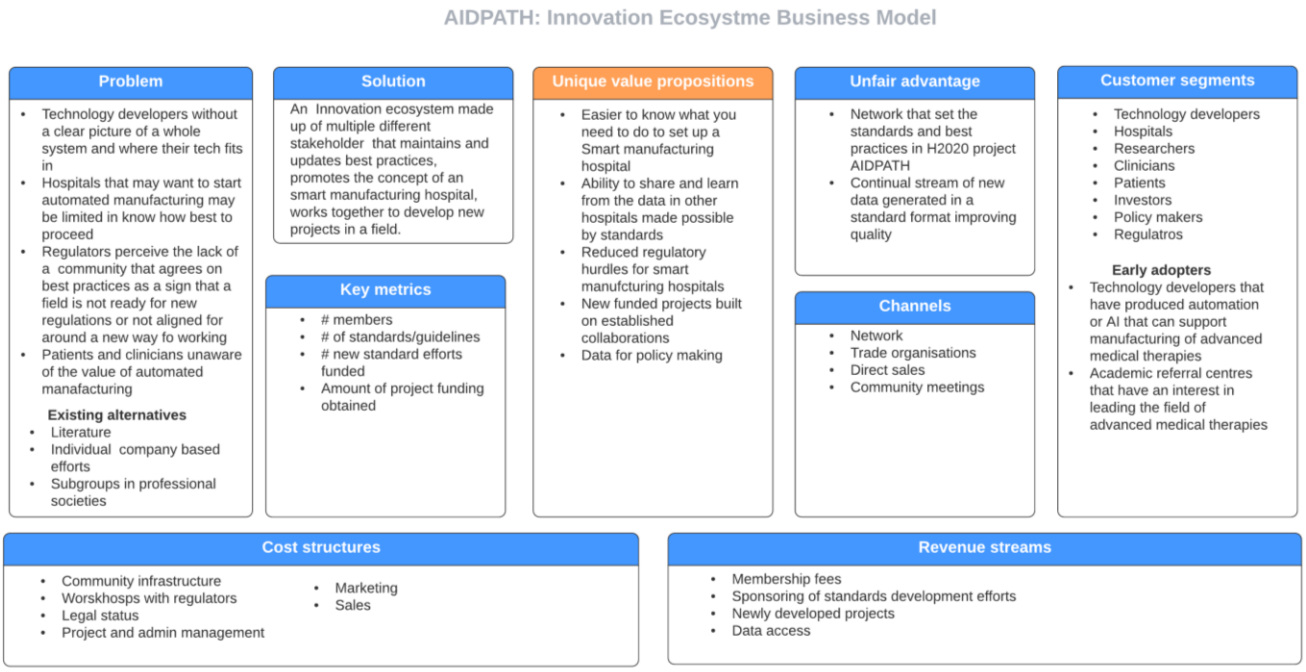


Figure 9: Innovation Ecosystem Business Model

manufacturing settings and indications). Given the novelty and complexity of the field, business model development will have to deliver a framework that is flexible enough to capitalize on the unique mix of innovative features provided by the AIDPATH platform, irrespective of the specific cell therapy product and therapeutic indication being addressed.

AIDPATH will produce a series of **commercialisable products**. This reflects the multiple disciplines involved in the project and the modular nature of our technology, which makes the re-use of our results in other applications and environments very attractive. Each of these apparatus can be sold separately, or as part of an end-to-end pipeline.

a) **AGL**, in collaboration with other partners, will optimize its novel ***bioreactor*** during the project, to manage multiple patient samples at once, each using its own selected chemically-defined xeno-free medium (FUJI), while being monitored with AGL's and UCL's new non-intrusive sensor and IPT's process management software. The original (novel) work will include modularization of the bio-reactor, software and hardware interfaces and validation, enhanced non-invasive monitoring using novel Raman and other bio-sensors (see below). This builds on AGL's existing Facer-1 product line.

b) **AGL** and **UCL** will develop new ***sensors*** for use within the pipeline. The sensors will utilize Raman spectroscopy to non-invasively measure concentrations of relevant substances (oxygen, cells, lactate, etc.) within cell-isolation bags, bioreactor cartridges, etc. Rights in the new sensors will be held by AGL, on whose existing research they will be built. UCLB, the innovation arm of UCL, expects to support the commercialisation and market launch of these new devices

c) **FORTH** will implement an in-house micro-bioreactor to develop potential novel biomarkers with metabolomics that will be linked to cell quality and potency. This system allows multiplexing of multiple process conditions allowing high throughput experimentation that is required for the screening of multiple donors. In addition, to enable automation regarding assay preparation together with IPT will automate sample preparation for extraction of metabolites from the CAR-T cell prior and after transfection. Finally, bioinformatics analysis will be used to link measurable metabolomics data sets to existing libraries within FORTH. These data will be validated also for the AGL system at larger scale and integrated with the data-library built by FORTH and UCL and will be available for SZTAKI for algorithm training and validation.

d) **FUJI** will work with UCL to define three variations of its existing **Xeno-free cell culture medium** line ('PRIME-XV'), each of which is optimized for specific patient sub-populations and will be made such that the medium is fully **chemically-defined**. This will benefit from the large-scale metabolomics experimental studies undertaken by FORTH and the process modelling from UCL, for patient stratification. The benefit for AIDPATH will be reduced variation in the production, and time-scale, of expanded T-Cell populations.

e) **UKW** will embed and host the physical automated AIDPATH manufacturing platform to enhance its development and commercialization of a series of CAR-T therapeutic programs. UKW has successfully launched the **first clinical trial with virus-free *Sleeping Beauty*-engineered CAR-T in Europe** (CARAMBA H2020 Project) and has a **rich pipeline** of additional **CAR-T products with novel target antigens and indications in haematology and oncology** (e.g. ROR1 CAR-T to treat lymphoma, breast and lung cancer; FLT3 CAR-T to treat acute myeloid leukaemia, etc.) that will serve as exemplary applications on the AIDPATH platform. **UKW's biotech spin-out T-CURX** is prepared to adopt these innovative CAR-T products for development to market once Phase I/IIa clinical development has provided a positive clinical signal and therapeutic index. IRIS will be responsible for implementing the AI process setup and monitoring/control (smart sensor) for the bioreactor. IRIS expects this will potentiate its existing PATBox system and a more general version will be applicable to different industrial processes, such as in the pharmaceutical and chemical/plastic recycling sectors. The integration of AI in industrial processes and decision support is a key future area and represent a pivotal goal for the Industry4.0 initiatives, environment of special interest for IRIS.

f) **RAL** will adapt its cybersecurity assurance framework to the medical industry. It will also extend its security by design and certification software platform to address the healthcare security requirements, evaluation criteria, standards and regulations, this will bring a great added value to the healthcare and smart manufacturing industry helping it fighting targeted cybersecurity attacks.

g) **SZTAKI** will develop a solution which can model personalized patient therapies. The solution will be able to handle general risks and uncertainties involved. SZTAKI will use its knowledge in the

development of solutions for resource constrained project management, production scheduling, and machine learning applications.

h) **ORTEC** will be responsible for the further developments of the Logiq Platform allowing AI solutions and their integration and productization. Various parties will customize their data models in a user-friendly environment to model their data as desired with support of ORTEC. This system will log data and support intelligent data processing for clinical care and Good Manufacturing Practices (i.e. LogiqCare and LogiqScience) with data availability at a need-to-know basis. ORTEC will **integrate** specialized data models to the **generic model** (with anonymous data) with authorization control will use the combination of role- and attribute-based access (RBAC, ABAC). The platform will also be used to facilitate **productization** of clinical-decision support tools in an MDR-compliant manner.

Collectively these outputs form **the AIDPATH manufacturing platform itself** – the modules, robot and software that enable the efficient and flexible combination of diverse technologies to achieve the specific aim of CAR-T cell production (but which can be readily re-configured for other products). The platform will be the result of collaborative work, led by IPT but involving each of the other technology partners in the creation of their respective modules. All elements of the platform (the design documentation, the process control software, the specification for sensors, robots and hardware) will be **published** by IPT, to encourage project partners and third parties to replicate the manufacturing platform. The physical platform developed during the project will be established at University of Würzburg, as a basis for a new product line based on AIDPATH.

Target Markets for AIDPATH IoT Infrastructure

The target markets for the AIDPATH manufacturing platform are include: (1) Clinical Centres of Excellence/large hospitals, (2) commercial Contract Manufacturing Organisations (CMOs) and (3) cell therapy product development companies

Direct target for the AIDPTAH technologies will be **hospitals and treatment centres**. Our expectation is that an AIDPATH manufacturing platform will be established in a laboratory setting (similar to a pathology laboratory), and used to generate cell therapies for one-to-several patients at a time. Each module will be linked to the same software and monitoring platform, but inputs, outputs, reagents and supplies for each module may be added or removed, via sterile bags, by human operatives. In addition, these hospital sites could be interlinked in the future, establishing a learning network that allows to optimize processes and AIs further based on the data that is generated from batch to batch over time.

For **CMOs**, we anticipate that the platform could be adapted to centralized manufacture of allogenic products, including a robotic handling of inputs and outputs with 24/7 operation, and many patients' therapies being produced in parallel. Full software and monitoring will be active, and human involvement will be minimal. It is envisaged that CMOs would be a significant beneficiary of such a platform as the AIDPATH enables extensive flexibility and would allow for interchangeable modules and technologies.

For cell **therapy product development companies preferring to develop manufacturing capacity in house**, we anticipate that they could license or buy the platform adapted to either a centralised or decentralised setting with robotic handling and manufacture of therapies for clinical administration, either as part of a clinical trial study, or indeed, for full manufacture. Similar to CMOs, the human involvement would be minimal and the exact nature and setup of the modules would differ based on the company's therapeutic application and manufacturing requirements. The intention would be to target those companies using existing SOTA systems (such as the Miltenyi Prodigy) and demonstrating the significant added value and benefits of moving to the AIDPATH system.

For both markets, training and support will be important added-value services. We shall build on UCL's MBI® CPD modular programme to train scientists, engineers, clinicians, regulators and hospital staff in the AIDPATH manufacturing platform. The training will highlight the benefits of the platform and equip them with the necessary skills and methods to take full advantage of the AIDPATH project outputs.

New business models are emerging, that may offer additional possibilities to innovators within the cell therapy domain. One configuration relevant to AIDPATH is that of the Collaborative Enterprise Network (CEN): a group of organizations, each a separate legal entity, united by at least one common (business) goal, who coordinate their activities through collaborative agreements according to agreed business practices. The CEN governing body, composed of members from each participant, takes strategic decisions concerning which manufacturing technologies to adopt (e.g. for production, QC, communications, inbound and outbound logistics). While remaining able to function independently, each member contributes a specific area of expertise to the overall manufacture and supply chain and is able to capitalize on business opportunities that

might not have been available if acting alone. The setup of CENs is one of the activities envisioned for the innovation ecosystem.

Legal entities: **ownership** of most of the intellectual property which will emerge from AIDPATH is quite clear, because the project builds on innovative foundations which partners contribute to the overall objective, as described above. Thus, AIDPATH will enable the project partners to develop new product lines, improve existing products, and explore integration in the platform, with the confidence that there will be no subsequent intellectual property difficulties. In the event that two partners are both substantially behind some new IP, case by case agreements will be agreed when the new technology has been developed and evaluated.

Sales and distribution: for partners' specific technologies, we anticipate that existing sales and distribution networks will continue to be used, but that AIDPATH will lead to stronger competitive position in the relevant sectors. For the integrated platform, we anticipate a roll-out programme (outlined above) that relies on organic growth and high-profile 'reference sites'.

b) Communication activities

AIDPATH will engage in extensive and appropriate communication activities both during and after project completion. To achieve this, clear messaging needs to be in place to ensure key message are communicated and have maximum impact. This will be achieved by ensuring all messaging is tailored to appropriate audiences (see below). Furthermore, a range of communication activities have been identified to ensure that there is broad acceptance of the system across the healthcare sector and amongst the general public.

Tailoring the channel and the message to audience is critical if the project is to communicate effectively. *Patients* and *organisations* need to know what we are doing, and what this may mean for them, so that they can engage with their healthcare providers and encourage them to consider wider use of cell therapies. For many patients, they rely on their organisations to keep them informed of scientific progress. *Clinicians* need to know that major obstacles (cost, risk, consistency) to clinical translation of promising research has been overcome, and that cell and gene therapies may become more routine and mainstream. *Media* are primarily interested in newsworthy events, such as the start of the project, major publications, the impact on patients, new industrial spin-outs, etc., but more coverage can be secured by making the message ready-to-print and easy for the media to use. *Researchers* take peer-reviewed publications most seriously, and these are the most important channel for the research community. The *website* provides all audiences with somewhere to go to 'find out more' and to get in touch with the team. Finally, our communication activities are an important support to securing new investment for further exploitation of the project results; such relationships will of course rely on meetings and direct contact, but good publications and high profile will certainly help.

Thus, our communication activities and resources (WP1) will include:

- A comprehensive **online** dissemination suite (website, Facebook page, Twitter account, YouTube account, ResearchGate). The website will be the main resource, and will have sections with media-ready materials and press-releases, downloadable project assets, publications and scientific papers (researcher area) and material for patients, organisations and the general public.
- Animations and videos that illustrate the science and technology involved in the project, to help to 'bring the science to life' and to reach new audiences. We will design these in collaboration with representatives of our target audiences, notably patients and clinicians.
- Outreach to **other projects** in cell therapy, in automation, in sensors, and other related domains. Automation and cell culture projects involving IPT, such as AutoCRAT, JointPromise and StemCellFactory are of course very relevant. Other projects funded under this call will be a particular target, because we expect that they may generate new technologies that could potentially be modularised in the future, thus adding value to both projects.
- Outreach to **industry** and **patient organisations**, by attending meetings, inviting them to our plenaries and symposia/workshops in each country, authoring articles for their websites and publications, etc. As established experts, and leading technology companies we already enjoy substantial personal networks into these communities, which we will leverage.
- Interaction with the **DIH Hero** e.g. *via* both participating as well as hosting conferences, meetings and other events with the goal to in**form, educate and exchange knowledge** between digital health care and robotics experts, thus ensuring visibility of the developed technologies and wide reach of the project results (see letter of Intent in the Appendix of Section 04-05 of this proposal)

Publication: Important new knowledge is expected from AIDPATH. This includes new technology development, new business models, decision tools and health economic analyses. As noted above, we intend to publish our AI concepts, platform software and technology outputs openly, in order to encourage the growth of the ecosystem. New intellectual property that belongs to a single partner (e.g. enhancements to existing Background) will of course be protected as appropriate, using patents and other methods. When such protection is in place, we will publish as much as possible, using open-access journals or paying for open-access in high-impact journals.

Beyond the academic/research publication milieu, the new knowledge will be made available outside the project through meetings and various media supports (publications in scientific journals, or general public newspapers, Internet, etc.). Furthermore, to further spread excellence and disseminate acquired knowledge the consortium plans to organise *public workshops in the third year and again in the final year* of the project. These events, to which industry and academic R&D teams will be invited, will serve to gather inputs and external perspectives, as well as promoting our worn work. Major breakthroughs will be made public through press releases and newsletters. Project newsletters will also be available for download on the public website of the project, to allow the general public to access the information.

Communication Venues

The venues in which the consortium has a strong record of publication, and/or which are targets for results from AIDPATH, include the following

Journals

- Nature Biotechnology
- Science Translational Medicine
- Biotechnology & Bioengineering
- British Journal of Cancer
- Gene Therapy – Nature
- SLAS Discovery/Technology
- European Journal of Operational Research
- Nature Biomedical Engineering
- Blood
- Biochemical Engineering Journal
- Cancer Gene Therapy
- IEEE Robotics and Automation
- BioProcessing International
- Biosensors and Bioelectronics;
- Sensors and Actuators
- Cancer
- IEEE Sensors Journal
- Biotechnology Advances
- Cell & Gene Therapy Insights
- Value in Health

Conferences

- ECI Scale up & Manufacturing of Cell-based Therapies VI (2019)
- European Society for Animal Cell Technology (ESACT)
- European Society for Biochemical Engineering Sciences (ESBES)
- British Society for Gene and Cell Therapy
- SelectBio Advances in Lab Automation & Robotics
- Society for Laboratory Automation and Screening (SLAS) Conferences
- IEEE Sensors Conference 2020
- EuroNanoMed
- Termis-EU (and World)
- International Society for Cellular Therapy (ISCT)
- Tissue and Cell Engineering Society (TCES) Conference
- Cell Therapy Manufacturing & Gene Therapy Congress 2019
- World Advanced Therapies & Regenerative Medicine Congress
- International Conference on Smart Manufacturing Technologies (ICSMT)
- World Advanced Therapy Congress
- American Society of Haematology
- International Conference on Significant Advances in Biomedical Engineering
- The European Laboratory Research & Innovation Group (ELRIG)
- The International Society for Health Economics and Outcomes Research (ISPOR)

Industry Groups

- BioPhorum Operations Group
- BioIndustry Association
- British Standards Institution
- ISCT Communication Committee
- Centre for Commercialization of Regenerative Medicine
- IChemE Biochemical Eng. SIG
- BioProcess International
- Alliance for Regenerative Medicine
- ASEBIO Spanish Bioindustry Association
- Phacilitate Automation SIG
- Parenteral Drug Association
- BIO Deutschland
- Cell and Gene Therapy Catapult
- DIH Hero

2.3 Data Management, Publication and Open Access

AIDPATH has a strong emphasis on exploitation and commercialisation, as outlined above. The consortium includes large industry, SMEs and research centres, which have a strategic focus in this area; as a result, the new data and foreground knowledge, which the project generates, will be carefully managed and protected.

However, we of course respect the broader interests of the scientific community, and appreciate the direct value that having other research teams review, replicate and validate our work will bring. As soon as is feasible within our intellectual property constraints, we are committed to *placing our research data on an open access research data repository*, as outlined below, with full supporting metadata and process/protocol data. This will enable other research teams to replicate our in vivo validation work, to explore how our CAR-Ts can be used in clinical applications, and to address complementary research questions such as using our platform to produce different types of cell therapies. This new knowledge will benefit both this consortium and the cancer and cell therapy communities, as well as patients, families, clinicians and healthcare providers.

Scientific publications: The generation of scientific publications is a key performance indicator for academic partners and remains a priority for the project, subject to intellectual property restrictions and our publication management model. We aim to make all our publications '**gold open access'** and will commit funds from our 'other' (dissemination/exploitation) budget lines to pay any associated fees from the journals, which publish our work. Our experience of this process to date has been positive, with costs low enough to be acceptable in the context of the overall project budget.

Data availability: we are committed to make our design documentation, software, process description, experimental and validation data available in an open-access repository with the minimum delay. However, we must also respect our commitment to the management and protection of foreground knowledge and intellectual property; the precise details of what we will make available, and when, will emerge only during the project and after we have published our results. Thus, validation data will be made available only after the project ends. Naturally, commercially-sensitive information will not be published or made freely available.

Data Repository Curation and Preservation: The data will be deposited at the open access research data repository. We will carefully identify and select a suitable repository with servers in Europe and which ensures long-term storage and availability. Options include utilization of (1) databases in cooperation with the intergovernmental organisation ELIXIR, which is a distributed infrastructure for life-sciences information, (2) data repositories from Fraunhofer Fordatis, which is an institutional research data repository providing Open Access for further re-use, or (3), 'UCL Discovery' which has a long-term strategic mandate to **preserve, curate**, manage access to, and add value to research results and data created by teams where UCL plays a key role. The costs of this curation and preservation are paid by UCL on an ongoing basis, as part of it strategic library costs. We will take every care that the data which we make available is complete and correct, subject to data protection legislation. It is in our interests to enable others to validate our work, as noted above.

3 Implementation

3.1 Work plan — Work packages, deliverables

The AIDPATH project is structured in **three major strands** of development activates – **(1) Artificial Intelligence**, **(2) bioprocess automation** and **(3) IoT infrastructure.** All three strands converge in final **implementation and validation**. All strands are informed by and subsequently feed into the business planning, dissemination, regulation and project activities of the project.

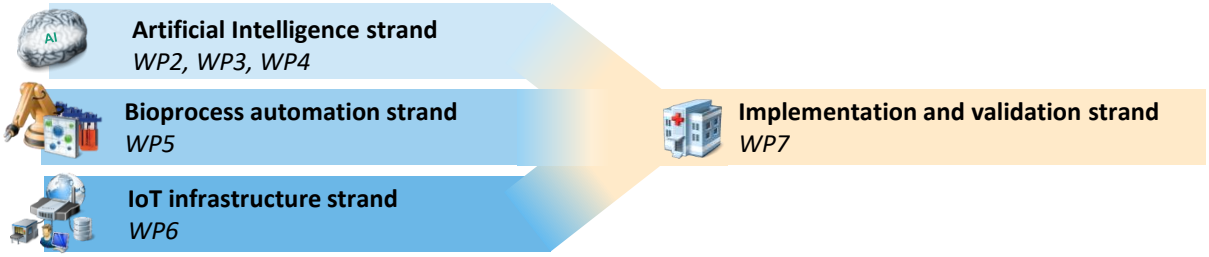


Figure 10: AIDPATH project strands: three concurrent technology strands converging in on-site implementation and validation

Artificial Intelligence strand

The Artificial Intelligence (AI) strand realizes the application of efficient AI solutions in the health care environment including definition, data preparation, AI training, development and validation.

1. Use Case Definition: To develop use case specific AI approaches a concrete idea, vision and measureable goals have to be defined and documented. Therefore, use case requirements need to be collected in strong collaboration with domain experts.
2. AI data structuring and preparation: Based on the use case description a work shop gathering and structuring all the available data will be conducted. After the data quality assessment and the data sampling, the most relevant data with the needed data quality is assigned to the different use cases. The data will be prepared and pre-processed in an automated manner in order to minimize processing time and to generate suitable data sets.
3. AI Development: Due to the broad application of AI in the AIDPATH infrastructure different AI solutions are developed focusing on the following areas:
    i) An *AI-powered digital cell twin* will mimic growth and media consumption rates for patient-individual CAR-T cell cultures allowing the prediction and optimization of process parameters
    ii) A *reactive online process control* of the CAR-T production plant will enable adaptive feeding strategies as well as a Rama-spectroscopy-based bioreactor control.
    iii) Production scheduling will be developed with the goal to use AI to predict production capacities and patient therapies. Production scheduling will focus the accurate planning of patients' therapies and a cost effective exploitation of the bio reactor.
    iv) In the *resource management*, the patient is the focus. Therefore essential equipment, materials and staff will be managed using AI. AI supports here to reduce complexity and the effort of human support needed without compromising transparency and traceability.
4. AI evaluation and benchmarking: The last step is to evaluate and benchmark the AI approaches developed before. Therefore, a concept has to be developed, which enables the evaluation of the different AI use cases and approaches regarding performance, fulfilled requirements and reached prediction targets. With this concept, the AI approaches are evaluated. To receive a more general impression of the AI approaches a use case independent benchmark has to be developed which is able to benchmark the AI approached and identify optimization potentials.

IoT infrastructure strand

The IoT infrastructure strand will develop and implement an overall framework that ensures connectivity, data exchange and security between all interacting entities in the context of Internet of Things (IoT).

1. IoT infrastructure definition: An IoT architecture will define all hardware and software components and their functional connections to each other as well as the data storage and integration.
2. Preparation of regulatory complaint development procedures: Before the defined infrastructure is developed and implemented, all relevant regulations (e.g. GAMP, GDPR) are determined and applied to the use cases. The business security and privacy requirements are taken into account in order to identify and handle sensitive personal data.
3. Deployment of data-driven manufacturing and resource planning systems: All processes linked to the CAR-T cell production plant will be governed by AI-supported software components assuring an efficient process control, manufacturing scheduling and resource management deployed directly in the IoT infrastructure. Processed data will be centralised handled by the LogiqCare software enabling a continuous data integration throughout the whole project and in the completed production plant.

Implementation of the IoT infrastructure: The different data, software and AI components of the IoT infrastructure will be connected for testing in order to assure the functionality for the on-site pilot validation.

Bioprocess and automation strand

Concurrently to the other strands, the modular AIDPATH manufacturing platform is realized in one bundle with the corresponding automated process protocols. This includes:

1. Protocol and bioprocess definition: Standard operating procedures (SOPs) and information on existing protocols for manufacture are gathered and one process for automation is consolidated. On this basis, requirements for automated system and AI development are derived and fields of action are identified.

2. Automated system development: On the basis of the protocol and requirements definition, the automated platform consisting of a manufacturing module integrating the Facer bioreactor and a QC module are constructed and built in the shop floor ready for installation.
3. Process analytics: Analytics play a key role in the entire process because they provide the data on which basis the AI can make process decisions. Therefore, suitable process analytical technology including online sensors and at-line / end-point assays are prepared for integration into the automated workflow.
4. Automated and AI powered protocols: Based on the protocols compiled in the first step, the process is transferred to a fully closed and automated process that is deployable on the automated platform. This includes both manufacturing protocols and analytic assays. The protocols have to ensure that they integrate the output of the bioprocess control AI from the AI strand.

Implementation and validation strand

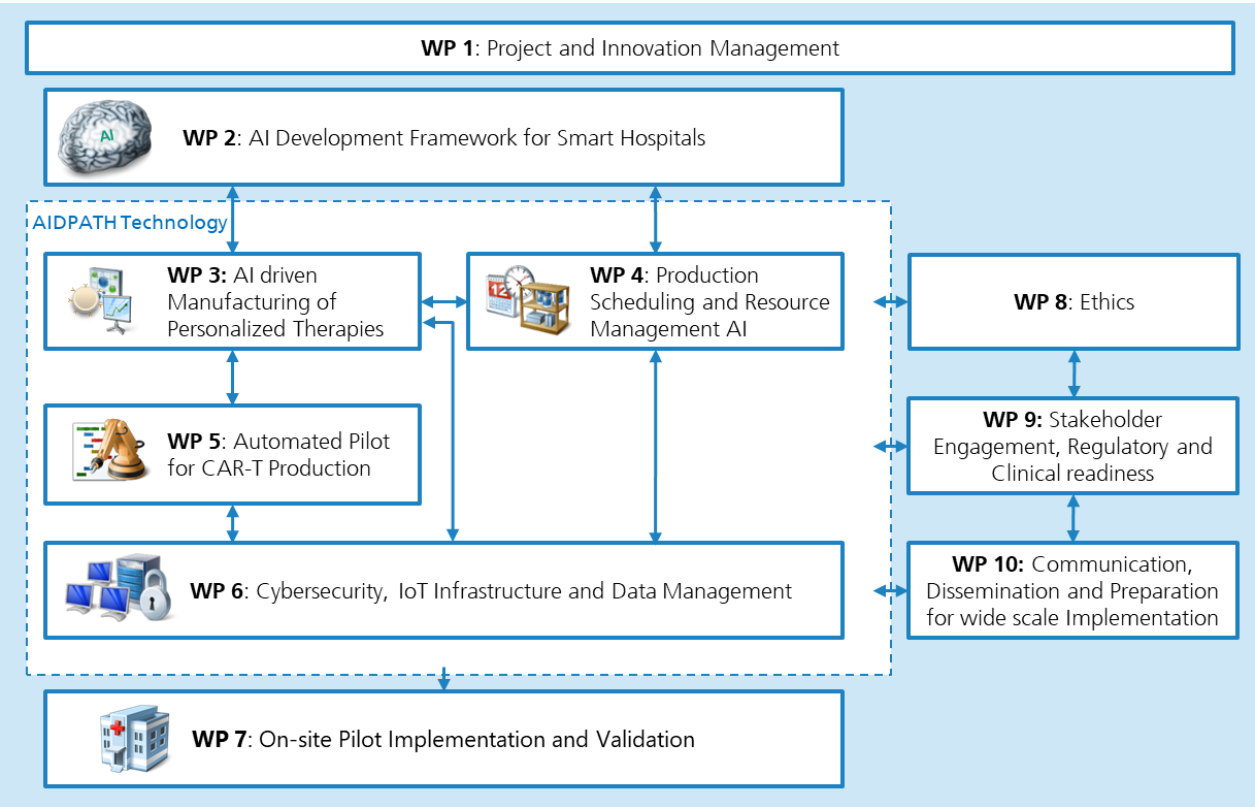
In this strand, outputs from AI, IoT infrastructure and bioprocess strand are concurring and all developments are integrated and deployed on-site in the hospital. This strand includes:

1. On-site pilot installation and integration into the hospital: Once the individual strands (AI, process and infrastructure) have been completed and integrated to one cyber-physical system, the system will undergo a technology transfer and is implemented on site at the hospital (UKW).
2. Technology validation: Following up to the system installation, the AIDPATH approach fully tested and validated by the end-users for the production of personalized CAR-T cells in the hospital (see WP7), with a focus on preparing and executing a training plan for manufacturing operations and quality operations personnel to enable the successful execution of the manufacturing process under GMP conditions in addition to preparing a manufacturing documentation and validation reports in preparation for GMP manufacture (see WP10).

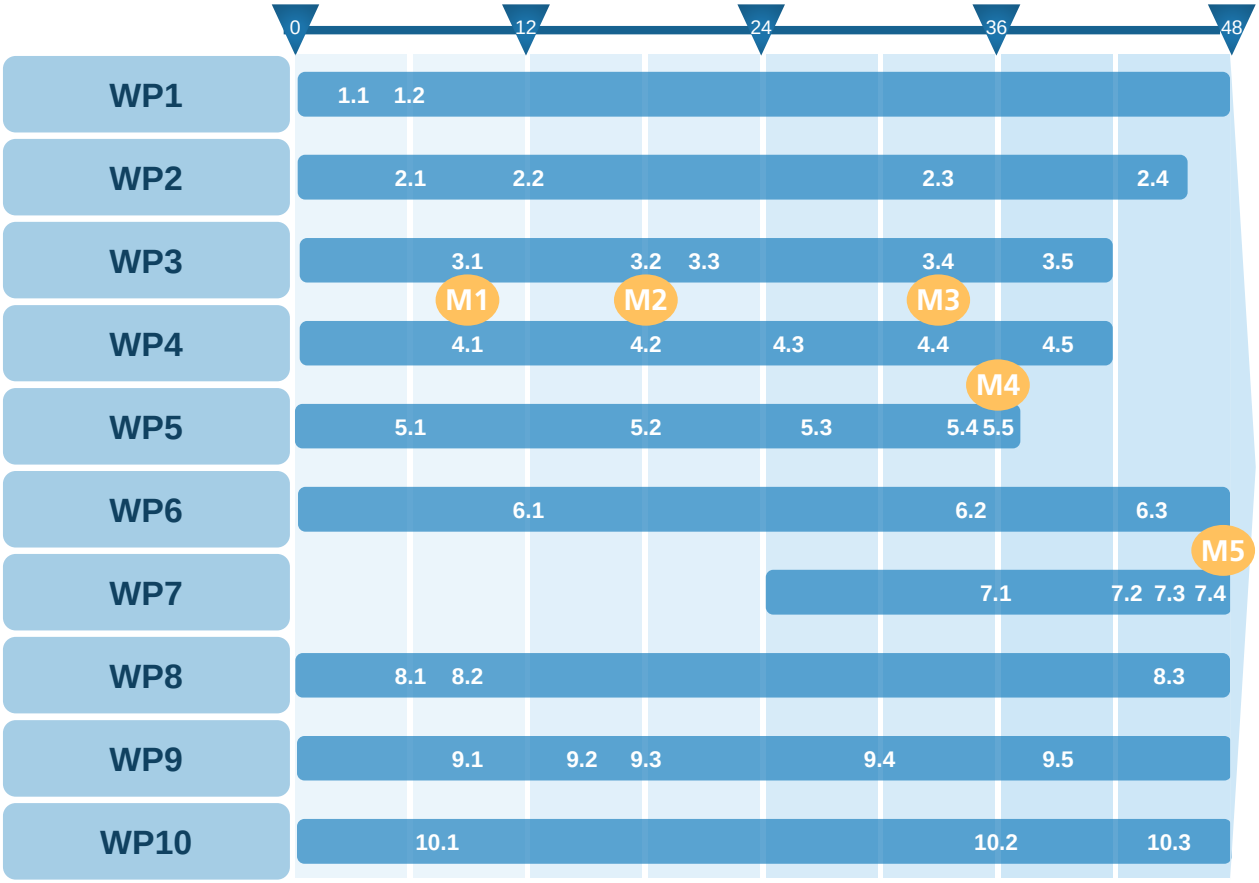
Concurrent support activities

Throughout the project, horizontal 'support' activities occur to ensure that the project results are on point and in alignment with legal, regulatory and stakeholder requirements. These tasks include ethics, stakeholder engagement, health economic evaluation, business modelling, regulatory affairs monitoring, communication and dissemination, exploitation planning and project management.

3.1.1 PERT chart



3.1.2 GANTT chart



3.1.3 Table 3.1a List of Work Packages

WP No.	Work Package Title	Lead No.	Lead	PMs	Start M	End M
1	Project and innovation management	1a	IPT	42.25	1	48
2	AI Development Framework for Smart Hospitals	1a	IPT	93	1	45
3	AI driven manufacturing of personalized therapies	2	UCL	248	1	42
4	Production scheduling and resource management AI	4	SZTAKI	117.7	1	42
5	Automated Pilot for CAR-T production	1a	IPT	143	1	36
6	Data security, IoT infrastructure and data management	13	ORT	116	1	48
7	On-site pilot implementation and validation	5	UKW	136.6	25	48
8	Ethics	1a	IPT	35	1	48
9	Stakeholder engagement, regulatory and clinical readiness	5	UKW	43.2	1	48
10	Communication, Dissemination and preparation for wide scale implementation	1a	IPT	54	1	48
		Total		**1028.75**		

3.1.4 WP 1

WP No.	1		WP title			Project and innovation management								
Lead beneficiary	IPT		Start month			1			End month			48		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	30		2	1			2	1		1	1		2.25	2

Objective

This work packages oversees and manages the entire project, for 1) project definition and progress control, 2) collection and monitoring of IP, 3) ensuring smooth project implementation through monitoring and proactive handling of risks and issues and 4) establishing an efficient and effective process for regular to the commission

Description of work

Task 1.1 Project Management and Coordination (IPT)

The coordinator will manage and closely monitor project schedule and progress of all WPs to ensure that all tasks and major milestones are delivered on time. At the beginning, a project charter will be drafted that summarizes scope, key objectives as wells as roles and responsibilities of the participants and identifies key stakeholders to ensure mutual understanding of all partners. A project plan will be laid out e.g. in MS project that captures timelines and dependencies between tasks and partners. The project plan, progress and interfacing tasks will be reviewed regularly at least during the biannual consortium meetings and as necessary in online meetings. A long-term perspective, seeking to maximise impact and project value, will be applied at all times. Major milestones will be tightly adhered to.

Task 1.2 Patent research and screening (IPT, ALL)

Relevant areas for the technologies to be utilized and developed within the project will be identified. Related patents, IP and protected know-how in the field will be searched and gathered. The resulting IPR scan file will be made available for the consortium partners to raise awareness on IP protection and patents.

Task 1.3 Data management (IPT, ALL)

AIDPATH data will be aggregated, managed and utilised as described in the innovation work packages, during the project. In collaboration with all partners in AIDPATH the used research data during and after the project will be defined in a Data Management Plan (DMP). The plan will be reviewed annually and updated, as required. In order to make this data findable, accessible, interoperable and re-usable (FAIR) it will be determined which data will be collected, processed or generated. The related data sources and types will be listed. A high emphasis will be given to the integration and storage of AI training and application data in order to gain an efficient AI handling during and after the project. Also, it will be clarified how the data of the project solutions will be curated and preserved. This is described in detail in section 3.2.

Task 1.4 Management of risks, issues and knowledge gained (IPT, ALL)

Intellectual property (IP) of the partners, issues and knowledge gained will be identified and gathered on a regular basis. Therefore, A review of risks, issues and knowledge will be permanent items on the agenda of each plenary meeting. The risks in this proposal will form the foundation for the risk register. The periodic risk review will seek to identify any new or emerging risks and how to mitigate them and upgrade or downgrade existing. In addition, other issues, learning and new knowledge gained is queried annually and collected in a joint database. It will be identified whether any items are also to IP protection or dissemination and treated accordingly. All activities and outcomes will be assessed for potential IP at all plenary meetings; any innovations or inventions that emerge between meetings will be fast-tracked to the IP sub-committee (see Innovation Management section in this proposal). The IP-assessment will direct each new piece of knowledge to either publication or protection (e.g. via patent), and will also clearly establish ownership (or kick-start a joint-ownership discussion). All IP management will build on the consortium agreement.

Task 1.5: Reporting to the European Commission (IPT, ALL)

In addition to the deliverables, periodic (18-month) reports and a final Report will be delivered to the Commission. These will be accompanied by Financial Statements, which will be validated in collaboration with all partners, prior to submission.		
Deliverables		**Date (M)**
D1.1.	Project Charter and IPR scan file	3
D1.2.	Data management plan	6

3.1.5 WP 2

WP No.	2		**WP title**			AI Development Framework for Smart Hospitals								
Lead beneficiary	IPT		**Start month**			1			**End month**			45		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	28		3		19	12				1	20			10

Objective

Assurance of an efficient collaborated preparation, conduction and result presentation of the AI development including: 1) structured datasets 2) development strategy 3) deployment plan and 4) guidelines & building blocks for AI in smart hospitals.

Description of work

Task 2.1 Data structuring and domain knowledge exchange for AI development (IPT, UCL, SZTAKI, IRIS, UKW, FCRB, ORTEC)

This task focusses on structuring, integrating and preparing the data for the later training and testing of the AI models in WP3 and WP4. The focus lies on the activities that are necessary for all considered AI use cases. It includes data acquisition and classification of all available data for AI development in liaison with domain experts (based on 3.1 and 4.1). The data basis is assigned to the AI applications from WP 3 and WP 4 and commonalities and overlaps are identified. Data structuring and preparation (Selection of relevant data, elimination of redundant data, assessment of data quality, and presentation of data structure in a transparent manner) is done subsequently. Also, a list of requirements for IoT infrastructure is generated. Evaluation of candidate software applications (statistical and data mining workbenches and libraries) based on data processing requirements, types of data and modelling will be done. Furthermore, the knowledge transfer between production experts and experts for bio processes, data scientist and engineers to generate a common knowledge basement (based on 3.1 and 4.1) is done.

Task 2.2 AI concept, interaction and development strategy (SZTAKI, IRIS, IPT, UCL)

In this task a technical concept will be developed, how the AI applications can be realized based on data structure, boundary conditions, use-cases (Task 3.1, 4.1) including model transparency, model complexity, expected training times. For that a comparison of properties and objectives of the AI applications and derivation of interaction opportunities and synergy effects (where is a collaboration essential, which AI application requires input from another AI application) will be done. This includes the interlinkage of the Cell Growth Model (Digital Twin) developed in WP3 as an input for WP4. Central aspect will lie on the decision what programming frameworks and libraries will be used by focussing on open source systems. To scale the development a concept how to integrate Automated Machine Learning (AutoML: Automating steps within data integration, preparation, modelling and deployment) will be developed.

Task 2.3 Definition of an End-to-End AI pipeline including deployment plan (IRIS, SZTAKI, IPT, UCL, ORTEC)

To align the implementation of the developed AI systems from WP3 and 4 this task aims at defining an End-to-End AI pipeline with a deployment plan. This includes the definition of data sources for the application phase. The retraining strategies and corresponding data streams are defined. Furthermore, the AI systems

interfaces and communication for deployment in the IoT infrastructure are defined. The deployment plan includes a concept for productionizing the model, e.g. serializing the model into HDF5/pkl or to implement it in c++/Java.

Task 2.4. Guidelines and building blocks for AI in the smart hospital of the future (IPT, UCL, SZTAKI, IRIS, UKW, FCRB)

Based on the results especially from WP 2, 3 and 4 building blocks for AI in the smart hospital of the future will be developed. This includes comparison and evaluation of the AI models, their performances and transferability within hospitals. Furthermore, the identification of transferability to other domains and elaboration of building blocks will take place – as well as the identification of challenges in AI development in smart hospital context and derivation of guidelines.

Deliverables		**Date (M)**
D2.1.	Integrated data structure for AI development	6
D2.2.	AI concept, interaction and development strategy	12
D2.3.	Defined End-to-End AI pipeline including deployment plan	33
D2.4.	Guidelines and building blocks for AI in the smart hospital of the future	45

3.1.6 WP 3

WP No.	3		**WP title**			AI driven manufacturing of personalized therapies								
Lead beneficiary	UCL		**Start month**			1			**End month**			42		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	3		75	73		12	25	16		15	29			

Objective

Develop an AI-driven manufacturing solution for the improved production of personalized therapies, including (1) AI System and Bioprocess Definition, (2) AI-driven bioprocess setup /optimisation through high-throughput (HT) metabolomics and analytics, (3) intelligent, adaptive, AI-based bioprocess control strategy for CAR-T cell (4) creation and implementation of an AI CAR-T digital twin and (5) validation, testing and evaluation of AI bio manufacturing platform.

Description of work

Task 3.1 AI System and Bioprocess Definition (FORTH, UCL, AGL, FUJI, UKW, FCRB, IRIS)

Task 3.1.1 System Design and AI Use Case Description (IRIS, FORTH, UCL, AGL, FUJI, UKW, FCRB)

This task will define the AI use case, including the characteristics of the final AI-driven process, AI elements for each process step, interaction and information exchange in terms of data streams, and specific task and goal of each AI element. This will be linked to tasks in WP2, specifically Task 2.2 which will establish the basis for the interaction of the AI applications.

Task 3.1.2 Definition of Bioprocess Metrics & Analytical Technologies (FORTH, UCL, FUJI, AGL, UKW, FCRB, IRIS):

This task will review the current process, literature, studies, and available data to establish metrics for defining a successful process (quality markers, throughput, cell numbers, efficiencies, etc.), identify factors influencing process outcomes (risks, sensitivities) and compile all data sources including analytical data to better inform the process. This will include definition of a panel of Optimization Objectives that should be ranked in importance, and Key Performance Indicators / Metrics. Additionally, this task will deliver a set of standardized protocols and instructions that define the analytical technologies and their quantitative range implemented in this project. The technologies in the toolbox will provide the necessary data for intelligent process control in

the final automated process and at the same time define a standardized approach, where further targeted complementary data sets are generated to enhance and enrich the existing data sets. Typical analytical technologies including online/at-line sensors will be employed, such as pH, dO2, glucose, lactate and biomass, in addition to an intelligent Raman-spectroscopy based soft sensor. Automated at-line flow cytometric analysis will also be established linking with WP5.

Task 3.2 HT data generation for bioprocess optimisation and development of AI data-driven models (FORTH, UCL, FUJI, FCRB, AGL, UKW, IRIS):

This task will focus on high-precision screening of critical parameters related to this process. These analyses will be carried out on diseased and healthy donor samples as well as on a series of customised and tailored culture media developed by FUJI. This work will enhance knowledge and provide novel data sets to drive the bioprocess mapping and optimisation in a HT and donor-tailored manner. Cell populations will be produced based on starting material from existing biobanks from the respective partners. Studies by FORTH will include targeted metabolomics to investigate metabolites related to specific pathways based on the physiological integrated -omic analyses and other information acquired from the cultures (through sensors in task 3.3.2). Metabolomics will be used for large-scale metabolic profiling revealing overall metabolic shifts that are beneficial to various steps of the CAR-T cell culture growth and identifying the "detrimental" and optimal windows of operation. Metabolite profiling will mainly use GC-MS and reversed phase LC-MS to profile polar and non-polar excreted metabolites (using in-house libraries and bioinformatics) and will be coupled to further omics technologies. Additional HT studies by UCL to evaluate and optimise process control and feed addition modifications with respect to cell yield, CAR-T cell expansion, viability and T-cell subpopulations. This will involve (1) donor variability studies to identify and characterise the extent of heterogeneous donor material variation, (2) the evaluation and selection of different chemically-defined media depending on patient starting material, and an analysis of the electroporation settings. This will involve the implementation of HT bioreactor systems to generate sufficient data with experiments governed by Design of Experiment (DoE) methodologies enriching the available historical data to generate statistically significant cause and effect relationships between bioreactor operation, off-line analytics and clinical patient information.

Task 3.3 AI CAR-T Digital Twin

Task 3.3.1 Optimum configuration/setup of CAR-T Digital twin and adaptive bioprocess with AI techniques (UCL, IRIS, FORTH):

This task will develop a highly sophisticated AI CAR-T Digital twin (***AI-CAR$_{Twin}$***) combining innovative mechanistic modelling with advanced real-time data analytics. HT-data and available historical data curated in Task 3.2 will be used to develop the structure of the mechanistic model by extending previously published mathematical cell culture models. Multiple model structures will be evaluated through an iterative procedure governed by the fundamental principles of natural selection and reproduction using a Genetic Algorithm (GA). This will enable the creation of an optimised model of low complexity that accurately predicts the given output from inputs using the available data. ***AI-CAR$_{Twin}$*** model parameters will automatically update (gradient based parameter optimisation) throughout the manufacturing process based on availability of high-frequency time-series measurements (temperature, pH, dissolved oxygen…) and lower frequency off-line measurements (glucose, lactate, viable cell density...). The model will include statistical relationships (Task 3.2) between patient clinical information and cellular growth kinetics enabling ***AI-CAR$_{Twin}$*** to better predict the growth and media consumption requirements of a patient's individual cells. Different simulations/scenarios will be assembled and run, whose objective is to find the optimum configuration. It could act as a "what-if" tool: given a set of inputs, what would be the outcome, and the inverse, for a desired output what combination of inputs would be necessary to produce it. This can also use a "bootstrap" case based reasoning approach which will compare outcomes of previous similar historical runs and adjust the process accordingly to meet described outcomes. ***AI-CAR$_{Twin}$*** will integrate with the resource scheduling and management AI solver in WP-4 providing near real-time information on the CAR-T manufacturing status to clinicians enabling better management of patient treatments.

Task 3.3.2 Implementation of the AI model for advanced and adaptive bioprocess control/monitoring with AI techniques (IRIS, UCL, FORTH):

This task will combine the knowledge gained from the HT experiments in Task 3.2 with ***AI-CAR$_{Twin}$*** to develop advanced control applications minimising patient-to-patient variability of CAR-T cell expansions. A fully integrated system will be developed combining all on-line process data, off-line analytics and all

recorded information related to gas and feed additions of the cell culture with the ***AI-CAR$_{Twin}$***. The ***AI-CAR$_{Twin}$*** will implement a hybrid-based control strategy combining both statistical and data-driven models with mechanistically derived equations enabling robust on-line predictions of key process performance indicators such as viable cell densities or key nutrient concentrations. ***AI-CAR$_{Twin}$*** aims to increases the cell productivity and improve therapeutic efficacy of these CAR-T cells. This process will validated for a wide number of CAR-T cell patient samples enabling tailored feeding-strategies for individual patients. The soft (intelligent) sensors will be integrated with the AI model software to provide real-time data acquisition for incremental learning, case-based reasoning and control. This will form the basis of a decision support system, once the data is accumulated, and will be bootstrapped with already available historical data.

Task 3.4 Validation, testing and performance evaluation for AI driven manufacturing of advanced therapies (IRIS, UCL, FORTH):
The AI bioprocess control will be assessed by Monte Carlo using % error as an output target (inverse prediction) for a given input configuration, and on the output precision in the case of forward prediction. Task 3.3 and 3.4 assessment will be based on a DoE for different empirical scenarios, for example, normal, extreme minimum and extreme maximum. Further optimization (iterative process): Active learning will take advantage of human domain expert to provide the correct solution to the model when the prediction is incorrect and incorporating this information into the model by retraining. In the case of the smart sensor, it can be reprogrammed or calibrated based on performance with respect to a baseline or off-line analytics. The proposed bioreactor control strategy will be integrated with the automated CAR-T manufacturing platform outlined in WP-5.

Deliverables		**Date (M)**
D3.1.	Use Case Description for precision AI manufacturing including review of influencing factors optimization objectives	9
D3.2.	Pre-processed data sets for precision manufacturing AI training	18
D3.3.	Prototype of AI-CAR Twin suitable for bed-side hospital operation	21
D3.4.	AI CAR-T Digital twin for advanced and adaptive bioprocess control strategies trained on optimised bioprocess datasets	33
D3.5.	Performance report on implemented cell culture and bioprocess AI	39

3.1.7 WP 4

WP No.	4		**WP title**			Production scheduling and resource management AI								
Lead beneficiary	SZTAKI		**Start month**			1			**End month**			42		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	26	2.7	2		70	6				1				10

Objective

Development of a manufacturing and resource scheduling AI solver including 1) a comprehensive use-case description 2) the simulated production environment 3) structured datasets and standardized interfaces and 4) assessment and validation

Description of work

In **WP4** the two AI solver for 1) resource management and scheduling and 2) Manufacturing scheduling will be developed. With 1) the key resources of the autonomous CAR-T cell process will be managed in order to assure their availability and to avoid waiting times. The closely-linked AI solver 2) will schedule the different orders (patient-specific cells) on the production plant in order to assure adequate lead times and utilization.

Task 4.1: Description of the manufacturing and resource scheduling use-case (IPT, SZTAKI, UKW, FCRB, IZI, HCATS)

This task describes the use case scenarios for the manufacturing and resource scheduling in close collaboration with experts from clinical, automation and bioprocess side. Domain-specific workshops will determine the requirements and conditions for the conceptual design of the two AI solver. With the CAR-T cell treatment protocols and further expert knowledge, the process steps are analysed including bottlenecks, potential parallel processing and identification of the key resources (e.g. materials, cells) and their effect on the process. The contiguous logistic processes of the hospital environment will be analysed in order to display the material flows and further boundary conditions (e.g. special staff, hospital beds). With the gained insights and a research of existing AI models and algorithms the two AI solver are defined including the optimization objectives, properties and benchmarks.

Task 4.2: Data aggregation, data extension by simulation and data pre-processing (IPT, SZTAKI, ORTEC)

All the data for resource management, production scheduling and control will be obtained from the data sources to be developed in WP2. Likewise, the key decisions, concerning the timing of the patients' therapies, the usage of various resources will be stored in the database provided by WP2. In addition, datasets for testing will be generated to test the solvers in the simulation environment in Task 4.6.

Task 4.3: Building of a comprehensive simulation model for the production environment and the supply chains (SZTAKI, ORTEC)

For the representation and better understanding of the production environment as well as the validation and data extension, a simulation model is created on basis of task 4.1 that shows the interrelationships of the various components and illustrates the effects of parameter variations. In 1) the simulation focus lies on key resources including material availability and supply, staff availability **and** resource requirements of the scheduled treatments. The simulation model of 2) depicts the machines and devices of the production plant under consideration of volatile process times and machine availability. Especially the modelling of 2) needs to be conducted in close collaboration with WP3 in order to meet the requirements of the bioreactor. The simulation software is selected according to determined criteria, a good candidate could be Anylogic, which has proven capabilities in bio-process applications.

Task 4.4: Development of a manufacturing and resource scheduling AI solver (SZTAKI, IPT)

This task consists of two subtasks. In both subtasks, the main steps will be: (1) interfacing with data sources in WP2, (2) development of the solver, (3) fine tuning of the solver and (4) module tests using the test data pre-processed in Task 4.2.

Task 4.4.1: Resource scheduling and management AI solver

The main input for resource management comes from the hospital in the form of patient data specifying the treatment protocols and a rough timing, along with the availability of the main resources (beds, staff, CAR-T cell production capacity). To cope with the uncertainties in almost all steps of the treatments to which resources have to be allocated, the solver will be based on a Markov Decision Process supported by Reinforcement Learning. The solver will integrate also the schedules determined by the scheduler developed in Task 4.4.2. The result will be a feedback control policy which can handle uncertainties smoothly.

Task 4.4.2: Development of a manufacturing scheduling AI solver

The main input to production scheduling and control comes from resource management. The scheduler will determine a time slot for manufacturing the CAR-T cells for each patient. Since manufacturing lead time are uncertain, techniques for robust scheduling will be combined with prediction techniques based on AI driven manufacturing of personalized therapies (WP3). The resulting solver will optimize the chosen objectives, such as idle time, or production delays.

Task 4.5: Assessment, simulation-based validation and deployment preparation of the AI solver (SZTAKI, IPT)

Based on the results of Tasks 4.1-4.4, the manufacturing and resource scheduling AI solver will be validated with the simulation model. The optimization objectives and benchmarks, identified in Task 4.1, will be used to assess the performance of the system using unseen test data and the knowledge of technological experts. Based on the final performance assessment by using measures such as F1-score, precision and recall, the AI solver can be optimized. This optimization can comprise a hybrid modelling approach, in which the results of the physical model is used for training the AI solver. For the deployment of the AI solver to the final environment, a proper explanation of the results and the models' behaviour is needed to make it understandable for technological experts. Furthermore, an adaptive strategy is carried out to ensure the

applicability of the AI solvers in case of model degradation. Finally, continuous improvement process will be developed in order to keep track on the model performance. In order to optimize and guarantee the reliability of the results obtained, a systematic feedback loop iteration will be implemented.		
Deliverables		**Date (M)**
D4.1.	Use Case Description for production scheduling and resource management including view of material flow, dependencies and optimization objectives	9
D4.2.	Pre-processed data sets for production scheduling and resource management AI training	18
D4.3.	Comprehensive simulation model of the production environment and resource management	26
D4.4.	AI production scheduling and resource management model	33
D4.5.	Performance report on implemented production scheduling and resource management AI	39

3.1.8 WP 5

WP No.	5		**WP title**			Automated Pilot for CAR-T production								
Lead beneficiary	IPT		**Start month**			1			**End month**			36		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	36	9	23	7		12	55			1				

Objective

Construction of an automated, robot-assisted platform including 1) a manufacturing and a process analytics module, 2) a comprehensive control software governing production control and communication between devices, AI and the user, 3) an enhanced cartridge-based bioreactor device for autonomous CAR-T cell expansion and 4) CAR-T manufacturing and analytics protocols for fully automated, smart production

Description of work

Description of work

WP5 is dedicated to the design and build-up of the automated, modular CAR-T cell production system. This WP will closely interlink with WP 9 to ensure that regulatory requirements (e.g. GMP, GAMP) as well as the stakeholder perspective is taken into consideration. The end of the build-up at IPT will be the start of WP7, when in which the production plant is shipped to UKW for system integration and validation.

Task 5.1: CAR-T Manufacturing protocol assessment and requirement definition (IPT, AGL, UKW, FCRB, IZI, ALL)

IPT and AGL will gather all relevant information on manufacturing by screening existing standard operation procedures (SOPs), conducting expert interviews and on-site visits at partners UKW, FCRB, IZI, UCL and/or FORTH in order to assess the processes, protocols and hardware to be incorporated into the AIDPATH system. This task is accompanied by a partner workshop in which critical process steps, variables, features and requirements for both the bioreactor as well as the overall system are defined. IZI will consult the requirement definition progress in the context of GMP compliance. Additional requirements and suggestions from regulatory and stakeholder standpoint obtained from WP 9 will be taken into consideration. The information will be harmonized and translated into an overall process map including the material flow. In addition, user requirement specifications (URS) for the Automated system hardware and software will be prepared to be signed by the partners. Based on these documents, IPT will develop a production plant design including the two different modules for 1) autonomous bio production and 2) robot-assisted, end-to-end at-line process analytics. This includes a definition of dimensions, layout and functionality. For each module, a comprehensive automation concept and a 3D model is developed. Thereby IPT will closely collaborate with AGL to ensure that the bioreactor seamlessly integrates into the system. In parallel, a market research is done to select the hardware for the AIDPATH CAR-T cell production plant. Any necessity for in-house

development and customisations by the project team results from the gap between technical requirements and existing solutions and will be defined and prepared.

Task 5.2 Adaptation of automated cartridge-based CAR-T bioreactor module (AGL, UKW, UCL, FORTH, IPT)

One unit of the Facer bioreactor will be manufactured together with a new cartridge which will incorporate the process defined in task 5.1. The unit is formed by three modules (the storage module, the smart module and the culture module) and will have the capacity to process one cell batch -one patient- at a time. The Facer 2.0 is already a very flexible prototype that allows for internal modifications. Therefore, it can used to incorporate new hardware that enhances the sensing potential of the equipment. The resulting unit will be further used to test the potential of the Facer to work as cell expansion module governed by the AI process model in WP3. Subsequently, AGL will modify the Facer 2.0 to integrate the new hardware and sampling options (according to the requirements defined in Task 5.1) to be able to run the AI process control model. Therefore, the AGL engineering team will study how to integrate the new hardware components into the current smart culture modules of the platform. After integration, new components will be tested and calibrated within the platform. Also, software will need to be adapted by creating new routines for the new components (reading, calibration and data processing routines). The monitoring of critical parameters will also be included in the platform interface. Additionally, AGL software will be modified to receive the feedback from the AIDPATH control software and actuate accordingly.

In order to develop a tailored single-use cartridge, design modifications will be required in the circuitry of the cartridge in order to incorporate new components required in the single-use unit, as well as the sampling solutions for continuous off-line analysis. At this point a new design will be defined and prototypes produced by 3D printing. Constant feedback from the consortium members during the design iteration process will be requested to ensure that the cartridge truly takes into account all the requirements for the final project aim.

For additional sensors integrated to the Facer, the sensing characteristics of the biosensors will be analysed, such as average sensitivity, linearity, response time, decay rate, drift rate, and limit of detection. Therefore, for each sensor spot the maximum reading frequency will be stablished, based on sensor lifetime and assumed culture periods. The metabolite readings will be compared to off-line analysis in terms of linearity, precision and accuracy. Also, a sterilization study will be conducted, when required.

Task 5.3: Development and build-up of the automated production plant (IPT, IZI, ALL)

This task starts with procurement of devices once the hardware selection in Task 5.1 is completed (e.g. robot, flow cytometer, nucleofection device, controlled rate freezer, etc.). Following the concept development, the bespoke solutions (e.g. prototype for automated drawing of samples) according to the list defined in task 5.1, as well as necessary modifications of the devices are developed (e.g. adding axes and motors to facilitate automated operation). For each module, a frame is built, which carries the respective devices, electrical and pneumatic components. In addition to the hardware implemented, the devices will be integrated into the manufacturing control software following a service oriented architecture approach. The control software, which IPT is already using successfully for production control of automated bio production systems (e.g. in StemCellFactory, Autostem, etc.), will be adapted to the requirements determined in Task 5.1. A standardized interface for communication between the control software and the individual devices will be created. Standardized interfaces to contiguous IT systems (e.g. manufacturing planning system, database management system) will be implemented for efficient data integration. In order to realize intelligent manufacturing control, interfaces are developed that allow a direct integration of the AI models developed in WP 3 and 4. This ensures that the decisions and parameters suggestions proposed by AI are executed at the process level. The user interface of the control software will be extended by a dashboard that displays process parameters, comprehensible information from the CAR-T digital twin as well as AI based processing decisions and process predictions transparently. The automated production system including both modules and process software will be installed and tested prior to shipping with an agreed test protocol with the end-users in order to ensure functional and sterile system operation (see WP 7).

Task 5.4 Automation of CAR-T manufacturing and analytic protocols (UKW, IPT, IZI FORTH, UCL)

In parallel to the development of the automated system, all manufacturing and analytical protocols will be updated, transferred and optimized to be compatible with manufacturing in the fully automated AIDPATH platform. Manual GMP-compliant SOPs for the production of CAR-T at clinical scale are available at UKW, which will be revised and adopted in order to fit the fully automated manufacturing process. Whenever possible, handling steps (e.g. pipetting, liquid handling) will be reduced to facilitate the automated process. The whole process will be transferred to a closed, automated setting by UKW that limits manual handling to

an absolutely required minimum of human intervention. In addition, protocols for at-line process analytics will be automated. Therefore partners UKW, UCL and FORTH will optimize the manual QC protocols ready for automation. IPT will then transfer the protocols onto the AIDPATH system and implement liquid handling processes on the pipetting robot. Protocols for QC include processing of samples for metabolomics (FORTH), surface expression of phenotypic markers (UCL) as well as indicators for activation and exhaustion and transduction efficacy (UKW). Thereby the assays to be performed regularly on the AIDPATH system at-line will include flow cytometry for evaluation of T-cell product quality, cell counts and viability. In addition, samples will be prepared automatically ready for offline analyses such as mass spectroscopy and functional evaluation. All off-line assays will be adapted to assess CAR-T samples harvested and processed directly from the manufacturing platform. Partner IZI will thereby accompany the process transfer and optimization to ensure that procedures are in accordance with GMP

Deliverables		**Date (M)**
D5.1.	CAR-T Manufacturing SOPs, Production Sequence Diagram and Material Flow Mapping, URS for Automated Production System	6
D5.2.	Design of automated production System incl. Hardware, Software and Interface Definitions	18
D5.3.	Automated bioreactor device and standardized cartridges for autonomous CAR-T manufacture	27
D5.4.	Automated modular CAR-T production pilot	34
D5.5.	Optimized SOPs for automated CAR-T manufacturing and analytics	36

3.1.9 WP 6

WP No.	6		**WP title**			Data security, IoT infrastructure and data management								
Lead beneficiary	ORTEC		**Start month**			1			**End month**			48		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	36		1		1	6					1	20		51

Objective
Provision and management of the entire IoT infrastructure in AIDPATH including 1) data security and risk assessment, 2) comprehensive data integration for AI development and the digital cell twin 3) software solutions for manufacturing and resource scheduling and 4) secure and standardized connection of all AI models, software and hardware components.
Description of work
Task 6.1: Cybersecurity by design including project risk analysis, business and data security (RAL, ORTEC) This task aims to identify and give value to the most **sensitive data** and activities of the project. This analysis will focus, among other things, on various criteria such as confidentiality and privacy of the data to be protected and the availability of the most sensitive services and will be based on security standards such as ISA/IEC 62443, and Eurosmart IoT SCS. A **comprehensive risk analysis** will ensure that the IoT infrastructure will be implemented in a secure manner while ensuring the privacy of the data that will be manipulated as well as the processes. This will include business line security goals, validation of the proposed IoT architecture, identification of threats and attack scenarios, and provision of a security and privacy risk cartography. The infrastructure of LogiqCare / LogiqScience, Big Data Portal and clinical decision making tools (ISO 27001 and NEN7510 conform) are reviewed, together with the to be developed external communications. Finally, a list of AIDPATH-related security & privacy requirements will be provided to be used by all partners during the integration, development and validation phase of the IoT infrastructure and CAR-T cell production pilot in order to ensure an implementation of cybersecurity by design.

Task 6.2: Development of an architecture for the IoT infrastructure including interface and communication definition (IPT, ORTEC, UKW) At the beginning of the project the IoT infrastructure (consisting of all software and hardware components inside and outside the AIDPATH solution) is determined in a workshop and its components are put into relation. For standardized communication and integration of all resources and human beings, an IoT architecture for the planning and control of the bioprocess and the adjacent logistic processes is required. For this purpose, requirements and system conditions are determined in close coordination with AI and bioprocess experts. The IoT architecture is subdivided into data and software architecture and is developed on the basis of approved industry 4.0 approaches (e.g. RAMI, IIRA). Taking into account the data security regulations, the data architecture determines where the data is stored and how its access is regulated. The software architecture presents the necessary software components (control software, software agents, AI models, data base systems …) and their functional relationships and interfaces. Based on the IoT architecture, the interfaces of the components and their standardized communication (e.g. OPC-UA) are defined.

Task 6.3: Implementation of a data storage framework and AI data preparation pipeline (ORTEC, IPT, UCL, SZTAKI, IRIS)

Based on the data structure (task 2.1) and the AI data sets and models (task 3.2 and 4.2) the data storage framework and the integration of the AI data will be implemented. The *Logiq Platform* (provided by ORTEC) will be maintained and expanded in this task. For the different use cases from WP3 and 4 so called *LogiqCare Apps* will be developed with support of the domain experts. Each application will sync to the local data, either by direct automated or manual recording and/or by data-exchange with a local application and store them in a cloud-based system. The integrative use of data of these apps will be shared using attribute based or role based access control rules (ABAC/RBAC) as defined by the domain users, taking GDPR into account. Data models will be either directly integrated using user-friendly interface, or synced from a local application. Essential project data will be synced anonymously into the *Big Data Portal* for analytical purposes. Supporting both application-specific and integrative AI applications, *Big Data tools* for data analytics will be realized in cooperation with AI experts and domain owners to enable the AI data preparation pipeline. The *Logiq Platform* will be designed in such a way that a data exchange through different clinical sites can be achieved in order to realize a wide scale implementation of the AIDPATH system after the project.

Task 6.4: Adjustment and validation of the manufacturing control and resource planning software (IPT, ORTEC)

In this task the operation software modules for control and planning will be adapted to the interface and security requirements of the IoT architecture (6.1, 6.2) and the use-case conditions compiled in task 4.1. A resource planning software (e.g. a solution by ORTEC) will be selected based on the needed functions from WP4 (e.g. capacity planning, resource scheduling) and will be adapted to the scope of AIDPATH. The resource planning software needs to interact closely with the manufacturing control software, developed in task 5.3. Therefore, the two software modules will be connected in a simulation environment to validate their functionality.

Task 6.5: Implementation of the AI-integrated IoT infrastructure and acceptance testing (ORTEC, IPT)

In task 6.6 the different components of the IoT architecture, defined in task 6.2, will be put together and their functionality will be tested. Therefore, the AI solutions of WP 3 and 4 will be connected to the manufacturing control and resource planning software (task 6.5). For the AI integration the end-to-end AI pipeline will be connected through the defined interfaces and an intelligent planning and control as well as the digital cell twin will be realized. The overall functionality of data integration, operational control and planning tasks and AI decision support will be tested, before the pilot build-up in WP7.

Deliverables		**Date (M)**
D6.1.	Report on system architecture and interface definition	12
D6.2.	Validated manufacturing control and resource planning software, Data storage framework and data preparation pipeline for AI	34
D6.3.	Documentation of AI-integrated AIDPATH IoT infrastructure and their software components	45

3.1.10 WP 7

WP No.	7		**WP title**			On-site pilot implementation and validation								
Lead beneficiary	UKW		**Start month**			25			**End month**			48		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	21	22.6	2			48	10			1		6	16	10

Objective

Implementation of the AIDPATH manufacturing platform in the hospital. This includes 1) preparation of supplementary documentation such as operations manuals and technical documentation, 2) physical implementation of the AI supported AIDPATH IoT infrastructure from WP 6 and the automated manufacturing system from WP 5, 3) testing and validation of the system including a full manufacturing run under GMP conditions and preparation of testing reports, 4) economic evaluation of the AIDPATH system as built and implemented.

Description of work

WP7 focuses on technology transfer and implementation of the AIDPATH autonomous manufacturing system at the clinical site in a pilot scale suitable for efficacious and safe therapy production. This WP will cover technology transfer of the automated system into the hospital and validate the system, bioprocess and bioprocess control as well as the final end product in *in vitro* and i*n vivo* studies.

Task 7.1 CE Marking, operations manual, training and testing protocols (IPT, UKW)
This task prepares operation manuals for the AIDPATH automated system, as well as the preparation of technical files and preparatory documents required for CE marking. All formal verification and validation testing for the system will be carried out, as well as formal testing as required by regulatory Notified Bodies to allow the device be CE marked. There will be two parts of functional testing for the AIDPATH manufacturing platform. The first part will concentrate on the technical functionality of all components, while the second set of tests will be performed after reception at UKW (site acceptance tests). Therefore, test protocols will be developed and a process FMEA will be carried out to minimize the risks of failure. This will involve IPT working liaising with task 7.2 and WP 9 and partners IZI and UKW to define testing protocols and create documentation templates, which ensure that all functions defined in the URS document and the major risks identified in the FMEA are tested for. The second part of the functional testing protocols focuses on manufacturing functionality, including maintenance of product critical quality attributes. Protocols will be developed for testing performance of the system under GMP conditions. In addition, IPT will prepare a training on the platform to enable the personnel at UKW to operate the facility and provide remote support during the validation runs to be conducted.

Task 7.2 Preparation of a GMP documentation package (IZI, UKW, IPT)
Prior to the full biological validation run, GMP documentation supporting the transfer of the optimised process will be prepared. This includes a validation master plan (VMP) and drafts for appropriate documentation for the automated manufacturing platform i.e., Installation, Operational and Performance Qualification (IQ, OQ and PQ) in addition to draft SOPs, Batch Manufacturing Records, Quality Control Records and operations training qualification. Although the system is entirely automated, it is critical that all data is captured and recorded and where interventions (either manual or automated) are performed or instructed by operators these much be captured accordingly. The Batch Manufacturing Records and the overall manufacturing protocols, in which key manufacturing instructions are described, shall be created to allow full traceability and compliance according to GMP. Aseptic Process Simulation runs will be performed and documented to demonstrate process capability for maintaining a sterile product throughout the manufacturing process.

Task 7.3 Technology transfer and installation of the platform at the hospital site and site acceptance testing (IPT, AGL, UKW, RAL, ORTEC)

This task begins with the completion of successful acceptance testing including demonstration of functionality and sterile operation at IPT at the end of task 5.3 according to the protocols defined in task 7.1. Subsequently, the AIDPATH manufacturing platform will be disassembled and shipped to UKW, where it is reassembled and set up in the hospital. In parallel, the different parts of the IoT infrastructure developed and adjusted in WP 6 will be rolled out in the hospital environment and connected to the hospital IT on site and the production pilot. A special emphasis will be given to the integration of the AI models form WP3 and WP4 ensuring a reliable and secure connection to the hospital environment. After the successful installation of the pilot at UKW, the functionality and integration of the platform will be tested according to the protocols defined in Task 7.1 to confirm appropriate technical functionality. In addition, equipment and ancillary materials will be physically laid out and tested in accordance with the new layout and test results are documented in the site acceptance testing report.

Task 7.4 CAR-T manufacturing test runs on the autonomous manufacturing system and characterization of the AIDPATH CAR-T cell product (UKW, IPT)

After the technical functionality is proven, the first biological processes will be performed within the platform by UKW using the automated manufacturing protocols developed in Task 5.4. Test runs will initially involve using healthy donor cells. The tests will start with simple cell cultivation experiments to prove sterility and functionality of components before moving onto a fully completed biological run for production of CAR-T cells. *In vitro* characterization will be performed during optimization and for the final product and will contain analyses of growth potential, surface phenotype, quality and functionality of AIDPATH CAR-T cells. Cell expansion and phenotype will be monitored on-line by cell-counting, life-death staining with PI/Annexin-V, and flow cytometry for phenotypic markers including CD4, CD8, CD45RA, CD45RO, CD25, CD27, CD28, CD69, CCR7, CD62L, and PD-1. Transfection rates will also be measured by flow cytometry. For selected cell products, we will perform additional off-line analytics, including mass cytometry (CyTOF) to measure up to 42 phenotypic markers simultaneously. T cell receptor repertoires will be assessed by TCR spectratyping to ensure that AIDPATH CAR-T cells retain a wide TCR spectrum. Additionally, we will analyse the genomic insertion site profile and copy number of the CAR gene per genome by ddPCR and deep sequencing. To evaluate AIDPATH CAR T cells' effector functions, we will test the generated CAR-T cell lines for recognition of suitable primary target cells and cell lines by measuring: a) cytolytic activity by luminescence-based cytotoxicity assay, b) production of IL-2, TNF-α and IFN-γ by multiplex cytokine assay, and c) proliferation and induction of activation-induced cell death by CFSE and PI/Annexin-V staining. The assay procedures will be validated with existing CAR-T cell products prior to testing AIDPATH CAR-T cells.

To characterize AIDPATH CAR-T *in vivo*, efficacy and biodistribution will be evaluated using established *in vivo* models. We will use NSG mice engrafted with either primary cells or target expressing cell lines. Dose titrations will be performed to evaluate efficacy and safety of AIDPATH produced CAR T cells in comparison to CAR T cells generated with viral or non-viral non-automated methods. We will analyse velocity and depth of tumour regression, CAR-T cell migration, distribution, persistence, and animal survival. For subgroups of mice, we will harvest blood, spleen and bone marrow to measure T cell infiltration, phenotype (e.g. activation and exhaustion markers) and tumour clearance. The biodistribution of CAR-T and tumour cells will be sequentially analysed by state-of-the-art bioluminescence detection that allows live-imaging of cell populations at high resolution and in anatomical context. Therefore, T cells and tumour cells will be tracked periodically over the whole duration of the experiment for each individual mouse. We will also monitor co-localisation of tumour cells and CAR-T cells in the bloodstream and organs as well as the emergence of extramedullary tumour lesions and the co-localisation of CAR-T cells at these sites. By use of sensitive quantification software, tumour and CAR-T cells will be quantified to draw conclusions about the efficacy, expansion, and survival of AIDPATH CAR-T cells under experimental *in vivo* conditions.

Task 7.5 Cost of goods analysis and economic evaluation of the AIDPATH system (PAN, UKW, AGL, IPT, FCRB, IZI, UCL, HCATS)

In order to fully analyse and quantify the value for money of the platform, in this task an economic evaluation will be conducted comparing the innovative AIDPATH platform to the current standard of producing cells. For this purpose, a health economic model will be built that can integrate data from various sources to analyse the costs and effects of the new platform compared to the external production of cells. The costs of the patient treatment, including the costs for the cell production, will be analysed, as well as the impact on treatment outcomes (e.g. by ensuring a smooth logistical and timely process, by starting treatment sooner). The analysis

will account for potential cost-savings in the treatment, as well as the possibility to scale up the production so that patients can receive treatment for whom this previously was not accessible due to the shortage of cell products. Outcomes of the analysis can include the incremental cost-effectiveness ratio (ICER) of costs per quality-adjusted life year and the difference in costs of treatment. Sensitivity analyses will be performed to identify the key parameters that drive the analysis´ results. Moreover, analysis will be conducted to identify the results of several what if-scenarios. The results of the economic evaluation will support the implementation of the system in the hospital by providing information on its costs, potential cost-savings, and information on patients for whom previously no cell products were available but become available with the new system.

Deliverables		**Date (M)**
D7.1.	Integrated system training and operations manual	36
D7.2.	GMP documentation package (VMP, IQ, OQ and PQ documents, manufacturing batch records)	44
D7.3.	Cost effectiveness analysis of the AIDPATH system	46
D7.4.	Documentation on Site Acceptance Testing (SAT) and manufacturing results and functional analysis of the CAR-T product	47

3.1.11 WP 8

WP No.	8		**WP title**			Ethics								
Lead beneficiary	IPT		**Start month**			1			**End month**			48		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	12		1			3				1		18		

Objective

(1) Establish a process for independent ethical review of the project. (2) Assure that the consortium operates under the principles of responsible research and innovation. (3) Define and monitor the implementation of measures to protect privacy and data security

Description of work

Task 8.1 Establishment of an independent ethics board (IPT, ALL)
An Ethics Board (EB) will be established, with the mission to assure ethical conduct of the project in terms of informed consent, data protection and privacy, and research integrity. The board will consist of 3 internal and 3 external members. This will include members with knowledge of the issues surrounding data privacy and security, use of AI, and informed consent for treatment. The external members will decide upon issues that relate to consortium partner misconduct. A process for anonymous reporting of issues to the external Ethics board members will be established.

Task 8.2 Review of project research and innovations in context of ethical considerations (IPT, ALL)
A Responsible Research and Innovation (RRI) charter will be developed by the EB that will be signed by all consortium members. This charter will be built around four principles as defined by the FP 7 project RRI tools: (1) Being responsible: honesty, fairness, objectivity, reliability, accountability, and openness; (2) A transdisciplinary and multi-sectorial approach: allowing for confrontation and collaboration of heterogeneous scientific cultures; (3): Engaging all stakeholders; (4) Assessing the impact of the project: ethical, societal, environment impact. The RRI charter and Ethics issues risks will be reviewed with consortium members on an annual based at the occasion of the annual meeting

Task 8.3 Development of a privacy framework to support GDPR compliance and actors training (RAL)

Within AIDPATH, personal data will be gathered and manipulated in different countries for this project. This task will ensure the compliance of these data according to the General Data Protection Regulation (GDPR). The work will be performed through a risk assessment based on a questionnaire in order to identify measures that should be put in place in the context of the AIDPATH project. For instance, categories that will be covered by the GDPR assessment are data security protection, data audit, activity logs and real time detection. Following the outcome of previous actions, the different actors working on the integration, implementation and validation will be educated in the roadmap and trained in order to raise awareness according to the specific risks of the AIDPATH project. This will allow to ensure that all actors are taking into account the security and the privacy by design.
To understand the privacy risks apparent in AIDPATH a Data Privacy Impact Assessment (DPIA) will be executed. The DPIA will enable the identification of privacy risks and the measures to mitigate these risks. Then, a framework for embedding Privacy and Security by Default and by Design throughout the entire project duration will be developed. The framework will be based upon current standards in information security (management) as well as risk management/assessment (e.g., based on ISO/IEC 27018:2019 and ISO/IEC 29100:2011), and will be augmented by current agreed approaches on privacy by design and by default, as promoted a.o. by ENISA and GDPR.

Task 8.4 Structuring legal agreements for the use and transfer of data and samples (IPT, ALL)
Legal agreements around data and sample handling will be reviewed both by the Ethics Board and a legal expert. A template agreement will be produced and included in the consortium agreement that complies with GDPR requirements. Deviations from this template will be reviewed and a legal review triggered when appropriate. Whenever consortium members seek to use existing data or samples from other initiatives this work package will assist the consortium members in assessing that the proper consents have been obtained to allow for the transfer of data or documents. A provision will be made for subcontracting legal support for reviewing plans and documents as well as providing advice to the Ethics Board and the project Coordinator.

Deliverables		**Deliverables**
D8.1.	GDPR compliant privacy framework	6
D8.2.	Responsible research and innovation charter	9
D8.3.	Report on Ethics issues	45

3.1.12 WP 9

WP No.	9		**WP title**			Stakeholder engagement, regulatory and clinical readiness								
Lead beneficiary	UKW		**Start month**			1			**End month**			48		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	12	2.7	2			6				1			10.5	9

Objective

(1) Ensure that necessary regulatory requirements and considerations regarding GDPR, data security and privacy are taken into consideration in the development and implementation of AI, IT infrastructure and AIDPATH manufacturing platform. (2)Ensure that the preclinical research package for AIDPATH is designed in accordance with the EU standards (3) Submit a scientific-advice request to the EMA for the development path of AIDPATH

Description of work

WP9 focuses on the regulations and clinical readiness of the AIDPATH manufacturing platform and of the cell products generated from the system. As both the use of AI and genetically modified T cells is a highly

sensitive topic in society, this WP is designed to ensure that stakeholders such as a patients and hospitals are involved from the beginning to facilitate trust and acceptance of the newly developed platform. Additionally, as such therapeutic products carry unforeseen risks and often experience setbacks during both preclinical and clinical testing, this WP will make sure that AIDPATH addresses all relevant regulatory guidelines.

Task 9.1 Establishment of an advisory board and development of standing communication channels with associations and the stakeholder community (IPT, UKW, UCL, FCRB, PAN, ORT, HCATS, ALL)
To ensure that the AIDPATH project contributes to trust and acceptance building in the AI technology among stakeholders, Task 9.2 will develop and engage with an advisory board and establish a strategy to engage with stakeholders. The advisory board will be a primary mechanism for engagement with stakeholders and will comprise of key representatives from the clinical, commercial, patient groups, policy and academic sectors. Specifically, this will include Professor Bruce Levine, President of the International Society for Cell and Gene Therapy (ISCT) the leading global community for advanced therapies, Dr Vijay Chiruvolu, VP Manufacturing of Kite Pharma who have an approved CAR-T product is a multi-stakeholder project, (3) a representative from within the AI community as well as (4) from the European Haematology Association (EHA), the leading European Society which serves medical professionals and promotes excellence in patient care, research, and education in haematology, (5) Professor Alejandro Madrigal, Scientific Director of Anthony Nolan, a leading charity that works in the areas of leukaemia, representing patients' needs and interest and (6) a representative from the Digital Innovation Hub DIH Hero. The strategic aim of the advisory board is to provide independent input, feedback and strategic direction to the AIDPATH team and project. This critical composition of key stakeholders ensures that the advisory board can provide an array of perspectives and will ensure that key challenges and emerging opportunities for AIDPATH are captured. The consortium will review the membership and remit of the advisory board regularly to ensure the primary aims and objectives of the advisory board are met.
In addition, alliances with the wider stakeholder community will be established, leveraging networks and connections of the individual partners. This will include academic societies and national/international networks. The partners, with expertise in different areas, will ensure that the primary objectives and outcomes of AIDPATH are communicated and disseminated to their respective networks and wider communities.

Task 9.2: Trust and acceptance building regarding novel AI based technologies and integration of stakeholder perspective (PAN, UKW, IPT, ALL)
For the future implementation of the AIDPATH platform in treatment centres it is essential that stakeholder's incl. patients, clinicians, and hospitals accept and trust the innovative AI-based manufacturing platform. As AI applications still are relatively new, users and providers are hesitant to trust and accept this new technology without restrictions. Also, stakeholders will have questions on how data security is ensured. Therefore, after the advisory board has been put together in task 9.2, the stakeholder will be invited by the consortium for first workshop. In this workshop, stakeholder are introduced to the AIDPATH vision and concept. To build trust and acceptance regarding this new AIDPATH AI-based platform, the expectations and questions of relevant stakeholders will be identified to, in the next step, address these within the project and inform the various stakeholders how AIDPATH meets these. During the project a stakeholder community will be developed. This will be a self-directed, participatory community to ensure that the community has the potential to grow to achieve the critical mass necessary to have the intended impact to move the concepts of a 'smart manufacturing hospital' into wide scale implementation. Initially stakeholder participation will take the form of consultation and advice. Interviews will be conducted with several stakeholders (incl. patients and patient organisations, caregivers, clinicians, hospitals, insurers) to identify the factors that influence the acceptance and trust of stakeholders regarding the AI-based cell manufacturing platform. Expectations and requests concerning design and usability of the AIDPATH system will be collected in the stakeholder interviews. Based on the gathered information, stakeholder-centric design principles for the AIDPATH technology are defined, that ensure the technology is catered to their needs and expectations. The identified issues and gaps on a broader societal level as well at the technical level will be used to formulate a broader participatory community invited members to contribute to the work being done by the AIDPATH consortium, or to propose other new initiatives in line with the goals outlined in the Societal Guide to Implementation of Smart Manufacturing Hospitals. Towards the end of the project when the pilot platform is implemented in Wurzburg, the advisory board and where applicable other key representatives of the stakeholders (such as patient associations) will be approached again to inform them on how the factors that influence their trust and

acceptance in the technology were addressed. The stakeholder community will then also be transformed into an innovation ecosystem as described above in section 2.2.1.

Task 9.3: Establish a regulatory roadmap to ensure that regulatory aspects are taken into consideration when establishing the AIDPATH manufacturing platform (UKW, IZI, ALL)

A regulatory roadmap for the AIDPATH manufacturing system and the final cell product, including quality management and assurance systems, will be established. This will include several activities ranging from manufacturing environment, defining drug substance and product, starting material retrieval and control, in-process controls up to manufacturing process related subjects, testing and release and required documentation and registration procedures. In detail, for ensuring compliance to GMP-guided manufacturing within an automated closed system, such as AIDPATH, a combinatorial approach led by European Union's Annex 11 (EUDRALEX Rules Governing Medicinal Products in the European Union, Volume 4, Good Manufacturing Practice, Medicinal Products for Human and Veterinary Use) and complemented by FDA's Part 11 (21 CFR Part 11) regulation will be provided. Thus, ensuring compliance regarding efficient validation of computerized automated systems, including operations and controls for closed systems as well as electronic signatures and data/patient safety. In addition, we will implement an extensive guideline package to ensure full GMP compliance in the AIDPATH manufacturing model.

We will also consider GAMP 5 (Good automated manufacturing practice) guideline for manufacturers and users of automated systems in the pharmaceutical industry, adapted to ATMP manufacturing. Particularly, we will rely to the International Society for Pharmaceutical Engineering`s (ISPE) guide "*The Good Automated Manufacturing Practice (GAMP) Guide for Validation of Automated Systems in Pharmaceutical Manufacture"* (Version 5) describing a set of principles and procedures that help ensuring required quality standards. Herein, we aim to stick to one of the core principles of GAMP that is quality cannot be tested into a batch of product but must be built into each stage of the manufacturing process. Consequently, all aspects of production; beginning from the raw materials, facility and equipment until the training, qualification and hygiene of staff will be implemented. Standard operating procedures (SOPs) will be implemented for all processes within the entire AIDPATH manufacturing platform that may affect the quality of the finished product. Nevertheless, it has to be pointed out that, in contrast to previous mentioned EMA and FDA guidelines having legal bindings, the ISPE GAMP guides have no legal bindings. However, they represent the current "gold standard" for validation of computer-assisted systems within the pharmaceutical industry, thus enabling to support our regulatory framework regarding GMP compliance of the implemented AIDPATH manufacturing model.

By ensuring that the AIDPATH manufacturing platform complies with the minimum standard that a medicine manufacturer must meet in their production processes, this would potentially aid in both regulatory and industry acceptance of the platform when moving into clinical testing.

Task 9.4: Scientific-advice request to the EMA and preclinical research package for AIDPATH in accordance with the EU standards (UKW, IZI)

Appling for a scientific advice with the EMA seeks to ensure that the development program for AIDPATH will meet the evidence needs of the regulatory bodies to move forward into clinical testing scenarios. Therefore, we will implement scientific advice procedures with the EMA`s Innovation task force (ITF) aiming to guide the regulatory process at an early stage in product and process development. This includes liaison with EMA on the technologies developed in the different strands of AIDPATH (cell technology, platform concept and layout, preparation and input during validation). Herein, EMAs Innovation Task Force provides an optimal platform for early dialogue on scientific, regulatory and legal requirements. The advice meetings will also be essential to establish contact with the EMA to address any issues that may require further clarification. We will also consider ITF briefing meetings that are intended to be much earlier than when one would normally seek scientific advice, thus covering early regulatory, technical and scientific issues during initial stages of the development path in AIDPATH. This task represents provision of support for ITF briefing meetings and scientific advice meetings, including preparing, planning and participating in these meetings.

This task will ensure providing adequate regulatory guidance for the implemented AIDPATH preclinical validation package in compliance with EMAs scientific guidelines on non-clinical development of medicinal products. We will establish a framework of regulatory requirements regarding non-clinical needs for ATMPs

in close collaboration with WP7 (task 7.4) (responsible for practical realization of preclinical in-vitro and in-vivo studies). This will include primary pharmacodynamics (PD) providing data on *in vitro* studies for characterization of the AIDPATH cell product regarding yield, growth potential, surface phenotype and *in vitro* quality profile of the cells. The data set will be complemented by pharmacokinetics/ biodistribution studies providing additional data to support PD and evaluating preclinical in-vivo safety aspects. Herein, in collaboration with WP7 (task 7.4), an *in vivo* animal model (NSG mouse model) reflecting the intended clinical background will be established and evaluated by a set of distinct parameters such as velocity and depth of tumour regression, CAR-T cell migration, distribution, persistence and animal survival.

Deliverables		**Date (M)**
D9.1.	Stakeholder community integration and communication plan	9
D9.2.	Regulatory roadmap in compliance with EMAs scientific guidelines	15
D9.3.	Stakeholder centric design principles ensuring trust and acceptance	18
D9.4.	Scientific advice Meeting with EMA's Innovation task force	30
D9.5.	Societal Guide for implementing Smart Manufacturing Hospitals	39

3.1.13 WP 10

WP No.	10		**WP title**			Communication, Dissemination and preparation for wide scale implementation								
Lead beneficiary	IPT		**Start month**			1			**End month**			48		
Participant number	1a	1b	2	3	4	5	6	7	8	9	10	11	12	13
Short name of participant	IPT	IZI	UCL	FORTH	SZTAKI	UKW	AGL	FUJI	HCATS	FCRB	IRIS	RAL	PAN	ORT
PM per participant	9		1	2	2	3	4	1		1	3	2	16	10

Objective
The aim of this work package is to (1) generate wide visibility of the objectives, concepts and results of AIDPATH, (2) derive a viable business model and (3) make all preparations allow a seamless transition towards technology scale up/out after end of the project
Description of work
Task 10.1 Communication and dissemination infrastructure (IPT, ALL) This task will set up internal and external communication channels in order to ensure continuous and efficient exchange between the project partners, stakeholders and the public: <u>Internal communication:</u> The coordinator and the Project Executive Group (PEG) will monitor progress of the WPs closely. In order to ensure a constant flow of information between the partners, regular meetings in person and online will be held. The project will start with an initiation ('kick-off') meeting. Plenary meetings will be held twice yearly thereafter where some of them will take place in conjunction with technical workshops. Plenary meetings act as self-assessments and internal reviews, as well as reporting and planning opportunities. Every WP will present a progress report and short-term future plans at each meeting. Upcoming deliverables and milestones will be reviewed. A review of project risks, opportunities, new intellectual property and an assessment of any new developments in advanced therapy and related sectors will be permanent agenda items. In addition to meetings in person, regular online meetings will be held (e.g. monthly) for the partners to give updates on the progress and to facilitate communication between the WPs. The tools and technologies required for collaboration and data sharing will be put in place and maintained throughout. A shared documents service will be used for collaborative authoring. Email, skype and telephone will be used in between face-to-face meetings. Project knowledge will be managed and retained throughout the duration of the project, so that material generated by one partner is available to all, old versions of documents remain available, relevant research data is shared and easily accessed and emailed discussions are archived and

retrievable. This will facilitate reporting and help to deal with any changes in personnel or consortium, increasing the robustness of the project as a whole.

External communication and dissemination: An infrastructure will be installed ensuring that the project has a strong visibility and dissemination activities will generate a wide outreach. Therefore, a project website will be set up at www.aidpath-project.eu (or a similar do-main name). The website will include a description of the project and partners, news and events page, a media area, links to related projects, and contact details for the project. Overall, web material will be aimed at the general public, with additional specific pages for specialist audiences (clinicians, industry partners, other projects, potential collaborators. Twitter, YouTube, LinkedIn and Facebook accounts will be set up and managed by IPT. The different partners will distribute information within their communities. This includes for example the engagement in the Digital Innovation Hub DIH Hero for exploitation of robotic and digital solutions in the case for IPT.

Task 10.2. Dissemination campaign plans (IPT, ALL)

A dissemination campaign plan will be prepared by IPT to ensure a continuous flow of information on the AIDPATH project. The plan will include contents, timings and target audience for dissemination of project activities, results and milestones reached. Therefore, IPT will update the website regularly throughout the project life-time, with all partners contributing content. In order to ensure that it reflects the current project status, the website will be reviewed at every plenary meeting. The dissemination will be done through the channels defined above (website, social media, communication with stakeholder community, scientific presentations, publications, representation at conferences, tradeshows and fairs, active engagement with DIH Hero and other associations (ISCT, etc.). In collaboration with all project partners, IPT will develop a project brochure, poster and other awareness materials for circulation by the consortium at conferences and other public events. The materials will also be made available as downloads from the project website. At the end of the project, IPT will report on the results dissemination campaign plan.

Task 10.3: Business plan development utilizing a modelling tool (PAN, UKW, FCRB, HCATS, ALL)

For smart hospitals to create, deliver and capture value by means of the innovative AIDPATH platform, they need to innovate their business models. A business model describes the rationale of how an organisation creates, delivers and captures value[37]. This task will develop a business plan modelling tool that provides insights into the business model, financial analysis, external environment, and a risk analysis for hospitals interested to transform their current business model into a future oriented business model by means of the AIDPATH platform. To describe the business model use will be made of the Business Model Generation Framework[37]. This framework consists of nine building blocks: 1. Customer Segments; 2. Value Propositions; 3. Channels; 4. Customer Relationships; 5. Revenue Streams; 6. Key Resources; 7. Key Activities; 8. Key Partnerships; and 9. Cost Structure. Inputs for the financial analysis will be derived from task 11.1: economic evaluation. As business models are designed and executed in specific environments, additional data will be collected about four areas: 1. Market forces; 2. Industry forces; 3. Key trends; 4. Macroeconomic forces. Combined with a strengths, weaknesses, opportunities and threats (SWOT) analysis, the Business Model Canvas enables a more focused assessment and evaluation of an organisation's business model and its building blocks. This type of analysis provides a solid basis for decision making and innovation of an organisation's innovative business model. Data will be collected from various sources, incl. white and grey literature, interviews with key opinion leaders and stakeholders (incl. customers and partners), and the economic evaluation (task 7.4). This task will report on a viable business model that will serve as a basis for discussion with industry, investors and stakeholders and will help to shape an action plan for wide-scale technology implementation.

Task 10.4 Industrial engagement and commercialisation (IPT, UCL, WUR, AGL, FORTH, ALL)

A key dissemination activity for the AIDPATH project is to engage with hospitals, treatment sites and potential other customers as well as funding agencies and investors to ensure uptake and implementation of project findings. Based on the business model defined in Task 10.3, a sustainability plan will be developed to ensure project results will be further developed, exploited and marketed. This will include the exploitation of individual innovations that are part of the AIDPATH system as well taking next steps for the wide scale implementation for the overall AIDPATH system:

[37] Ostenwalder et al. 2020

As is expected from a project which will deliver multiple innovations and advances in the field, each partner with new intellectual property will develop plans for the further exploitation of project results. Partners with sole IP ownership (e.g. FISI's media formulations, AGL's bioreactor system, ORT's software platform) will fold this work into their existing business plans and product development plans. Partners who share some IP will discuss and agree how such shared IP is to be further developed. Possible strategies include one partner licencing the IP of others, or a new company being established with shared equity, or the breaking-up of the IP into solely-held units. The AIDPATH system is designed to address key bottlenecks in manufacture for hospitals and therefore will be relevant to both to clinical sites as well as companies that are developing therapies, acquiring and licensing new manufacturing plants. In order to market the system as a whole and prepare for 'wide-scale' implementation, engagement with potential joint venture partners and investors will be sought. This can include the formation of strategic alliances and the development of public private partnerships. Across Europe, clinical sites and treatment centres active in the CAR-T therapy field will be addressed to raise interest in joining forces for preparation of implementation of the AIDPATH technology across multiple sites. In addition, the consortium will seek engagement with funding agencies on regional, national and international level to leverage effect on additional funding. These activities will be stimulated through an Exploitation Workshop at M36, in which technologies and target groups are identified. This will require input from all consortium members to ensure a significant geographical and network reach and liaison with the advisory board from the innovation ecosystem set up in Task 9.2. Investors, companies and other institutions that are relevant for licencing and commercialization of the technology will be invited to an innovation day, in order to "pitch and match" for investment in technology developers.

Deliverables		**Date (M)**
D10.1.	Concept dissemination campaign plan	8
D10.2.	Results dissemination campaign plan	36
D10.3.	Business model, sustainability plan and engagement with investors and funding associations	44

3.1.14 List of Deliverables

Del #	Deliverable name	WP#	Lead name	Type	Diss'n level	Date
1.1.	Project Charter and IPR scan file	1	IPT	R	CO	3
1.2.	Data management plan	1	IPT	R	CO	6
2.1.	Integrated data structure for AI development	2	IPT	OTHER	CO	6
5.1.	CAR-T Manufacturing SOPs, Production Sequence Diagram and Material Flow Mapping, URS for Automated Production System	5	IPT	R	CO	6
8.1.	GDPR compliant privacy framework	8	RAL	R	CO	6
10.1.	Concept dissemination campaign plan	10	IPT	R	CO	8
3.1.	Use Case Description for precision AI manufacturing including review of influencing factors optimization objectives	3	IRIS	R	CO	9
4.1.	Use Case Description for production scheduling and resource management including view of material flow, dependencies and optimization objectives	4	IPT	R	CO	9
8.2.	Responsible research and innovation charter	8	IPT	R	CO	9
9.1.	Stakeholder community integration and communication plan	9	IPT	R	CO	9
2.2.	AI concept, interaction and development strategy	2	SZTAKI	R	CO	12
6.1.	Report on system architecture and interface definition	6	IPT	R	CO	12

9.2.	Regulatory roadmap in compliance with EMAs scientific guidelines	9	UKW	R	CO	15
3.2.	Pre-processed data sets for precision manufacturing AI training	3	FORTH	OTHER	CO	18
4.2.	Pre-processed data sets for production scheduling and resource management AI training	4	IPT	OTHER	CO	18
5.2.	Design of automated production System incl. Hardware, Software and Interface Definitions	5	IPT	R	CO	18
9.3.	Stakeholder centric design principles ensuring trust and acceptance	9	PAN	R	PU	18
3.3.	Prototype of AI-CAR Twin suitable for bed-side hospital operation	3	UCL	DEM	CO	21
4.3.	Comprehensive simulation model of the production environment and resource management	4	SZTAKI	OTHER	CO	26
5.3.	Automated bioreactor device and standardized cartridges for autonomous CAR-T manufacture	5	AGL	DEM	CO	27
9.4.	Scientific advice Meeting with EMA's Innovation task force	9	UKW	R	CO	30
2.3.	Defined End-to-End AI pipeline including deployment plan	2	IRIS	R	CO	33
3.4.	AI CAR-T Digital twin for advanced and adaptive bioprocess control strategies trained on optimised bioprocess datasets	3	UCL	DEM	CO	33
4.4.	AI production scheduling and resource management model	4	SZTAKI	DEM	CO	33
5.4.	Automated modular CAR-T production pilot	5	IPT	DEM	CO	34
6.2.	Validated manufacturing control and resource planning software, Data storage framework and data preparation pipeline for AI	6	ORT	OTHER	CO	34
10.2.	Results dissemination campaign plan	10	IPT	R	PU	36
5.5.	Optimized SOPs for automated CAR-T manufacturing and analytics	5	UKW	R	CO	36
7.1.	Integrated system training and operations manual	7	IPT	R	CO	36
3.5.	Performance report on implemented cell culture and bioprocess AI	3	IRIS	R	CO	39
4.5.	Performance report on implemented production scheduling and resource management AI	4	SZTAKI	R	CO	39
9.5.	Societal Guide for implementing Smart Manufacturing Hospitals	9	PAN	R	PU	39
10.3.	Business model, sustainability plan and engagement with investors and funding associations	10	IPT	R	CO	44
7.2.	GMP documentation package (VMP, IQ, OQ and PQ documents, manufacturing batch records)	7	IZI	R	CO	44
2.4.	Guidelines and building blocks for AI in the smart hospital of the future	2	IPT	R	PU	45
6.3.	Documentation of AI-integrated AIDPATH IoT infrastructure and their software components	6	IPT	R	CO	45
8.3.	Report on Ethics issues	8	IPT	R	CO	45
7.3.	Cost effectiveness analysis of the AIDPATH system	7	PAN	R	CO	46
7.4.	Documentation on Site Acceptance Testing (SAT) and manufacturing results and functional analysis of the CAR-T product	7	UKW	R	CO	47

3.2 Management structure, milestones and procedures

3.2.1 Management Objectives

The **management structure** for AIDPATH has been developed to ensure that the consortium **fulfils its aims** effectively and efficiently, in line with the EU Call Text and the following objectives:

- Deliver **an inclusive consortium** that brings together domain experts, companies and other stakeholders,
- Support an **excellent innovation programme** that extends the state of the art in multiple dimensions;
- Facilitate the work, and the **collaboration and communication**, of all the work packages**.**
- Maximise the **impact** of the project during and after the project lifetime;
- Deliver the **contractual obligations** of the project to the EU, by continuously monitoring progress, facilitating regular internal communication and fostering a shared understanding of roles and responsibilities.

3.2.2 Organisational Structure

Project coordinator

The AIDPATH project is led the Fraunhofer IPT, part of the Fraunhofer organization and leading experts in production technology. Their role includes strategic leadership, overall scientific vision, day-to-day orchestration of the various WPs, administration, reporting and periodic liaison with the EU. IPT has a great deal of experience in multi-partner and EU projects, including the successful AutoCRAT and JointPromise projects, on which this application builds. The coordinator can count on a strong team of actively interacting partners from leading institutions (see 'Consortium as a whole' and section 4 of this proposal).

Overall, the **role of the coordinator** is to ensure that the project is progressing smoothly, that partners are communicating and collaborating effectively, that the contract and associated annexes are being complied with, and that all steps are being taken to maximise the project impact for research and industry.

The coordinator will coordinate preparation and submission of ***periodic reports*** and a ***final report***, including financial reporting, as required by the *Description of Action.*

Project Executive Group (PEG)

The project coordinator chairs a small Project Executive Group (PEG), consisting of the coordinator, Prof. Dr.med. Michael Hudecek from University Clinics Würzburg (UKW), Dr. Qasim Rafiq from University College London (UCL) as well as Ortec (ORT) from the industrial side. This PEG provides day to day leadership and direction for the project, and ensures that the major innovation challenges are addressed in a coherent and integrated manner that takes into account both the Artificial Intelligence and the advanced-therapy viewpoints. The PEG provides vision and strategic orientation of the project, strategically liaising with the advisory board and the stakeholder community.

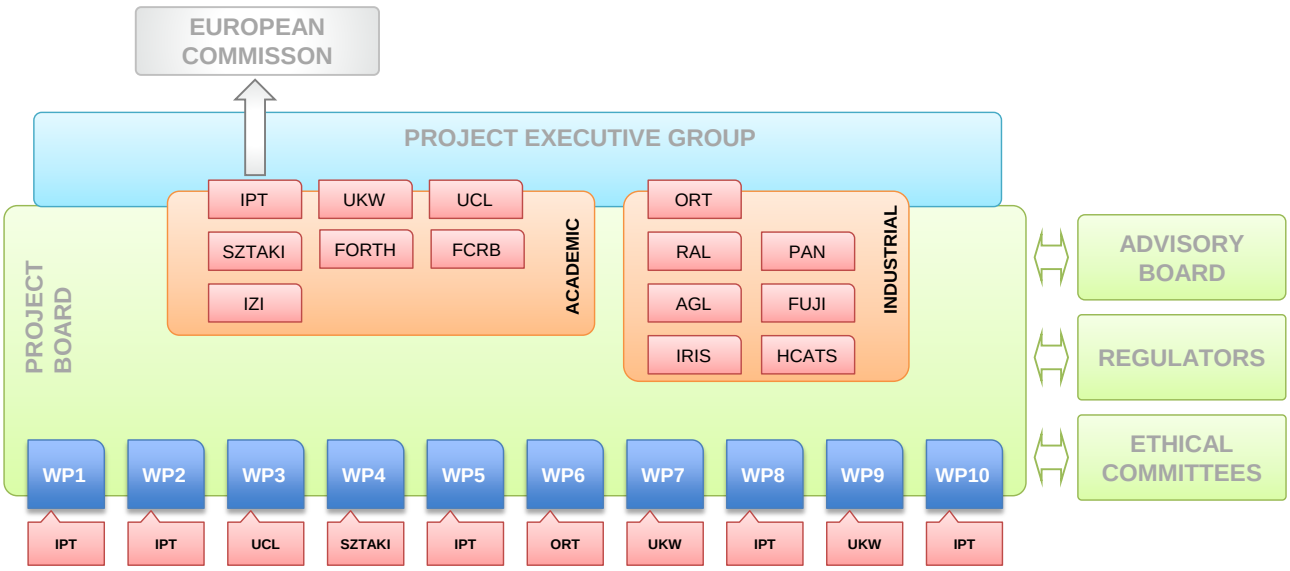


Figure 11: Overview of the organisational structure of the AIDPATH project

Project Board

Overall strategic and long-term oversight of the project will be the responsibility of **the Project Board**, which consists of one representative of each beneficiary (partner) and is chaired by the coordinator. This board will ensure, from a strategic perspective, that the project is proceeding according to plan and that the research being carried out meets the project objectives. The PEG, in collaboration with the work package leaders, will implement decisions taken by the Project Board.

The Project Board will meet every six months at the plenary meetings; it has the following main tasks:

1. To **monitor work progress** and suggest necessary redistribution of work plan and budget in case of unforeseen events that can affect the outcome of the success of the project, such as new opportunities, new ideas or unexpected failures of planned tasks;
2. To supervise the progress and **assess the quality** of the work, to follow up results and support any corrective measures accordingly, in order to attain the expected impact;
3. To **solve any emerging problems**, including conflict management, without delay;
4. To **take any necessary action**, if any partner fails to come with the expected deliverables, and thereby to ensure a successful completion of the project;
5. To handle **intellectual property** issues;
6. To manage and respond to **risk**.

3.2.3 Work-packages

AIDPATH is structured as a collection of work-packages (**WPs**), each of which contributes to the overall achievement of the project aims. Each WP has a nominated leader, and a team consisting of all beneficiaries that contribute to the WP.

Role of the WP leaders: WP Leaders are responsible for ensuring that the WP team reaches the objectives and milestones of the WP and produces timely deliverables of a high standard. The other beneficiaries involved in the WP (the WP team members) report to the WP leader in the context of this particular WP. Each WP interacts with the other WPs in their work, combining to deliver the project as a whole.

Process: At the project start, WP leaders will clearly establish the responsibilities of each member of the WP team. Throughout the project, the WP leader will report progress to the *PEG* and the *Project Board*. Where any issues or delays arise, the WP leader will work with the rest of the *Project Board* to resolve them in a timely manner. WP leaders may convene meetings of their WP team (additional to the plenary meetings) as needed. WP leaders will also present WP results at plenary meetings and provide any required material for periodic reviews.

3.2.4 Appropriateness of this Structure for AIDPATH

AIDPATH combines multiple complementary skills and disciplines, into a holistic AIDPATH IoT infrastructure, which combines several **innovative technologies** to meet the needs of **smart manufacturing hospitals**. Bringing these perspectives and skill-sets together, and keeping them in focus at all times, is the underlying reason for the creation of the PEG, as a strategic management group. Thus, while the PEG is an additional layer of management, over and above the usual H2020 'project board', it acts more as a three-part coordinator, with in-depth skills across all the domains where management is required.

The **Project Board** structure is a 'standard' Horizon 2020 approach to multi-partner projects, with which the team is very familiar, and which we have used effectively across several other projects. Many of the team have worked together in the past, and share the trust and confidence in one another's performance that is important in a project as ambitious as AIDPATH. The **WP leadership** model enables very clear and familiar lines of responsibility and reporting, so that all partners are aware of what is expected of them, what resources they can call upon, and how they relate to other WPs and the PEG.

3.2.5 Project Management Procedures

Plenary meetings

An initial kick-off meeting and workshop will be held to launch the project. All project board members will attend this meeting. The agenda will include presentations from all WP leaders, with a specific focus on the activities planned for the first six months of the project for WPs starting in M1. The plenary meeting will be followed by a knowledge-transfer workshop that builds the basis for the AI Use Case Definition, process definition and requirements analysis. These will be followed by irregular workshops with confined partner constellations in order to detail out use cases and specifications required for the different work packages. The

WP leaders are responsible for ensuring that the results of each workshop are shared with the consortium and coordinated with the other WPs.

Thereafter, consortium-wide ***plenary meetings*** will be held every six months. All partners will be represented at every plenary. The aim of the meetings will be to present, discuss and analyse research results, plan for upcoming activities and track progress against the project plan (milestones and deliverables). Additional technology workshops will be held throughout the first 18 months with larger and smaller subgroups of the consortium, to bring together all early-stage modules, under the leadership of the IPT engineering team.

Plenary Agenda: The *project executive group (PEG)* will set the plenary meeting agendas, in consultation with the wider *Project Board*. A number of items will be permanently on the agenda: WP updates, risks, issues and knowledge gained, market updates, project management and budgeting.

Minutes: The *coordinator* will circulate minutes of the meetings, including any action items, to the consortium. The minutes will be used to document progress and capture any key decisions, discussions and agreements reached.

Other meetings

WP leaders may choose to convene meetings at the WP level, as required. They may elect to hold meetings with other WP teams in order to organise collaborative activities. The majority of such meetings will occur via tele- or video-conferencing. Minutes will be taken for all meetings and circulated to the WP team, as well as the *Coordinator*.

Communication tools

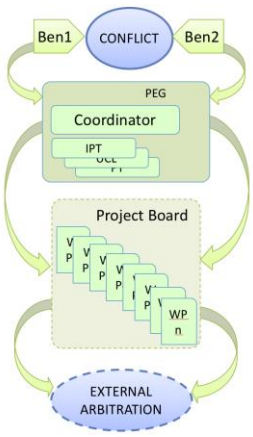
Clear and effective communications are essential to the success of any multi-partner research project. A number of communication tools used extensively by the consortium during preparation of the AIDPATH proposal will be employed for regular communication during the project:

- ***Tele- and web conferences*** will be facilitated using a suitable technology such as Microsoft Teams;
- A project wide ***mailing list*** and a ***list of key contacts*** will be established and maintained by the coordinator;
- Documents will be shared within the consortium using a secure document sharing solution such as Fraunhofer ***OwnCloud***;
- Scheduling will be facilitated using Fraunhofer-based Tool ***Foodle***.

Consortium agreement

A consortium agreement (CA) based on the DESCA 2020 model (http://www.desca-2020.eu/) will be drafted and signed by all beneficiaries before the project starts. The CA will govern a range of consortium interactions, including decision-making, conflict resolution and financial processes. All ***Intellectual Property*** issues, e.g. ownership and access rights, will be regulated in the CA. Given the significant number of cutting-edge companies involved in AIDPATH, and the high likelihood of further development and product breakthroughs, the CA will be carefully drafted and can be expected to provide a good deal of detail around ownership and access to IP.

Conflict resolution



The consortium is fully ***committed*** and aligned to the objectives of AIDPATH and to taking Smart Hospitals to the next level. All partners are excited to take part, and many expect to benefit by developing know-how and IP that will increase their commercial competitiveness and market share.

Many of the participants have worked together in the past, and are confident of fruitful and harmonious collaboration in this new project. Furthermore, the details of the research programme and procedures for interacting with each other will have been clearly defined before the project start in Annex I (*Description of Action*) of the Grant Agreement and the CA, respectively. In this context, *conflict is not anticipated* and the consortium will strive for consensus in decision-making. However, a conflict resolution process will be put in place should the need arise. The concerned parties will inform the *coordinator* of the conflict. The *coordinator* and PEG will engage in discussions with the concerned parties to ascertain if a mutually agreed resolution can be reached. If this is not possible the matter will be referred to the *Project Board*, who will discuss the matter in plenary session. This discussion may be held at a regular board meeting or at a special session, depending on the urgency of the matter, as decided by the

coordinator. If the board is not able to reach a satisfactory resolution, the issue will be referred to external arbitration, as detailed in the CA.

Progress monitoring

The PEG will track progress of the project versus the project plan, as laid out in the *Description of Action*. A project calendar, including milestone and deliverable dates, will be generated at the project start and maintained over the project lifetime. Each partner will submit a progress and budget summary report (1-2 pages) to the *PEG* every 6 months. A summary of these reports will be shared with the consortium. These reports are staggered with the plenary meetings, to provide a project management 'heartbeat' every three months. Contractual deliverables will be submitted to the *PEG* a month in advance of the due date, to allow for quality assurance and to ensure that all deliverables are submitted on time.

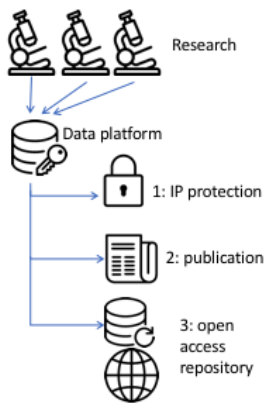
Quality assurance

All deliverables will be prepared using a common template. Each deliverable will be reviewed by a partner not involved in its preparation in order to ensure that high quality deliverables are submitted to the Commission. The *PEG* will also carry out a final review before submission.

3.2.6 Table 3.2a List of Milestones

MS #	Milestone name	WPs	Date	Means of verification
1	AI Use Cases Defined	3,4	9	Use Case Description Documents
2	Complete Data Sets for AI Training available	3,4	18	Complete data sets uploaded to a shared platform
3	AIs trained, validated and ready for deployment	2,3,4	33	Deployable program and reports on AI and validation
4	Cyber physical manufacturing infrastructure incl. Automated protocols completed and ready for deployment	5,6	36	Physical modules built, deployable software programs available and documentation on functional testing
5	Installation and validation of the cyber physical infrastructure at the clinical site complete	7	48	CAR-T manufacturing runs completed, and results of product validation available

3.2.7 Innovation Management



Advanced therapy development and production are among the most dynamic and rapidly-innovating of all industry sectors over the last several years. The AIDPATH project is primarily concerned with applying and integrating technologies from new companies, and breakthroughs from leading scientific teams, to the real-world challenge of producing advanced therapies for large patient populations.

We confidently expect that AIDPATH will lead to important product and technology innovations for several partners, and that these innovations will feed into new and improved products. Similarly, we anticipate important publications, new IP and potential spin-out and licensing opportunities for the academic and research partners, including IPT, UCL and others. Managing this innovation effectively, to protect its value while maximising its scientific and societal benefit, will be an important project management task, led by IPT's innovation team.

The innovation management process will build on the Consortium Agreement, which will list the key background that each beneficiary brings to the project, and that states any up-front agreement about IP ownership. As far as possible, we will allocate **ownership of each piece of new IP to a single beneficiary**; this beneficiary will typically be the one that brings the most related background IP to the project, and who will typically lead the work that generates the new Results ('foreground' in FP7 terms). This greatly simplifies its subsequent exploitation and translation into real-world impact.

Most new IP is expected to lead to **patents and protection** first, followed by scientific **publication** and open access to experimental and supporting **open data** wherever possible. Much of the new knowledge generated

by the project will relate to the products and IP of the partners; scientific and media publication will also serve as 'sales' or 'promotion' information, as discussed below under Dissemination and Communication.

Identifying and managing innovation: IP will be reviewed as a standard agenda item at plenary meetings. However, as soon as a new invention is noted, an Invention Discovery Form (IDF) will be completed by the scientist(s) involved. Copies of lab notebooks and supporting data will be made, so that the inventorship is clearly defined. Nature and ownership of the new IP will be formally recognised at the next plenary meeting.

Where **jointly-discovered Results IP** emerges as a result of collaborative work, the Consortium Agreement will be used as the basis for case-by-case agreements on ownership. The CA also states the processes for claims of co-ownership, in case a beneficiary believes it has contributed to some IP, and that this should be recognised. The relevant partners will work together with a third partner (nominated by the PEG) to agree a joint ownership agreement, including agreement on access, licensing to third parties, publication, etc.

Publications Management

AIDPATH will generate numerous scientific publications, including early papers (system design, AI and bioprocess concepts, methods, systematic reviews) and later results publications, in areas across smart hospitals, artificial intelligence, advanced therapy manufacture, decentralized therapy production, automation and advanced manufacturing and more. The correct management of this process is critical, if the team is to maximise innovation and publish the best possible work in high-impact venues.

A **publication plan** will be agreed at the kick-off meeting, and then further developed at the M6 and M12 plenaries, to identify which partner(s) will lead the work on which papers, on which topics. In general, many papers will focus on the work of a single work-package, with the WP leader acting as lead author, and all other partners being co-authors, subject to the ICMJE/Vancouver conventions. The PEG will note the publication plans of each partner and will track progress against these. Other opportunities for publications (e.g. workshops and conferences) will be taken, where they do not jeopardise 'big' publications. All publication decisions will be **subject to IP protection**; no publication will jeopardise or take away legitimate commercial aspirations. One year from project end, we will review progress on all publications, and focus on writing up results and submitting papers during the final year.

Data Management

AIDPATH will generate extensive data, which is a key requirement for AI training, process control and quality assurance activity, and a strict requirement for audit trail and quality data and in-vitro and in-vivo studies for validation and testing purposes. The generation, collection and storage of data will be recorded, reviewed and updated regularly throughout the project to ensure data is FAIR (findable, accessible, interoperable and re-usable). The DMP will include what data the project will generate, whether and how it will be made accessible for verification and re-use, and how it will be curated and preserved. As laid out in section 2.3 *Data Management, Publication and Open Access*, we will ensure that to generate scientific publications, data availability and preservation. Extensive data will be collected and generated throughout the AI work packages (WP2 – 4), which are used for AI training and application. The *in-vivo* and *in-vitro* validation work (WP7) will generate information about experiments and their results, which confirm the quality of the platform end-product and which may feed back into the platform for iterative refinement. All data storage and sharing will be in accordance with GDPR and anonymized as far as possible.

Open data: we are committed to making our results and our research available to other researchers to enable replication and further use of our work (subject to commercial intellectual property constraints). By sharing our data as openly as possible, we intend to make it feasible for others to verify the quality of our work and results, and thus to drive demand for the products and services offered by consortium members into the future.

Data sharing will be achieved by placing our aggregated and anonymised research data on the open-access data repository chosen at the beginning of the project, such as ELIXIR, Fordatis institutional research data repository of the Fraunhofer-Gesellschaft or the UCL Research Data Storage (URDS) service. These data repositories ensure curation and long-term preservation and thus long-term availability.

Future Planning

The AIDPATH project has been conceived with a very future-facing perspective. From day one, our plans and activities will reflect our aim to have AIDPATH systems adopted at wide-scale in large hospitals across Europe and beyond, to intelligently and consistently produce new therapies (some not even invented yet). The project as proposed is a stepping-stone towards creating smart hospitals of the future which act as treatment and therapy production hubs. With AIDPATH, we will lay the foundation towards this future not only by

prototyping and demonstrating a holistic, smart manufacturing IoT infrastructure in the hospital, but also by preparing exchange channels (both on data and communication side) to allow exchange of knowledge and expertise between the hospital sites. We anticipate that his technology will generate a fertile breeding ground in which future AIs, technological developments and other advanced therapy production processes can be seamlessly integrated, thus further increasing product quality, benefits for the patient and decreasing costs of personalized treatments.

As this dynamic innovation and commercial area evolves during the project, we will adapt and respond to it. Full-day innovation-review workshops will be held at M18, 36 and 48, immediately preceding the plenary meeting. These workshops will review changes in the state of the art (scientific and commercial) and reflect on whether this evolution requires any change in AIDPATH, or in its future or business planning. Strategic alliances and joint forces will be sought with other clinical partners and across the stakeholder community. In order to provide a starting point for further funding, licencing and sponsoring through funding agencies, large companies and investors, we will hold an innovation da where technologies developed in AIDPATH can be presented and pitched to interested parties in the above mentioned group.

Each partner (both academic and commercial) has its own perspectives on how this project will lead to exploitation and progress (see WP10 and section 2.4 Exploitation for details). However, we also share a common view that further shared R&D and further projects are likely to offer more stepping-stones along our shared research agenda. This reflects the collaboration of some of the partners already in projects such as AutoCRAT and Jointpromise, which have directly informed and enriched the plans for AIDPATH.

3.2.8 Risks

The AIDPATH project is exposed to typical risks for multi-disciplinary project that combines hardware (modules, bioreactors and sensors), software (to control the hardware), Artificial Intelligence and cutting-edge bio-technology. These include technical compatibility issues, software interoperation and maintenance, the development or evolution of innovative technologies, and the need for a shared understanding, across specialist fields, of the capabilities and constraints inherent in each discipline.

However, the consortium has the required expertise and experience, with strong track records in the specific areas required. Automation/manufacturing and advanced therapies are both represented in the PEG; the overall project board holds a uniquely strong mix of real-world expertise, brand-new technologies and breakthroughs, and scientific excellence.

The following table explores the **most relevant risks** and how as a team we have mitigated against them (and/or, if they arise, how we will overcome the problems they represent).

Table 3.2b: Critical Risks for implementation

Description of risk	Likeli-hood	Involved WPs	Proposed risk-mitigation measures
Delay within the work packages, in achieving milestones, or partners remiss in the execution of assigned obligations	Medium	All	Critical paths of dependencies among actions have been identified. Ensure to keep the schedule by regular checks on cost-performance schedule, supported by bi-monthly progress meetings. Regular phone conferences and updates via email. In serious cases, coordinator will accord one month for corrective action and if no sufficient improvement has been made, backup work force will be allocated and work redistributed.
Conflict regarding IP ownership	Medium	All	The Consortium Agreement will explicitly name the beneficiaries who have ‘first call’ on IP in specific areas. These will reflect the background IP brought to the project by each beneficiary. Where conflicts arise, conflict resolution process will be followed as is outlined above
Lack of data availability to build data driven AI models	Medium	2, 3	(i) Use publicly available data repositories, (ii) simulate data
Problems with data quality (erroneous values, missing data etc.)	Medium	2, 3, 4	Fix problems by requesting new data and/or revising data processing.

Insufficient data coverage	Medium	2, 3, 4	(i)Request more data, (ii) reduce the scope, (iii) engage with stakeholders to generate data
Development of the use cases are delayed	Medium	3,4	The progress of the use cases will be continuously monitored in order to recognise any discrepancies and deviations from the original plans.
Missing interfaces of devices	Low	5	Market research to identify best device solutions, engagement with device vendors, Integration via software agents possible even for heterogeneous interfaces
Lack of automatability of devices	Medium	5	Market research to identify best device solutions, engagement with device vendors, IPT can also prototype additional devices / functionalities and amend technology
Unexpected process complexity	Low	5	Consortium consists of multiple process experts (e.g. UKW, UCL, IZI). Process definition at the very beginning, conducting of workshops and expert interviews to address this early on, reducing complexity in the process as a demonstration process
Deployment of AI solutions fails or delays	Low	6,7	Including deployment strategy and productionizing tests in the development of the AI solutions. Continuous deployment tests over the course of WP 2,3 and 4.
Sustainability of AI solutions is not given (e.g. new data cannot be successfully included)	Low	6,7	Including retraining strategy and data flows into the AI development. Varying data formats and values during test phase to emulate upcoming changes and variances.
Unexpected changes to cybersecurity & privacy standards and regulations	Low	8, 9, 10	Creation of non-official liaison with the relevant EU standards and regulations, designate a project officer to follow closely these organisations and report to the group.
Incompatibility of AI with IT-infrastructure in the hospital	Medium	9	Early communication between AI-Partners and IT-department at UKW
Dissemination fails to engage audiences	Medium	10	We have extensive experience in scientific dissemination and communication. The value of our results to our target audiences is potentially revolutionary – lack of interest is not to be expected

3.3 Consortium as a whole

3.3.1 Consortium

The AIDPATH consortium has been assembled to ***meet the specific requirements*** of the project. These objectives need an ***outstanding team***, with a broad mix of quite specific skills. In order to deliver a project of this broad scope and substantial size, leaders from several disciplines need to work together effectively. The AIDPATH consortium consists of fourteen partners from six different countries involving research institutes, SME's and CMO's making it a consortium with broad knowledge and expertise perfectly aligned with another. With an extensive involvement of SME's (RAL, IRIS, PAN, AGL) the AIDPATH projects gives these companies the opportunity for an advancement of their products and an engagement manifold community. This consortium will built on and encourage further integration and leadership of women

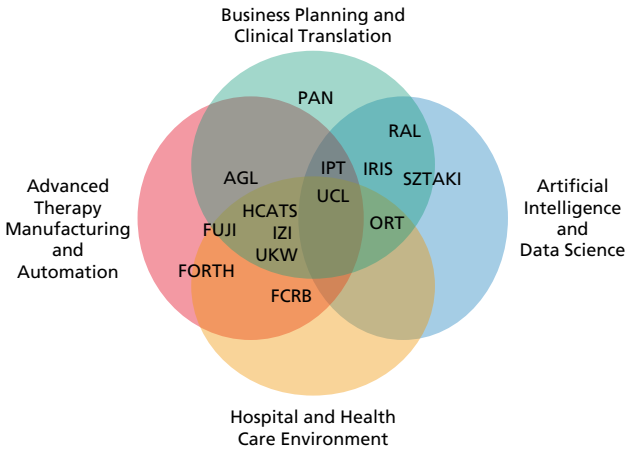


Figure 12: Consortium Venn diagramm

towards achieving equal gender balance. For example, Dr. Katrin Mestermann from UKW and Jelena Ochs from IPT will take over leading roles in the project and WPs on a strategic and day-to-day basis.
Our vision to bring an AI-driven autonomous CAR-T cell production into the hospital environment under compliance of all regulations can only be fulfilled by a consortium having a deep knowledge and experience in (1) AI and data science, (2) bio processing and automation, (3) hospital and healthcare environment and (4) business and clinical translation.

Here we demonstrate how the AIDPATH consortium meets its **requirements**:

- **Leadership in Artificial Intelligence and data science**

With IPT, SZTAKI, IRIS and UCL the AIDPATH consortium has an excellent group of AI partners, which are all experts in their specific fields. The SME IRIS will take advantage of their expertise in the application of AI algorithms for digitalized tailored solutions applied to different sectors and will deliver ML algorithms for bioreactor process prediction and control, data analysis and the optimization of the AI pipeline. UCL complements this expertise with specific knowledge of the use of AI technology for cell production processes. As the leader of the 'Centre of Excellence in Production Informatics and Control' SZTAKI provides the necessary knowledge of the application of AI technologies for manufacturing and resources processes.

- **Leadership in cellular cancer immunotherapy with CAR-T**

With UKW and UCL, AIDPATH brings together the two leading institutions in cellular cancer immunotherapy in Europe. UKW has a more than decade-long track record in bringing innovative T-cell based cancer immunotherapy into first-in-man clinical trials (e.g. the T-cell engaging bispecific antibody Blinatumomab was clinically developed from Phase I through Phase under the leadership of UKW PIs: At present UKW has the most active translational CAR-T program in Europe (based on 2019 publication impact points) and is visible world-wide for its innovations in virus-free gene-transfer (Sleeping Beauty, e.g. Querques et al. Nature Biotech 2019); target identification and validation (SLAMF7, Gogishvili et al. Blood 2017; FLT3, Jetani et al. Leukemia 2018; ROR1, Stüber et al. JITC 2020) and CAR-T technologies (Remote control, Mestermann et al. Science Translational Medicine 2019). UCL currently has the largest IIT CAR-T clinical trial program in Europe with innovative CAR-T products to treat B-cell leukaemia and lymphoma (CD19), T-cell lymphoma (TCRbchain) and paediatric neuroblastoma (GD2).

- **Advanced therapy manufacturing and automation**

IPT is the leading organisation in Europe, perhaps worldwide in this field. They have specific hands-on experience of building automated stem cell production platforms, via their StemCellFactory, JOINTPROMISE and AutoCRAT projects. UCL has a proven and extensive track-record in the manufacture and clinical development of ATMPs as evidenced by the fact that over 70% of all clinical trials in the UK involving the use of ATMPs takes place at UCL. UCL is considered a world-leader with respect to bioprocessing and bio manufacture and leads large UK and EU consortia in these areas. The senior staff of IZI and its research and cooperation partners have a high level of expertise in GMP, especially regulatory consulting, prepares preclinical and clinical development. Also, IZI has a GMP Process Development Unit responsible for transferring manufacturing processes from the lab into a clinical setting. FORTH is the leading research institute in Greece and is a leader in metabolomics and also stem cell engineering and bio manufacturing holding expertise in both national and EU consortia in the field. The consortium also consists of excellent commercial companies in this area (HCATS, FUJI, AGL). This means that they tend to have a product or technology already on the market (or close to market); this level of technological maturity is important, if integration and automation is to be realistic in the timeframe of the AIDPATH project.

- **Comprehensive knowledge in hospital and health care environment**

For AIDPATH to be successful, it needs to be accepted by the end-site users, i.e. hospitals. In order to guarantee compatibility with the hospital's requirements and for AIDPATH to meet clinical needs, UKW, FCRB and ORT are partners in the project. UKW is internationally recognized as a "Centre of Excellence in Stem cell Transplantation and Cell Therapy", and is therefore a hotspot of clinical translation. UKW has both access to patient, patient data and material as well as the knowledge and infrastructure for preclinical in vitro and in vivo analyses of the final product. FCRB is leading different research projects with focus on cell immunotherapies and has a large experience in immunotherapy, included clinical trials and research projects and thus can provide insights health care and treatment considerations. ORT applies its IT and data management solutions already successfully in the hospital context making it to an important partner for the implementation of the AIDPATH infrastructure into the hospital.

- **Business planning and clinical translation**

Avoiding a 'solution that nobody wants' is essential. The project must generate outputs that are acceptable to healthcare providers and funders, and that will transition successfully to a clinical environment. While cost is a major factor here, issues such as regulatory approval and certification of the technology and of its outputs are also critical success factors. PAN has extensive expertise in business modelling and economic evaluation of healthcare innovations from early onset until market access in various EU countries/ markets.

- **Regulatory expertise in clinical and cybersecurity issues**

UKW are very expert in ATMP regulation and clinical approvals, in GMP manufacturing, in technology certification and in the translation of research into the clinic. UCL have also built up substantial track record in the regulatory aspects of advanced therapies in the course of their ongoing research activities. With RAL the consortium has an expert in the field of cybersecurity addressing business privacy and security issues through comprehensive risk analysis and cybersecurity by design during the project. With regard to cyber security, IZI can rely on appropriate very high level working in-house IT personnel and on specialized Fraunhofer Institutes within the Fraunhofer Society with which numerous joint projects are being worked on.

- **Coordination and Management**

AIDPATH is a complex project, with several quite different disciplines from engineering and science, business and regulatory affairs, all playing critical roles. The Project Executive Group (PEG) has all the skills and resources needed to combine multiple disciplines and perspectives, to effectively orchestrate the project and to deliver its aims. The PEG members have managed many excellent R&D projects at national and EU levels.

- **Marketing, communications and outreach**

IPT, UKW and UCL all have significant professional communications teams, the resources of which will be at the disposal of AIDPATH for outreach and dissemination.

The table below summarises the skills which the partners bring to the project.

	Artificial Intelligence	Automation and advanced production	Cybersecurity	Data science and management	Cell production technology	Regulatory issues	Stakeholder engagement	Communication & Dissemination	Commercialisation	Management & Administration
IPT	☑	☑		☑	☑		☑	☑	☑	☑
IZI		☑		☑	☑	☑	☑	☑	☑	☑
UCL	☑	☑			☑	☑	☑	☑	☑	☑
FORTH		☑			☑		☑	☑		
SZTAKI	☑							☑		
UKW					☑	☑	☑	☑		☑
AGL		☑			☑				☑	
FUJI					☑					
HCATS		☑			☑	☑		☑	☑	
FCRB					☑		☑	☑		
IRIS	☑			☑						
RAL			☑			☑				
PAN							☑	☑	☑	
ORT	☑		☑	☑				☑	☑	

Table 1: Skills Matrix - the skills and experience needed to deliver the project is fully available within the consortium

3.3.2 Industrial / Commercial commitment

The AIDPATH project is rich in commercial partners and technology companies. All of these companies, from SMEs such as AGL and IRIS to large multi-nationals such as HCATS and ORT, are excited about the AI-driven, autonomous CAR-T cell production, the opportunities that it presents, and the value of overcoming key challenges to large-scale, lower-cost availability and dissemination. For all of these companies, AIDPATH

represents an opportunity to enrich their own skillset and product development by collaboration with top research teams and with other industrial leaders. Each industrial partner will demonstrate the effectiveness and value of its own offering, but also gain access to a market for in-clinic advanced therapy production, which we anticipate will appeal to hundreds or thousands of hospitals across Europe and beyond, as well as to CMOs and research groups. The following table summarises the industrial partners and the benefit that will accrue to them from AIDPATH; these benefits underpin and explain their commitment to the project.

Company	***Benefit***
AGL	Extension of existing Facer bioreactor technology to CAR-Ts. Integration of standalone technology into modular platform opens routes to new markets and the clinic. Access to major technology companies for new strategic relationships.
FUJI	New technologies for personalisation and optimisation of FUJI existing xeno-free media products. Substantial added value to existing products. Access to new markets (clinical domain)
HCATS	Novel manufacturing platform for improved quality CAR-T product. Opportunity to massively increase current production of CAR-T, and access to new clinical markets.
IRIS	Bioreactor process setup, simulation and control (smart sensors) including the use of AI algorithms, techniques, and associated data processing and decision support.
RAL	A tailored set of cybersecurity and privacy by design requirements and evaluation criteria tailored to the Healthcare Smart Manufacturing industry
PAN	Extension of its Health Technology Assessment services to AI as new technology.
ORT	Supporting clinical and manufacturing data registration system with logged data and intelligent data processing tool (LogiqCare/Science). Integration of data for generic modelling. An MDR-compliant sustainable platform for clinical-decision support tools. Building a relevance platform to engage the user community for future developments.

Overall, the **benefits to the companies** come in two components: (a) in the immediate term, they get the opportunity to combine their own products with those from other leaders in the field, and to enhance their own product lines and competitiveness; (b) in the medium term, they are embedded in an excellent clinical platform which can made advanced therapies available to many more patients, at much lower cost, in many more hospitals and other production environments.

3.3.3 Other countries and international organisations

Hitachi Chemical Advanced Therapeutics Solutions (or "**HCATS**", formerly known as PCT) is a global contract development manufacturing organization (CDMO) with headquarters in New Jersey, USA, with extensive expertise in the area of CAR-T manufacturing. HCATS has been an external cell therapy optimization, development and manufacturing partner for more than 21 years. With this background, they have been working from the beginning with the first developers of CAR-T technologies and their clients now include a number of the key players in the industry. Their T cell-specific experience and expertise includes both cell therapy manufacturing, and technical understanding of their customer's products. With a portfolio of client programs that span pre-clinical to commercial, HCATS can provide insights on industrial T cell manufacture beyond a single process or product as most other companies in CAR-T manufacture today could. Therefore, they provide an invaluable contribution to the AIDPATH project, where they will be able to share insights on manufacturing, logistical and cost aspects of CAR-T manufacture, thus complementing the clinical view. As a multinational company, they will be able to provide a truly global manufacturing perspective with operations also in EU and Japan. **HCATS** will be contributing **on their own costs**, and thus will not be claiming any contribution from the European commission.

Consequently, within this project, there are **no countries** that are **requesting EU funding** and based in a country or **international organization** that is not automatically eligible for funding.

3.4 Resources to be committed

3.4.1 Table 3.4a: Number of required person months

#	Short name	WP 1	WP 2	WP 3	WP 4	WP 5	WP 6	WP 7	WP 8	WP 9	WP 10	PM per partner
1a	**IPT**	30	28	3	26	36	36	21	12	12	9	213
1b	**IZI**				2.7	9		22.6		2.7		37

2	**UCL**	2	3	75	2	23	1	2	1	2	1	112
3	**FORTH**	1		73		7					2	83
4	**SZTAKI**		19		70		1				2	92
5	**UKW**	0	12	12	6	12	6	48	3	6	3	108
6	**AGL**	2		25		55		10			4	96
7	**FUJI**	1		16							1	18
8	**HCATS**											0
9	**FCRB**	1	1	15	1	1		1	1	1	1	23
10	**IRIS**	1	20	29			1				3	54
11	**RAL**						20	6	18		2	46
12	**PAN**	2.25						16		10.5	16	44.75
13	**ORT**	2	10		10		51	10		9	10	102
Total PM		42.25	93	248	117.7	143	116	136.6	35	43.2	54	1028.75

3.4.2 Table 3.4b: Other Direct Costs

01/ FHG	**Cost (€)**	**Justification**
Travel	64.040	Travel to consortium meetings (2 per year), additional workshops, conferences and fairs
Equipment	565.000	Licences for ERP program, Automated manufacturing equipment including flow cytometer, cell counter, robot, analytic devices, transportation system, bioshaker, nucleofection device, cell separation device, controlled rate freezing device, clean room cabinet, cooled storages, PCs, PLC-PC
Other goods and services	175.000	Consumables and lab material, Frame, PLC terminals, electrical cabinet, cladding, sensors, grippers, motors, pumps, pneumatics, axes and components for individual automated solutions, Costs for shipping of the modules, licences (windows, inventor, plc), open access publications, catering, conference and fair fees, CE certification support, audit, dissemination material, costs for travel of the persons on the advisory board, rental of AI cluster computing power
Total	804.040	

02/ UCL	**Cost (€)**	**Justification**
Travel	53.332	Travel and Subsistence costs for 2 investigators and 4 researchers to attend 8 plenary meetings and 1x national and 1x international conference per year over project duration
Equipment	110.244	Analytical sensors and cell expansion systems
Other goods and services	125.436	Laboratory consumables (plasticware, reagents, antibodies, media, FBS, single-use bioreactors), Cost of core research facilities (ACBE pilot plant, Flow Cytometry Core Facility), Audit costs, catering costs)
Total	289.012	

03/ FORTH	**Cost (€)**	**Justification**
Travel	20.000	Travel costs for PhD student, postdocs and PIs to attend conferences and AIDPATH project meetings. Also this covers registration expenses for conferences and symposia.
Equipment	85.000	(i) A liquid handling robotic device for automating the metabolomics protocols and assay preparation, (ii) a mini-bioreactor system

Other goods and services	170.000	Laboratory consumables for cell culture, plasticware, culture media , growth factors, GC-Mass Spec consumables, bioreactor spare parts, software licenses for modelling, costs for auditing and for open access publications and for catering expenses for team members
Total	275.000	

05/ UKW	**Cost (€)**	**Justification**
Travel	40.000	To consortium meetings, meetings with regulators, patient advocates and policy makers.
Equipment	200.000	Equipment for automated CAR-T manufacturing and AI implementation incl. IT hardware.
Other goods and services	210.000	Consumables and reagents for CAR-T manufacturing runs, functional validation experiments, sudit
Total	450.000	

06/ AGL	**Cost (€)**	**Justification**
Travel	5.000	Travels for meeting with the partners and for installing the Platform
Other goods and services	433.000	Gases, Media for cell culturing, Sensors, materials needed for assembling the system, Audit, Facer Case Manufacturing, Cartridge Injection moulding
Total	438.000	

07/ FUJI	**Cost (€)**	**Justification**
Travel	8.000	Travel to plenary meetings and technical WP meetings @ UCL and IPT.
Other goods and services	74.870	Reagents consumables, test kits and ingredients for cell culture media.
Total	82.870	

08/ HCATS	**Cost (€)**	**Justification**
Travel	-	
Equipment	-	
Other goods and services	-	
Total	-	

09/ FCRB	**Cost (€)**	**Justification**
Travel	8.000	1 meeting per year per 2 people
Equipment	82.000	Electroporator
Other goods and services	10.000	Reagents, Publications
Total	100.000	

10/ IRIS	**Cost (€)**	**Justification**
Travel	21.000	Considering KO meeting, review meetings, General Assemblies
Other goods and services	29.000	Licensing of software and cloud computing services, GPU for fast data processing, Cost of publications, dissemination costs for conferences and events
Total	50.000	